Contract No. FA864923P1125
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Executive Summary

Background. The Department of Air Force (DAF) generates hazardous waste from mission critical operations, including aircraft repair, emergency/fire training or response activities, and fuel storage. Historically, United States hazardous waste management involves landfilling or incineration. Recent moratoriums preventing Department of Defense (DoD) incineration of emerging contaminants[1] indicate that historical practices are unsatisfactory and presents an opportunity to invest and implement sustainable solutions. Per- and polyfluoroalkyl substances (PFAS) represent a class of emerging contaminants, some of which were recently classified as a hazardous substance by the EPA (2024). DoD estimates its PFAS environmental liabilities to be as much as $9 billion, a 1,500% increase since 2022[1-3].

AxNano's proprietary waste management process uses supercritical water oxidation (SCWO) to eliminate hazardous waste disposal risk, including PFAS. This technology can be deployed for public and private entities across the globe. While "forever chemicals" are making headlines, DAF needs a waste management overhaul for both emerging and existing hazardous wastes.

Objectives: The overall goal of this AFWERX Phase II project was to demonstrate that SCWO when combined with AxNano's automated feed management system (AFS) creates additional value for DAF hazardous substances treatment beyond SCWO alone. This Phase II program addressed a key technical assumption of our value proposition by testing continuous-flow operations with multiple DAF hazardous wastes on AxNano's pilot SCWO system.

Methodology: AxNano's prototype AFS was integrated into a pilot SCWO system at our partner General Atomics' facility. Industry-relevant solid and liquid wastes containing PFAS were secured from AF bases. AFS processing parameters were established to create a slurried waste feed capable of supporting continuous SCWO operation. Analysis of influent versus effluent including PFAS, chemical oxygen demand, total organic carbon, metals, percent solids, and other wastewater-relevant parameters were measured. A technoeconomic analysis was generated from empirical data.

Key Successes: Over the course of this 21-month program, the following successes were realized

- built and operated a prototype AFS
- processed of approximately 1,445 gallons of five different PFAS-containing waste including solids and liquids
- reduced waste processing time for typical SCWO pre-treatment from tens of gallons per day to 0.33 gallons per minute
- achieved 99.995% PFAS reduction efficiency in a granular activated carbon (GAC) waste that was previously unable to be processed by SCWO
- established a return on investment model demonstrating 83% cost reduction over 10 years for on-site treatment of at least 351 tons solid and 3162 tons liquid waste per year compared to off-site incineration and landfilling
- created a technical data sheet for environmental contractors to use when applying for permits, reduces their time and labor costs
- secured two purchase orders for AFS lease for private and federal contracts
- received two requests for proposals for AFS lease in the private sector

Conclusions: DoD selection criteria for PFAS waste disposal include both technical and economic considerations. With our AFS, SCWO is now a viable alternative to landfilling and incineration with a 83% cost reduction over 10 years, when installed on-site and operating under continuous operations to treat mixed solid/liquid waste. Additional testing and vetting are required for SCWO to become an approved PFAS disposal method by the EPA, after which it could be adopted as a standard practice by the Air Force/DoD. Upon regulatory approval, SCWO would be used for disposal of remediation-derived waste by AFCEC-CZO (Air Force Civil Engineering Center Environmental Operations Directive), at bases within concurrence and coordination with CEG/CEIE (Civil Engineering Group, Compliance) for on-site, or through Defense Logistics Agency (DLA) for off-site at a Treatment, Storage, and Disposal Facility (TSDF) (both AFCEC and CEG).

Next Steps: AxNano has filed for trademark of AxSure™ as well as patent applications on the underlying technology. We have already achieved a path for Phase III transition through two POs for leasing the AFS and two Request for Quotes (RFQ) from private sector. AxNano has applied for Tactical Funding Increase (TACFI) funding in collaboration

with Robins AFB and Air Force Specialty Code (AFSC) to support additional waste testing and improved PFAS analysis to address remaining questions from this Phase II test. Private investment funding will support expansion of our Sales & Marketing team, Intellectual Property filings, and manufacturing. A Defense Innovation Unit program is supporting additional AFS scale-up work and engineering reliability testing.

Figure 1. AxNano's Phase II (Contract # FA864923P1125) successfully demonstrated the Automated Feed System (AFS) in a relevant environment (pilot SCWO testing facility), processing waste that was otherwise not treatable by SCWO. A successful Phase III transition is demonstrated by securing two purchase orders supported by non-SBIR funds.

1.0 Project Background

The Department of Air Force (DAF) generates hazardous waste from mission critical operations, including aircraft repair, emergency/fire training or response activities, and fuel storage. Historically, United States hazardous waste management relies on landfilling or incineration, which are not sustainable solutions for mitigating the risks of harmful wastes. When hazardous compounds are not fully destroyed, the risk of human exposure remains - most commonly through consumption of contaminated water. Ensuring safe, clean, and reliable water supplies is critical to supporting the military Operational Environment (OE). Recent publications highlight the importance of hazardous waste management for the DAF and broader Department of Defense (DoD). Recent moratoriums preventing DoD incineration of emerging contaminants[1] indicate that historical practices are unsatisfactory and present an opportunity to invest and employ sustainable solutions.

Per- and polyfluoroalkyl substances (PFAS) represent a class of emerging contaminants that are unique due to their recalcitrant nature and the strength of the carbon-fluorine (C-F) bond. A small subset of PFAS, including perfluorooctanesulfonic acid (PFOS) and perfluorooctanoic acid (PFOA), were classified as a hazardous substance by the EPA (2024) and have very low water cleanup levels (i.e., part per trillion levels). PFAS are present in environmental media at nearly every DoD site, stemming, in part, from DoD's past use of aqueous film-forming foam (AFFF). DoD estimates its PFAS environmental liabilities to be as much as $9 billion, a 1,500% increase since 2022, and costs will likely continue to rise as the expected life cycle of remedies like pump & treat technologies can last for more than a century[1-3].

AxNano's AxSure™ system combines our proprietary automated feed management system (AFS) with supercritical water oxidation (SCWO) to effectively destroy contaminants that persist through traditional wastewater treatment methods. The ability of SCWO to mineralize PFAS has been previously demonstrated.[1–3] Improper management of the feed into SCWO reactor leads to faults and shutdowns, ultimately causing incomplete waste destruction, decreased throughput and increased operational costs. The test aimed to demonstrate the utility of our AFS to pre-process various

PFAS-containing waste streams, feeding them into SCWO, and support stable reactor conditions. The SCWO reactor uses high temperatures and pressures to completely break down organic contaminants into carbon dioxide, water, and inorganic salts, providing a clean and effective waste treatment solution as an alternative to landfilling and incineration. The AFS is a key component of our proprietary SCWO process, offering a unique value proposition beyond SCWO reactor alone.

The overall objective of this Phase II program was to build, design and test our AFS prototype and demonstrate that it is a valuable component of an integrated SCWO system by supporting continuous operations with a range of waste feed types.

1.1 CLIN010 Description

This document serves as CLIN010 which is a progress report toward Task 10. "Strategize for Phase III." Phase III of the Small Business Innovation Research (SBIR) program focuses on commercialization. Phases I and II are funded by the SBIR program. Phase I focuses on feasibility, bringing the solution to Technology Readiness Level (TRL) 4, where proof of concept is demonstrated through laboratory testing. Phase II focuses on prototype development, bringing the technology to TRL 6 or 7 by refining the solution and testing it in a relevant environment. Phase III projects receive funding from non-SBIR sources, such as government agencies or private sector investments. These projects extend or complete work from previous SBIR phases and aim to bring innovations to market.

Much of the work in Phase II focused on the design, building and testing of the prototype AFS, but effort was also devoted to other de-risking activities needed to support a Phase III transition. Our work in Task 10 specifically involved creating a strategy for integrating SCWO systems into waste management practices for our customers, considering the results from the prototype test, DAF stakeholder surveys, and permitting requirements. This report describes nonproprietary components of our current Phase III transition strategy.

Other work in Task 10 included designing of a full-scale system following lessons learned from testing. CLIN010 is the final deliverable for this program, therefore we have also included a final report summary as Appendix A to this report. Appendix A includes results and conclusions previously submitted as CLIN007, with the addition of results and conclusions for DQO 3 regarding waste categorization and cost analysis. Appendix B includes a presentation on stakeholder mapping conducted as part of the program to understand DAF workflow in preparation for Phase III transition. Appendix B was previously submitted as CLIN008. Appendix C includes a Technology Specific Report to support future permitting during Phase III. Appendix C was previously submitted as CLIN009.
The acceptance criteria for CLIN010 is that "Technical Point of Contact (TPOC) confirms reasonable effort to Task 10."

1.2 Overview of Phase II Project Results and Conclusions

Our Phase II program was designed to address key technical de-risking needed to support a Phase III transition. The test evaluated three Data Quality Objectives (DQOs), which guided the testing process by setting clear parameters for what needed to be measured and assessed. This ensured the results were useful for determining the feasibility of an AFS to process relevant volumes of Air Force-derived wastes through AxNano's SCWO process under continuous operation at an acceptable cost and efficacy. A high-level summary and conclusions of the DQOs are explained in the following paragraphs. The test was conducted in August 2024 at General Atomics Electromagnetic Systems (GA-EMS) facility in San Diego, using an AxNano-built prototype AFS and GA's existing pilot SCWO system.

In this document, we refer to the term "boundary conditions" as an acceptable range within which an identified parameter must fall for a certain condition to pass. For example, the boundary condition for particle size of solids in the waste feed for current SCWO operation is 0-250 micron. This means that the minimum particle size is zero (i.e., no solids present) and the maximum particle size that the current SCWO system can process is 250 microns. Particles larger than 250 micron would cause the SCWO system to shut down.

Table 1. Summary of Tests and Data Outputs and Major Outcomes for DQO 1.

DQO 1 Goal: Identify effects of the waste feed on PFAS destruction efficiencies and reactor stability.					
Test Number	Objective Description	Wastes Comparisons	Data Inputs	Data Outputs	Major Outcomes
1	Liquid PFAS Waste vs. Solid PFAS Wasted	AFFF wastewater vs. GAC	• Particle size • Solids wt% • BTU • PFAS concentration • Reactor temperature • Reactor pressure	• PFAS removal efficiency • Reactor temperature stability • Feed rate	• The form of waste (solid versus liquid) did not negatively impact PFAS removal efficacy because our AFS system was able to successfully produce a stable waste feed regardless of waste type. • Neither solid weight percentage (0-15%) nor particle sizes within the SCWO-operable range (e.g., 0-500 microns) noticeably affected PFAS removal efficiency or reactor stability due to AFS usefulness.
2	Low PFAS Concentration Solids vs. High PFAS Concentration Solids	Resin vs GAC	• PFAS concentration	• PFAS removal efficiency	• No correlation was found in starting PFAS concentration on eventual removal efficiency in this test. • No conclusions could be made on removal efficiency of resin versus GAC because analytical limitation on measuring starting PFAS concentrations on resin.
3	Low British Thermal Unit (BTU) PFAS Waste vs. High BTU PFAS Waste	AFFF wastewater vs reverse osmosis reject	• BTU • Ratio of waste:diesel:water in feed	• PFAS removal efficiency • Reactor stability • Cost per volume treated	• A positive correlation between BTU and PFAS removal efficiency was observed. • Reducing the flow rate by eight times (e.g., decreasing waste and increasing diesel) did not prove effective in enhancing PFAS removal efficiency, indicating that flow rate optimization may not be the key lever for improving efficiency, and attention should be given to other operational parameters, in particular maximizing waste feed BTU • Optimizing BTU of waste by mixing waste streams to decrease diesel requirements can decrease operating costs up to 93%.
4	Low Salt PFAS vs. High Salt PFAS	AFFF wastewater vs reverse osmosis reject	• Salt concentration • PFAS concentration	• PFAS removal efficiency • Metals in effluent	• High salt concentration did not negatively impact PFAS removal efficacy nor does it result in reactor corrosion (for the 1,445 gal duration of this test)

1.2.1 DQO 1 Overview

DQO 1 aimed to determine whether AxNano's process could effectively degrade recalcitrant compounds in waste that were resistant to current treatment methods while maintaining controllable SCWO reactor conditions and achieving complete degradation. The goal was to identify the effects of waste feed on PFAS destruction efficiencies and reactor stability during SCWO tests using various real-world wastes, including liquids and solids. The tests designed for DQO 1 are summarized in Table 1. DQO 1 was important for Phase III transition because future customers will select waste management approaches based on cost and technical feasibility (see Appendix 2). Proving that we can achieve PFAS destruction supports discharge goals. Improving reactor stability enhances waste destruction while also decreasing operating costs.

1.2.2 DQO 1 Test Summary

In Test 1, we aimed to evaluate the effect of PFAS form (solid versus liquid waste) on removal efficacy. We compared dissolved PFAS in AFFF wastewater to sorbed PFAS on spent GAC. Our hypothesis was that PFAS sorbed onto solid particles might be less susceptible to oxidation and destruction. However, the results showed a slightly higher removal efficiency for solid PFAS compared to liquid waste. SCWO treatment of GAC waste resulted in 99.995% removal efficiency, compared to 95.91% for liquid AFFF wastewater. This could be due to two factors: first, the starting PFAS concentration for GAC was much higher than for AFFF, allowing for greater destruction potential; second, the PFAS effluent of GAC was lower than that of AFFF, which may have been caused by residual GAC or ash sorbing PFAS in the effluent or by incomplete destruction of the PFAS in the AFFF. Other differences included the processing conditions, with AFFF wastewater fed at a lower rate and higher pressure than GAC. No correlation was observed between PFAS removal efficiency and solid versus liquid form or particle size. In conclusion, our results suggest that PFAS sorbed onto solids does not negatively impact removal efficiency in SCWO treatment, and the AFS system successfully prepared stable slurry feeds for SCWO treatment of solid-phase PFAS without affecting reactor stability.

In Test 2, we aimed to examine the effect of PFAS concentration on SCWO treatment of solids by comparing spent GAC and spent resin, which differed by an order of magnitude in starting PFAS concentration. PFAS removal efficiency was higher for spent GAC (99.995%) than for resin (98.52%). However, the two waste materials had distinct chemical and physical properties, such as particle size, modulus of elasticity, COD, and BTU, which complicated definitive conclusions. Resin's strong PFAS binding made desorption difficult, likely causing inaccuracies in measuring the starting PFAS concentration. Future work will address these analytical limitations by collaborating with external laboratories. Despite these challenges, we observed that resin had a higher modulus of elasticity than GAC, affecting the AFS's ability to achieve the target particle size, although the resin was mostly destroyed in the SCWO process. No correlation was found between starting PFAS concentration and removal efficiency.

In Test 3, we aimed to evaluate the impact of waste BTU on SCWO treatment efficacy but found that comparing AFFF wastewater to RO reject provided clearer insights. AFFF had a very low BTU (<1 lb/L) and a total PFAS concentration of 10.1 μg/L, while RO reject had a much higher BTU (4,400) and a total PFAS concentration of 13.4 μg/L. The PFAS removal efficiency for AFFF was 95.91%, compared to 99.66% for RO reject. AFFF had lower PFAS concentration, COD, TOC, and BTU, and despite being tested at a lower feed rate and higher pressure, it still showed lower removal efficiency than RO reject. These results suggest that waste with higher BTU and PFAS concentrations tends to have better removal efficacy, regardless of flow rate and pressure. A positive correlation (R^2=0.9829) was observed, indicating that higher BTU supported improved PFAS removal efficiency, likely due to the additional energy released during oxidation in the reactor.

In Test 4, we evaluated the impact of low versus high salt concentration in the waste feed on PFAS removal during SCWO treatment. Salt concentration is critical in SCWO, as elevated levels, especially chloride and carbonate, can cause salt precipitation, leading to reactor corrosion and plugging. However, the General Atomics SCWO reactor's design makes it less prone to these issues. We found a loose correlation (R^2=0.8108) between higher salt concentration

and increased PFAS removal efficiency, although this may be coincidental rather than causal. RO reject, with nearly 1,000 times higher salt concentration than AFFF wastewater, did not negatively impact PFAS removal efficiency, nor did it increase the levels of titanium, tungsten, or nickel in the effluent, suggesting no significant corrosion. Further monitoring of this parameter is planned for future tests.

1.2.3 DQO 1 Conclusions

In conclusion, Tests 1 through 4 provided valuable insights into the factors influencing PFAS removal efficiency in the SCWO process. Test 1 demonstrated that PFAS sorbed onto solid waste (GAC) did not negatively impact removal efficiency compared to liquid waste (AFFF), with solid PFAS showing slightly higher removal efficiency. Test 2 revealed that while spent GAC achieved higher removal efficiency than spent resin, no clear correlation was found between PFAS concentration and removal efficiency, due to differences in waste properties and analytical challenges. Test 3 showed that higher BTU waste, such as RO reject, generally resulted in higher PFAS removal efficiency, indicating that energy release during oxidation could support better PFAS destruction. Cost modeling discussed later in DQO 3 also indicated that optimizing BTU by mixing waste streams to reduce diesel consumption can lower operating costs by up to 93%. Finally, Test 4 found no significant negative impact of high salt concentrations on PFAS removal, suggesting that salt levels did not cause reactor corrosion or decrease treatment efficacy. Together, these tests support the robustness of the SCWO process in handling a variety of waste types, with waste energy content and salt concentration emerging as key performance factors.

1.2.4 DQO 2 Overview

DQO 2 evaluated whether AxNano's process could sustain continuous operation with identified waste streams, thereby reducing high startup energy requirements and establishing a more sustainable waste treatment process compared to existing methods. The goal was to design and test the AFS prototype equipment to produce a feed that met the known boundary conditions for GA's SCWO process. The tests for DQO 2 are summarized in *Table 2*.

1.2.5 DQO 2 Test Summary

In August, Tests 5 through 10 were conducted to test and calibrate essential equipment for the AFS prototype design. This included the colloidal pump and screw conveyor motors, load cells, and flow meter. Following these tests, slurry preparation tests were carried out to determine whether to use a single or dual colloidal pump for continuous feeding based on the particle size of the solid waste material.

Test 5 involved the control and performance of colloidal pump motors using a programmable logic control (PLC) panel. The pump's speed, noise, vibration, and leaks were monitored to ensure smooth operation. Test 6 focused on valve actuators, testing their ability to control the proportional addition of water and antifoam C emulsion to the slurry preparation tank via the PLC. Test 7 tested the slurry tank's weight loadcell, ensuring it met accuracy standards for measuring solid waste transfer. Test 8 evaluated the flow meter's performance, ensuring accurate water measurement in the slurry preparation tank to achieve the target solid content.

In Test 9, the screw conveyor's performance was tested to ensure proper material flow without spillage or blockages. Power consumption and temperature were also monitored. Test 10 assessed the colloidal pump's ability to grind solid particles to a specified particle size (< 250 micrometers) at a continuous flow rate. If the colloidal pump did not meet this criterion, Test 11 was considered as an option to enhance grinding efficiency by using two colloidal pumps in series to achieve better particle size reduction.

Table 2. Test and Data Outputs for DQO 2.

DQO 2 Goal: Design and test AFS prototype equipment that can produce a feed that meets known boundary conditions for GA's SCWO.				
Test Number	Description	Data Input	Data Outputs	Major Outcomes
5	Motor controls	• Programmable logic control	• Power on • Run/stability • Power off	• System remained on and operated through the PLC, with the exception of the mill aperture being changed manually
6	Valve actuation	• Programmable logic control	• Power on • Power off	• Operated successfully and promoted the automated processes.
7	Slurry tank load cell test and calibration	• Volume	• Electronic conversion to weight	• Load cells damaged in shipping, need better packaging protection to support Phase III.
8	Flow meter test and calibration	• Volume per time	• Power on • Power off	• Successful control of slurries throughout AFS during processing, recirculation, and pumping out to SCWO reactor.
9	Screw conveyor speed	• Volume per time	• Mass transferred	• Successfully able to meter out two types of solids from hopper into slurry tank.
10	Grinding test - 1	• Process time	• Particle size	• Successfully reduced GAC and resin down to particle sizes that were processable by SCWO without clogging. • Reduced processing time from gallons per day to 0.33 gallons per min.
11	Grinding test – 2 (Optional)	• Number of pumps • Process time	• Particle size • Capital Cost • Operational Cost	• Grinding test 2 was not needed as all material was able to be processed with the original design.

1.2.6 DQO 2 Conclusions

Overall, the DQO 2 Goal tests for the AFS, have been successfully completed. Test 5 and Test 6 demonstrated successful operation through the PLC. Test 7 was not completed due to damage to the load cells during shipping, but slurries were manually prepared and added to the hopper while awaiting load cell replacement. Test 8 was successfully completed, showing the ability to control slurry flow through the circulation pump and particle reduction mill, which contributed valuable intellectual property. Test 9 confirmed successful transfer of solids from the hopper to the mixing tank, while Test 10 was completed during pre-testing in New Jersey, and Test 11 was finalized at the GA facility without requiring a secondary grinding pump.

Our resin tests confirmed that the SCWO process maintained high PFAS removal efficiency and stable operation, even with larger resin particles (>250 microns). These results indicate that the AFS is an effective solution for SCWO treatment of solid waste, achieving excellent PFAS removal. Optimizations and hardware improvements have reduced grinding time to approximately 0.33 gallons waste per min, supporting continuous operation at a feed rate of 3 GPM, with solid waste contributing to this flow. (Note that 3 GPM is the total waste feed volume of the slurry). Prior to this Phase II, solid wastes were processed at gallons per day (estimate) and in some cases, pelletized GAC was not processable by SCWO at all. The maximum waste throughput achieved was 1.8 GPM.

Table 3. Test and Data Outputs for DQO 3.

DQO 3 Goal: Test if AxNano's process (AFS prototype + SCWO reactor) meets technical (throughput, discharge limits) and economic requirements (cost, logistics) for integration into DAF workflow.				
Test Number	Description	Data Input	Data Outputs	Major Outcomes
12	Effluent concentrations versus disposal requirements	• PFAS concentration, metals concentration	• Waste disposal categorization	• Effluent met all Industrial Wastewater Control Program Current Limits the City of San Diego • Currently there are no discharge limits on PFAS and WWTPs may be reluctant to accept any non-zero PFAS waste. Effluent from this test contained PFAS concentrations ranging from 43 ppt to 1700 ppt. Therefore all effluent was disposed of as hazardous waste. Baseline measurements were 219 ppt PFAS, suggesting background contamination was likely. Future studies need more QA/QC sampling to identify PFAS sources.
13	Cost per volume treated versus current methods	• Volume waste treated per unit time • Chemical consumption • Utility usage • Equipment costs • Labor effort • Effluent chemical composition	• Capital costs • Operating costs • Adherence to current DAF disposal guidance	• The AFS adds only 3% to total capital and operation expenses (CapEx) but provides up 93% reduction in operating costs. • Current OpEx range from $88.46-$1,346.16/ton depending on waste stream. • Diesel consumption and labor are the highest cost drivers for OpEx. • For customers generating sufficient wastewater to support continuous operation, a government-owned and contractor-operated AxSure™ system can reduce costs by 77–83% over a 10 year period compared to landfilling. However, for customers producing only solid PFAS waste, landfilling remains the most economical option, though long-term liability considerations were not factored into the model. • This work supports DoD's evaluation of emerging technologies, such as SCWO, based on applicable criteria identified in EPA's April 2024 guidance technology evaluation framework.

Moving forward, our next steps include integrating the AFS into the SCWO process under the Defense Innovation Unit/Environmental Security Technology Certification Program (DIU/ESTCP) program. Additionally, longer-term testing for resin and further process optimization may be required. We plan to seek Phase III funding to support these optimization efforts.

1.2.7 DQO 3 Overview

DQO 3 assessed the potential integration of AxNano's process into DAF hazmat treatment workflow. The goal was to test whether AxNano's process (AFS prototype + SCWO reactor) meets technical requirements (throughput, discharge limits) and economic criteria (cost, logistics) for integration into the DAF workflow. Details of the DQO 3 Goal Tests are summarized in *Table 3*.

1.2.8 DQO 3 Test Summary

In Test 12, we evaluated whether the effluent from the AxNano process - consisting of water, salts, and minerals - could be discharged into the city stormwater system leading to a wastewater treatment plant (WWTP) or whether it must be managed as a hazardous waste. Key data inputs included effluent analytical results for metal and PFAS concentrations.

In Test 13, we assessed the operating cost per volume treated for AxNano's process (AFS + SCWO). This test was essential to addressing the overall objective: determining whether AxNano's AFS combined with SCWO could create additional value for Air Force hazardous substance treatment beyond SCWO alone. Key data inputs included actual parameters from AFWERX-supported tests on various solid and liquid waste streams.

The cost estimation process involved measuring the quantities of chemicals used, such as diesel, pH adjusters, and additives. These costs were synthesized with the waste treatment volume to determine the cost per unit of waste treated, informing the Cost of Goods Sold (COGS) and pricing structures for AxNano's process. Performance and pricing were then compared to incineration, using the EPA interim guidance document as a reference due to the lack of an existing DoD comparison, and to landfilling costs, which were sourced from Shaw AFB, Biddle ANG (AFCEC, Environmental Flights, and contractors).

1.2.9 DQO 3 Conclusions

In conclusion, effluent samples contained low concentrations of metals, all of which were below the discharge limits set by the City of San Diego. PFAS concentrations were measured across all effluent samples. Currently, there are no discharge limits for PFAS, though WWTPs may be reluctant to accept PFAS-impacted waste. The baseline effluent sample, which was collected before introducing waste to the SCWO reactor, had low levels of PFAS compounds (1.1–6.9 ng/L), except for PFBA, which was measured at 200 ng/L. These findings suggest potential background PFAS contamination from tap water mixing with the effluent in the quench tank, making it difficult to draw a definitive conclusion on the true effluent PFAS concentrations.

The conclusions from Test 13 indicate that for customers generating sufficient wastewater to support continuous operation, a government-owned and contractor-operated AxSure™ system can reduce costs by 72–77% over a 10 year period compared to landfilling and incineration. However, for customers producing only solid PFAS waste, landfilling remains the most economical option, though long-term liability considerations were not factored into the model. AxNano's Automated Feed System is critical for AxSure™ technology adoption, as integrating solid and liquid waste streams is necessary to make SCWO cost-competitive with landfilling.

On-site treatment proves economically viable for customers generating at least 930,000 gallons of liquid waste per year, ensuring continuous operation. For Air Force restoration activities, systems could be procured or leased by AFCEC and operated by contractors, while recurring wastewater treatment systems for Air Force bases could be

procured or leased by the base host Wing and managed by Civil Engineering Squadrons or contractors. Air Force customers unable to support an on-site AxSure™ system could still utilize SCWO if a system were available at a TSDF. In cases requiring off-site disposal, specifications would be outlined in a restoration Performance Work Statement (PWS) or as a contract line item (CLIN) under a DLA contract.

The DoD is currently reviewing the EPA's April 2024 "Interim Guidance on the Destruction and Disposal of Perfluoroalkyl and Polyfluoroalkyl Substances and Materials Containing Perfluoroalkyl and Polyfluoroalkyl Substances-Version 2". This interim guidance encouraged permitted Class 1 non-hazardous industrial or hazardous waste underground injection wells, permitted hazardous waste landfills, and permitted hazardous waste combustors that operate under certain conditions, and interim storage with controls. The EPA also provided a technology evaluation framework to help practitioners evaluate safety and effectiveness of new destruction and disposal technologies. The guidance focused on four approaches, including SCWO as well as mechanochemical degradation, electrochemical oxidation, gasification and pyrolysis. The results of this Phase II provide data inputs in the "Materials" Category of Information by expanding the inputs to:

- What PFAS-containing material has been used to test the performance of the technology of interest?
    - Results from this Phase II expand testing to reverse osmosis reject, PFAS-laden granular activated carbon, and PFAS-laden ion exchange resin
- Which PFAS, and at what concentration ranges, have been used to test the performance of the technology of interest?
    - Results from this Phase II expand concentration ranges to those of spent sorption media.
- Are there non-PFAS constituents that may affect the performance of the technology of interest?
    - Results from this Phase II expand inputs to show impacts of solid particle size and calorific value (BTU/lb) that will impact performance. Our results also show that salt concentrations, previously believed to negatively impact SCWO, does not have a performance impact on our system.

1.3 Lessons Learned from Phase II and Impacts on Full Scale Design

In addition to the above conclusions learned from testing described above, this SBIR Phase II program also supported our technology development efforts and commercialization in the following ways:

- Importance of tracking waste sample containers. One of our waste samples was initially believed to be part of an AFFF stockpile. The waste had been sampled by base personnel for PFAS analysis prior to shipping the container. However, when the container arrived, it was clear that the container of waste received was different from what had been originally sampled. Future work should ensure that samples taken for analysis match the container that is received for testing. This can be done by requesting photos of the containers at the time of sampling and providing customers with barcoded labels to attach to the container.
- Importance of measuring all baseline and background sources of PFAS. The effluent analysis from our tests across all waste streams, including baseline, indicates that PFAS levels were non-zero in the tap water. As a result, any introduction of tap water into the effluent after the reactor negatively impacts our results. Future testing should account for this by measuring all tap water, chemicals, or other inputs added after the reactor that could impact results. We strongly suggest placing a PFAS treatment system on the source water and/or as a polishing step for the effluent to increase likelihood of being able to discharge all effluent into a municipal wastewater treatment plant.

The specifics of our full-scale design remain proprietary; however, here we provide high level overview of the process including some improvement components to meet the requirements of this final report.

Additional components added based on Phase II prototype testing and lessons learned:

- Sampling port design – This change in design will help us to get more uniform samples of the slurry as it is being processed.
- Flow meter upgrades – This change in design will help us to better control the particle size reduction work and slurry preparation.
- Temperature controls – This change in design will help us to better monitor and control heat generated from the particle size reduction components.
- Vapor and foam controls – This change in design will help us to better control the contents inside of the slurry mixing tank and help make the system more amenable to handling a wider range of waste types. This is especially important for high PFAS concentration waste streams that may tend to foam.
- Waste transfer controls – This change in design will help us to automate transferring waste from containers (drums) into the AFS, minimizing human exposure.
- Additional safety components – This change in design will help us to increase the automation of the system to minimize direct human interaction and to integrate safety interlocks to protect system components.

1.4 Nonproprietary Strategy for Phase III Transition

This Phase II SBIR has provided significant benefits as we bring our AxSure™ technology to market for PFAS waste destruction, specifically supporting the testing of our prototype AFS. The AFS is a key proprietary component of the overall system. As shown in Figure 1 above, we have already succeeded in small scale Phase III transition by securing two purchase orders for our AFS system. We will expand our Phase III transition by following our commercialization strategy that balances technical feasibility of next generation systems, regulatory compliance, market demand, and financial viability. A nonproprietary outline of our commercialization strategy is:

1. Market Analysis and Positioning
    a. Continue to test willingness to pay for target markets including, industrial waste processors, military installations, industrial waste generators, and municipal wastewater treatment facilities that receive wastes from industrial users.
    b. Continue to conduct competitive analysis of our system against both traditional methods (landfilling and incineration), innovative processes (hydrothermal alkaline treatment, electrochemical oxidation, plasma, and gasification), as well as other SCWO-based vendors (374water, Revive Environmental, and Aquarden).
    c. Continue to validate our value proposition through technical and market testing demonstrating AxSure™'s ability to effectively eliminate PFAS in a wider range of waste streams than any other innovative technology while complying with increasing stringent environmental regulations on landfilling and incineration.

2. Regulatory and Compliance Strategy
    a. Continue engaging with environmental regulatory agencies to ensure compliance with PFAS disposal regulations and providing updates on technology performance to add in creation of guidance documents. Agencies include EPA and state environmental departments at states that either have or are planning to have budget associated with PFAS treatment and mitigation within the next 5 years. Our work with EPA and NC DEQ will support this effort.
    b. Obtain necessary permits and certifications for the use of AxSure™ to treat waste and/or work with first adopters to support this work. The technology-specific data sheet prepared in this program will

help support this work, as well as working with engineering consulting firms that routinely support the application process. We have also expanded our team to include personnel with prior experience in writing wastewater permits for the NC DEQ.

c. Collaborate with government agencies (e.g., DOD ESTCP and EPA Office of Research and Development) and research institutions (NC State, Duke, UNC Chapel Hill) to validate AxSure™'s effectiveness and safety. Our work with the NC Collaboratory program will support this effort.

3. Technology Development and Demonstration
    a. Continue testing our system at pilot scale demonstrations in operating environments with collaborators at municipalities, military bases, and industrial partners to show case our capabilities. Our work with the Defense Innovation Unit and demonstration at Clean Earth TSDF supports this effort. We have a program proposal under review with the AF SBIR TACFI program to support demonstration with Robins AFB in Georgia.
    b. Continue addressing technical challenges such as throughput and system scalability.
    c. Publish case studies and technical articles in peer reviewed journals to build credibility and attract early adopters. The work from this Phase II program will be submitted as a manuscript to the Journal of the Air & Waste Management Association.

4. Business Model and Revenue Streams
    a. We have a "hardware as a service" model where customers can lease our units to decrease upfront capital costs and receive technical support in operations and maintenance.
    b. We are expanding our portfolio of equipment offerings available to capture and concentrate PFAS to both meet our customers' needs of the ever changing PFAS treatment landscape as well as aid in concentrating waste streams down to meet AxSure™ throughput limitations.
    c. Explore government funding opportunities, including DoD, DOE and EPA to support both contracts for PFAS remediation as well as to support R&D with non-dilutive funding.

5. Strategic Partnerships and Scaling
    a. Continue to strengthen our partnerships with engineering firms (Geosyntec, TetraTech), waste management companies (proprietary), and engineering contractors (Bay West) to integrate SCWO into existing waste treatment infrastructure.
    b. Establish international partnerships to expand into global markets with high PFAS contamination concerns (Greenthesis in Italy and proprietary partnerships in Europe).

6. Market and Public Awareness
    a. Develop educational contents to inform stakeholders about PFAS risks and AxSure™'s effectiveness through lunch-and-learns and white papers
    b. Engage in industry conferences and trade shows to showcase our technology (AWMA PFAS conference in Minneapolis May 2025 and more).
    c. Expand digital marketing (LinkedIn, newsletters, blog posts on website), case studies, and testimonials to build credibility and attract customers.
    d. This area will be our major focus over the next phase as we leverage the AF SBIR TACFI program to secure investment capital that can be used to fund sales and marketing and business development efforts that are not allowable on SBIR contracts.

Appendix A
Final Report

Sustainable, Non-Incineration-based Destruction of Air Force Hazardous Waste Stockpiles

Contract FA864923P1125
AxNano, LLC

Automated Feed Management System (AFS) Prototype Test Report

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Acronyms and Abbreviations

°C	Degree Celsius
%	Percent
AF	Air Force
AFB	Air Force Base
AFFF	Aqueous Film-Forming Foam
AFS	Automated Feed System
ANGB	Air National Guard Base
BTU/lb	British Thermal Unit per Pound
Cd	Cadmium
CIC	Combustion Ion Chromatography
CO	Carbon Oxide
CO_2	Carbon Dioxide
COC	Contaminant of Concern
COD	Chemical Oxygen Demand
COGS	Cost of Goods Sold
Cr	Chromium
DoD	Department of Defense
DQO	Data Quality Objective
EPA	Environmental Protection Agency
EFF	Effluent
GA	General Atomics
GA-EMS	General Atomics – Electromagnetic Systems
GAC	Granular Activated Carbon (or Charcoal)
GC	Gas Chromatography
GPM	Gallons per Minute
HF	Hydrofluoric Acid
HP	High Pressure
ICP	Inductively Coupled Plasma
INF	Influent
kg	Kilogram
L	Liter

LC	Liquid Chromatography
L/min	Liter Per Minute
MCL	Maximum Contaminant Level
MDL	Method Detection Limit
mg/L	Milligram per Liter
MS	Mass Spectrometry
NaOH	Sodium Hydroxide
ng/L	Nanogram per Liter
ND	Non-Detectable
NO_x	Nitrogen Oxides
O_2	Oxygen
PFAS	Per- and Polyfluoroalkyl Substances
R&D	Research and Development
RL	Reporting Limit
RO	Reverse Osmosis
SOP	Standard Operating Procedure
SO_x	Sulfur Oxides
TOC	Total Organic Carbon
TF	Total Fluorine
PPM	Parts per Million
SCWO	Supercritical Water Oxidation
µg/L	Microgram per Liter
VOC	Volatile Organic Compound

1.0 Introduction

1.0.1 Background

AxNano's AxSure™ system combines our proprietary automated feed management system (AFS) with supercritical water oxidation (SCWO) to effectively destroy contaminants that persist through traditional wastewater treatment methods (Figure *1*). The ability of SCWO to mineralize PFAS has been previously demonstrated.[1–3] Improper management of the feed into SCWO reactor leads to faults and shutdowns, ultimately causing incomplete waste destruction, decreased throughput and increased operational costs. The focus of this test is to demonstrate the utility of our AFS to pre-process various PFAS-containing waste streams, feed them into SCWO, and support stable reactor conditions. The SCWO reactor uses high temperatures and pressures to completely break down organic contaminants into carbon dioxide, water, and inorganic salts, providing a clean and effective waste treatment solution as an alternative to landfilling and incineration. The AFS is a key component of our proprietary SCWO process, providing a unique value proposition beyond SCWO reactor alone.

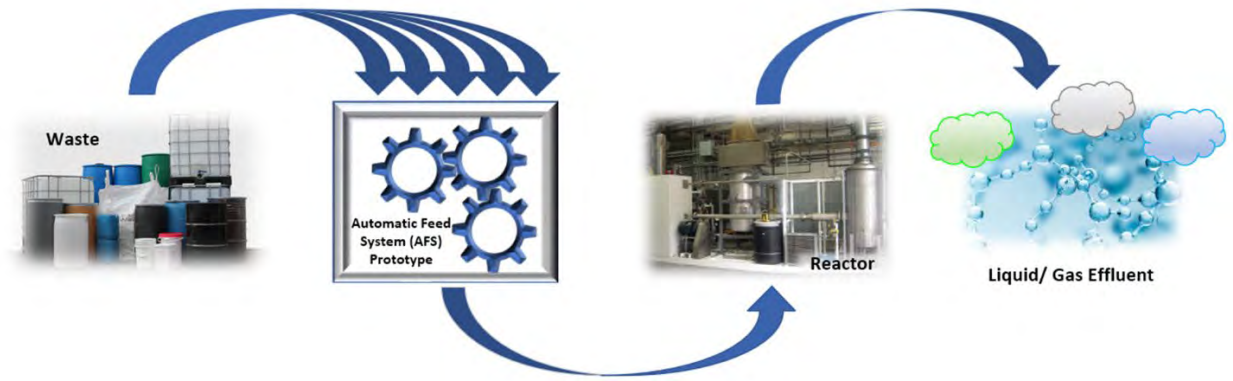

Figure 1. Schematic of AxNano's PFAS-waste destruction process. The focus of this SBIR Phase II program is the development of the Automatic Feed System (AFS) prototype.

This report provides detailed information on the results from pilot testing of the AxSure™ system components. The test was conducted in August 2024 at General Atomics Electromagnetic Systems (GA-EMS) facility in San Deigo, using an AxNano-built prototype AFS and GA's existing pilot SCWO system. This work was supported by a Small Business Innovative Research contract FA864923P1125 where the objective of the program was to build, design and test our AFS prototype and demonstrate that it is a valuable component of an overall SCWO system by supporting continuous operations with a range of waste feed types. This document has been created to present the results of the testing and provide conclusions on the current state of the technology in meeting program objectives.

In this document, we refer to the term "boundary conditions" to refer to an acceptable range that an identified parameter must fall in in order for a certain condition to pass. For example, the boundary condition for particle size of solids in the waste feed for current SCWO operation is 0-250 micron. This means that the minimum particle size is zero (i.e., no solids present) and the maximum particle size that the SCWO can process is 250 microns. Particles above 250 micron would cause the SCWO system to shut down. The boundary condition for particle size is 0-250 microns for SCWO operation.

1.0.2 Objective of the Test

Our primary goals for the pilot test were to:

- test our prototype AFS on real wastes,

- demonstrate ability of the AFS to produce a feed that supports continuous SCWO operation, and therefore reduce operational costs,
- evaluate the efficacy of SCWO to break down PFAS compounds in various wastes (liquids and slurries), and
- obtain data on operational parameters to inform a cost model.

As such, data quality objectives (DQO) goals specifically include:

- DQO 1 Goal: Identify effects of the waste feed on PFAS destruction efficiencies and reactor stability.
- DQO 2 Goal: Design and test AFS prototype equipment that can produce a feed that meets known boundary conditions for GA's SCWO.
- DQO 3 Goal: Test if AxNano's process (AFS prototype + SCWO reactor) meets technical (throughput, discharge limits) and economic requirements (cost, logistics) for integration into DAF workflow.

1.0.3 Scope of the Test

The test aimed to process waste types with unique characteristics that are difficult to destroy with other destruction methods. High salt concentrations can cause corrosion of electrodes in electrochemical treatment and can cause salt plugging in some SCWO systems. Solids and slurries are not currently able to be processes on electrochemical processes or hydrothermal alkaline treatment. Our ability to process these difficult to treat wastes demonstrates our value to the Air Force and future customers. We also sought to select waste that would allow us to test the effect of different waste characteristics on the overall process. For example, comparing high versus low salts; high versus low PFAS concentrations; high calorific value vs low calorific value; or solid phase PFAS waste versus liquid phase PFAS waste. We note, however, that it is difficult to keep all things constant and change only one variable when working with real-world wastes.

The wastes used in this test include:

- Liquids:
    - Reverse osmosis (RO) reject with high salt content
    - AFFF wastewater with low salt content
- Solids:
    - Biosolids with 10% solids content and low BTU
    - Spent granular activated carbon (GAC) with high BTU and PFAS concentration
    - Spent ion exchange resin with low PFAS concentration

The tests started on August 5th and ended on the 14th.

1.1 Materials and Methods

1.1.1 Wastes Description

The waste selection was based on their high priority for Air Force disposal needs and/or wide-ranging physical characteristics needed to test our AFS. The wastes selected included, biosolids, resin, GAC, AFFF and RO reject.

1. The RO reject was received from an industrial wastewater treatment plant in North Carolina, which processes a wide array of waste streams from military facilities, landfills, manufacturing plants, and industrial sites, utilizing various membrane filtrations at the end of their treatment process, with RO being the last. RO reject was selected for industry relevance and the opportunity to test a high salt content waste stream that can cause operational and treatability issues for other treatment systems.

2. The AFFF wastewater was initially believed to be 3M AFFF concentrate by the base personnel. However, analysis showed a PFOS concentration of 1.4 μg/L, which is lower than the typical 3M PFOS concentrations

of 10 mg/L. After further communication, the base HAZ Waste Manager located in California indicated that the contents was 3M AFFF surplus of unknown use or origin. It is assumed as diluted 3M AFFF wastewater. AFFF wastewater was selected for its industry relevance and high PFAS concentration.

3. The biosolids came from a wastewater treatment plant on base to treat industrial wastewater (including a flight line) and domestic wastewater across the Air Force Base in California. Biosolids were selected as a low BTU and solid plus liquid form.

4. The GAC was used in a pump-and-treat remediation system treating groundwater for PFAS in California. This was selected as a solid-form PFAS waste with a different PFAS concentration than the other solids to observe effect of starting PFAS concentration on AxSure™ processing. GAC also has a different abrasion than resin.

5. Ion exchange resin was used in a water treatment train system to remove residual PFAS, particularly PFOS, from the groundwater infiltrating into a drainage basin at an Air Force base in Pennsylvania. GAC is used in the primary treatment vessel, with resin serving as a polishing step after GAC. This waste was selected because it has high PFAS concentration and is a solid with a different BTU content than GAC.

Images of all raw as-received waste samples are presented in Figure 2. The sampling protocol was previously discussed in the Sampling Design Rationale. Physical properties, such as total PFAS, total salt, and percent solids were measured following methods as described in Section 1.1.5.

Table 1. Characteristics of as received wastes.

Waste Code	Waste Description	Starting total PFAS (1633)	Total Salt (mg/L)	Percent Total Solids as Received	Particle Size of Solids (microns)
Biosolids	Biosolids collected from wastewater treatment system	5.0 µg/L*	99.0	10%	80% < 0.2 mm
AFFF wastewater	Stockpiled AFFF waste	10.1 µg/L	2.7	0%[a]	N/A
RO reject	Reverse osmosis reject from industrial wastewater treatment plant receiving wide range of wastes	13.4 µg/L	2508	0%[a]	N/A
GAC	Spent GAC from groundwater remediation treatment system	56.1 µg/g	257	62%	98% < 2,000
Resin	Spent Purofine™ PFA 694E resin from groundwater treatment system	90.1 µg/g	268	87%	99% 500-1,100

* The biosolids were 90% liquid and were analyzed and reported by the lab as liquid.

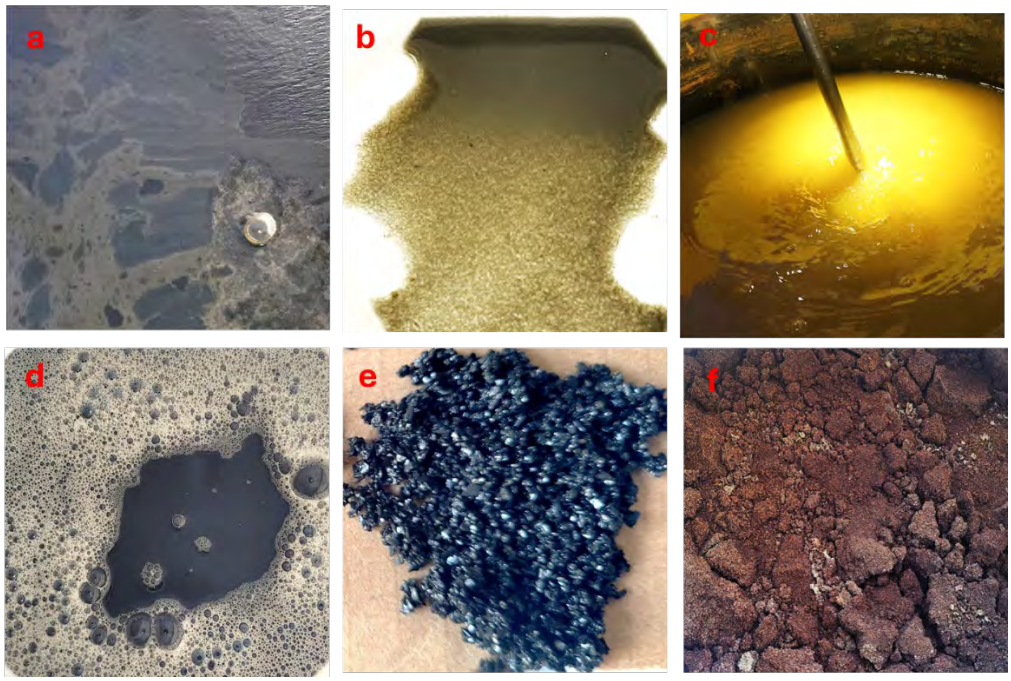

Figure 2. Raw wastes as received from different resources: biosolids (a and b); AFFF wastewater (c); RO reject (d); GAC (e); and resin (f).

1.1.2 Automated Feed System (AFS) Process

The current GA-EMS SCWO reactor has a series of boundary conditions for the feed. All wastes must be aqueous and pumpable, so any solids must be formed into slurries. Feeds can have no more than 15 wt% solids for the current SCWO orientation. Solid particles larger than 250 microns may clog the high-pressure pump, valves or the reactor injectors, therefore it is necessary to do particle size reduction in the AFS for wastes with larger particle sizes. Liquid waste streams may require thickening or anti-foaming agents depending on properties. Some liquids can be directly injected into the SCWO reactor through a high-pressure liquid feed pump. Further detail on waste preparation is proprietary.

The AFS (Figure 3 and Figure 4) was designed by AxNano specifically to address the need of processing solids-containing wastes to meet the boundary conditions of SCWO. Ultimately, solid wastes may be mixed with liquid waste to increase the amount of waste per unit time processed. But for this test, all slurries were prepared with clean water. Additives were incorporated to satisfy the specific SCWO input boundary conditions that ensure efficient and stable operation. AxNano has filed for IP protection on the AFS system and process. Below we describe non-proprietary components of the AFS process.

A predetermined amount of wet or dry GAC was placed into a hopper. This hopper meters and delivers the waste material, via screw feeder into a mixing tank where it is combined with a specified amount of water to form a slurry with a known solid-to-liquid ratio. The slurry is continually circulated through a homogenizing mill and returned to the tank. This continuous process ensures uniform particle size reduction and even distribution within the slurry. The mixture circulates according to the current settings and adjustments are made to further process the waste until the desired specification is achieved. The timing, pressure, and feed rates are influenced by the size and composition of the starting materials. Once the target slurry composition is reached, the slurry is transferred to a holding tank, then introduced into GA-EMS's SCWO process (Section 1.1.2). The components of the AFS were selected in order to reduce GAC with a starting granular size of 4 mm down to the target upper limit of the boundary condition (250 microns). Mill specification,

residence time, circulation pressure, GPM, and temperature control were regulated to achieve greatest grinding efficiency. Details on this process are proprietary.

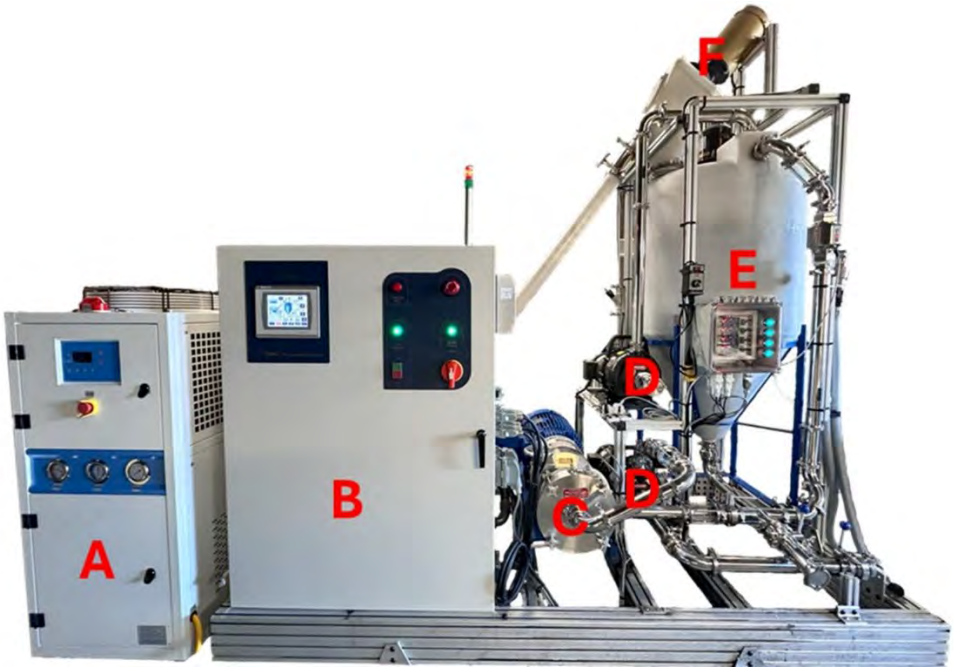

Figure 3 The complete Automated Feed System (AFS): A-chiller; B-main PLC control; C-particle size reduction mill; D-circulation(top)/transfer(bottom) pumps; E-slurry mixing tank; and F-screw feeder.

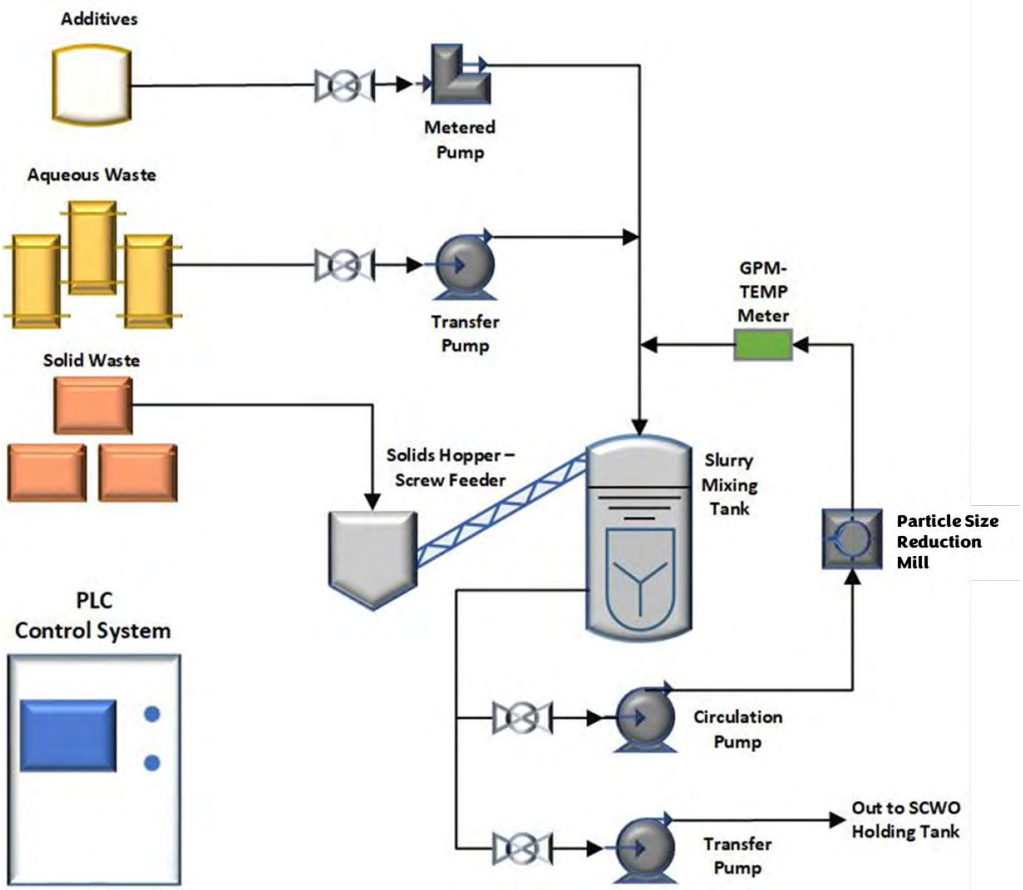

Figure 4 AxNano's Automated Feed (AFS) system concept design for solid waste transferring, grinding and slurry preparation.

1.1.3 SCWO Process

As illustrated in Figure 5, the SCWO process includes a high-pressure air compressor, a startup fuel-fired preheater, and a high-pressure SCWO reactor. The setup also features a pressure letdown system, a low-pressure gas-liquid separator, chemical-resistant and pressure-rated conveyance piping and connections, and necessary holding tanks for the treated effluents.

Influent wastes (from the AFS) were fed along with a co-fuel (diesel) into the high-pressure reactor through a nozzle at top of the reactor, using a chemical injection pump for liquids and a slurry pump for wastes containing 10-15% solids. This system was designed to mix the air with the influent waste and diesel in a reaction zone in the reactor maintained at 650°C. The temperature of the reaction zone is regulated through an automatic closed-loop system that adjusts the diesel flow as needed. Diesel flow is determined automatically as a function of calorific value of the feed waste by monitoring temperature fluctuations as the feed is introduced. For wastes with high calorific value (e.g., high BTU), then less diesel is needed and vice versa. This control is proprietary to GA's system. SCWO reactor inlet pressure was set to 230 bars to maintain reactor pressure above 200 bars. Actual reactor pressure can be affected by factors such as feed flow rate, reaction kinetics, temperature fluctuations, etc.

Quench water is introduced around the exterior of the high-pressure reactor near its base to lower the temperature of the effluent to approximately 300°C before it enters the pressure reduction system. A 5-30% solution of sodium hydroxide (NaOH) was manually added to the quench water to neutralize any expected inorganic acids formed. Acid

formation, including hydrofluoric acid (HF), was estimated based on the prior analysis of the waste feeds and the real-time monitoring of the effluent pH levels. In some cases, where wastes had low acid-forming organic content, then no NaOH was added. The cooled effluent is then directed to a gas-liquid separator operating at atmospheric pressure, where gases and water vapor are vented. The remaining liquid passed through a heat exchanger, further reducing the temperature to around 55°C before exciting the process. Effluent is collected in a storage tank for testing prior to discharge or disposal according to City of San Diego regulations.

Critical process parameters such as waste feed rate, reactor pressure, reactor temperature, and diesel feed rate are listed in Table 2. These parameters were determined based on trade secret and operational experience of GA. Table *2* also lists liquid effluent generation rate, process water and quench water consumption rates.

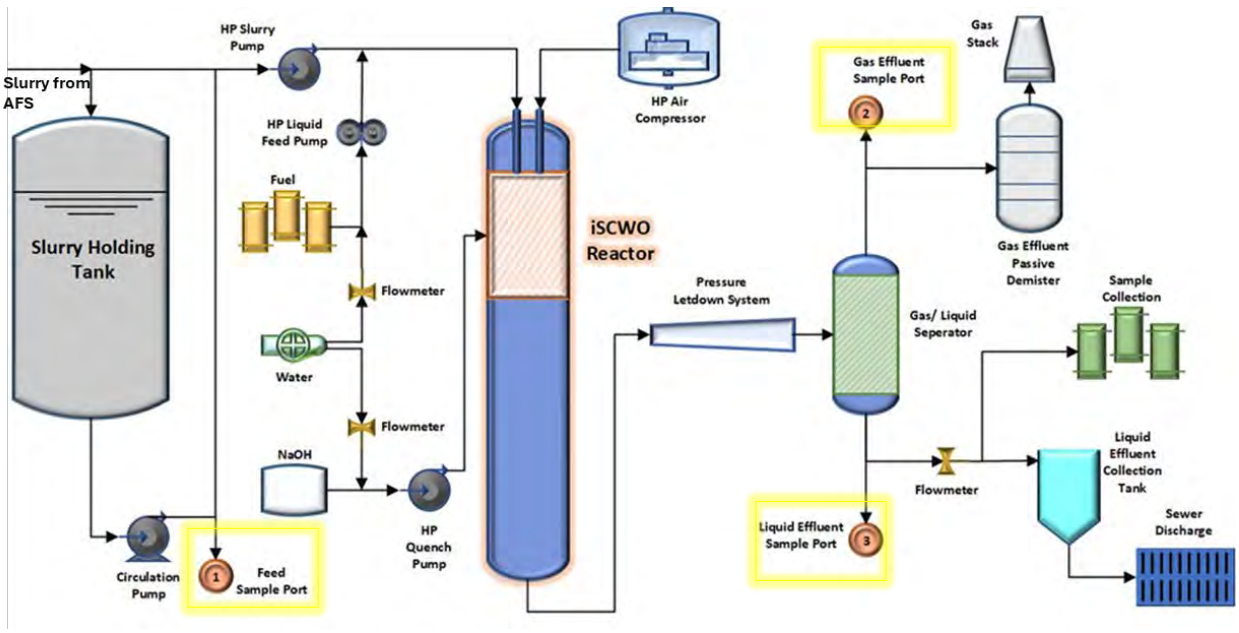

Figure 5 Block flow diagram for current GA SCWO process with three sample ports indicated in yellow boxes: 1-influent; 2-gas effluent; 3-liquid effluent.

Table 2 Process parameters monitored during the SCWO tests.

Waste Type	Waste Feed Rate (lpm)	Steady State Run (mins)	Reactor Pressure (Bar)	Reactor Temp (°C)	Diesel Flow Rate (lpm)	Liquid Effluent Flow Rate (lpm)	Process Water (lpm)	Quench Water (lpm)
Biosolids	7.0	165	215.8	645.7	1.8	3.9	3.7	15.1
AFFF wastewater	1.0	135	227.9	649.1	1.1	2.0	4.3	10.7
RO reject	8.2	240	200.1	645.0	0.6	8.4	0.6	20.6
GAC	7.0	45	204.8	646.5	NA	8.2	3.7	21.0
Resin	7.0	20	191.9	647.3	NA	8.3	5.6	21.3

1.1.4 Sampling Methods

The feed samples (post-AFS but pre-SCWO) were collected from sample port 1 situated after the slurry holding tank and before the high-pressure slurry pump. Liquid and gas effluent samples were collected from sample port 3 and sample port 4, respectively, which are situated after the SCWO and liquid/gas separator. Sampling ports are highlighted in Figure 5. Effluent samples were taken after the SCWO reactor reached steady state operation. Sampling was only conducted when the SCWO was operating in steady state. Sampling during the start-up and shut-down phases was not done in this test, but may be considered in future testing. Full sampling design and rationale as well as sampling materials and procedures provided by contract laboratory services were followed accordingly (Attachment 2).

1.1.5 Analytical Methods

1.1.5.1 Particle Size Analysis

Sieves were used to monitor particle sizes on-site. A particle size distribution analysis was conducted at Particle Technology Labs (Downs grove, IL) using Malvern 3000 liquid analysis for final post-AFS slurry particle size analysis.

1.1.5.2 Liquid PFAS, Total Fluorine and Fluoride Analysis

Table 3 lists the standard analytical methods used for analyzing chemical compositions of influent and effluent samples. The liquid/slurry samples (feed) were tested for 40 PFAS compounds using EPA Method 1633 (LC-MS/MS). Gas samples were analyzed for 48 compounds using a modified EPA Method 537 (LC-MS/MS). PFOA and PFOS are designated as hazardous substances by EPA under CERCLA. Note that some of the PFAS compounds analyzed in this test are now proposed as hazardous substances by the EPA under Title 40 of the Code of Federal Regulations Part 261 Appendix VIII.

Total fluorine (TF) was analyzed using direct combustion ion chromatography (CIC) to capture both organic and inorganic fluorine. The standard operating procedure (SOP) for TF method was developed by Eurofins Lancaster Laboratories Environment (ELLE) and can be found in Attachment 3. The ELLE method involves directly combusting solid and liquid mixtures at temperatures between 900-1,000 °C in a humid, oxygen-rich environment. Under these conditions, the samples oxidize, breaking the strong carbon-fluorine bonds, and the resulting vapors pass through an absorption solution. The solution absorption is then injected into ion chromatography using conductivity detection with chemical suppression. This CIC method is assumed to capture all fluorine in the sample regardless of what form it is originally. Fluoride (F-) concentrations were measured to quantify the inorganic, soluble forms present in the samples using method 9056 (Eurofins Scientific). Normalized concentrations for effluent are calculated by accounting for dilutions that occur within the system through mixing with the quench water (Table 5).

Table 3 Analytical methods for liquid/slurry analysis

Analyte	Method	Analytical Method
PFAS (targeted)	USEPA Method 1633	LC-MS/MS
Volatile organic compounds (including 1,4-dioxane)	USEPA Method 8260C	LC-MS/MS
Anions (salts)	USEPA Method 9056A	GC-MS
Total fluorine	Direct CIC using ELLE SOP	CIC
Total organic carbon	USEPA Method 9060A	Flame ionization detection
Metals	USEPA Method 6010C	Ion chromatography
Chemical oxygen demand	Hach TNT plus COD	UV-Vis spectroscopy
Percent solids	USEPA Method 1684	Mass

1.1.5.3 Other Waste Analysis Methods

Volatile organic compounds (VOC), including 1,4-dioxane, were measured using SW846 Method 8260 (GC-MS). Anions were tested via SW846 Method 9056 ion chromatography (IC). Total organic carbon (TOC) was measured using SW846 Method 9060. Metals were analyzed using SW846 Method 6010C using inductively couple plasma-atomic emission spectroscopy. Chemical oxygen demand (COD) was measured inhouse at the AxNano lab using a Hach TNT plus COD test kit, where the sample is heated for two hours with potassium dichromate, and the change is measured colorimetrically. Total solids were also measured at the AxNano lab following EPA method 1684.

1.1.5.4 Methods for Calculating PFAS Removal Efficacy

Total PFAS removal efficiency (%) is calculated by first converting the measured effluent PFAS concentration into a normalized total PFAS concentration in the effluent. Normalized value accounts for dilutions that occur inherent in the process. The following equation was used:

$$Normalized\ total\ PFAS\ in\ effluent\ (ng/L) = Measured\ total\ PFAS\ in\ effluent\ (ng/L)\ X\ \left(\frac{Liquid\ effluent\ flow\ rate\ (lpm)}{Liquid\ influent\ feed\ rate\ (lpm)}\right)$$

$$Total\ PFAS\ removal\ efficiency\ (\%) = \left(\frac{Total\ PFAS\ in\ influent - Normalized\ total\ PFAS\ in\ effluent}{Total\ PFAS\ in\ influent}\right)\ X\ 100$$

1.1.5.5 Gas PFAS Sampling and Analysis

Gas samples were collected following a modified TO-13 method, developed and validated by Eurofins Scientific. A CORBO 1500 sampling apparatus, shown in Figure 6, was prepared by Eurofins Scientific and used as received according to the provided SOP. This setup included a rubber stopper with ¼" glass tubing adapter, polyurethane foam plugs, and an XAD-2 resin bed. The glass tubing adapter was connected to the gas flow from source (line from the gas separator) using a ¼" tube. This non-isokinetic field sampling procedure was used to collect PFAS compounds for the EPA Method 537 analysis. This method follows the same rigorous quality control criteria and extraction method as the EPA OTM-45. The sampling process generally spans 1 to 3 hours, with a nominal flow rate of 0.5 to 5 lpm. Non-isokinetic stack sampling means that the velocity through the sampling system is not equal to the stream velocity through the stack. The velocity through the TO-13 sampling apparatus is estimated based on the velocity through the Testo system, which is situated on the same side-stream sampling line. More details on the gas collection are provided in Table 4.

1.1.5.6 Other Gas Analytes

Other gas effluents such as CO, NO_x, and SO_x were measured in real-time (once per second) using a calibrated Testo 350 gas analyzer. Monitoring these gases was essential to ensure air quality compliance following the Maximum Achievable Control Technology standards for a typical incinerator found in the Code of Federal Regulations, Title 40, Part 63. HF was measured using Niosh 7902 once per test.

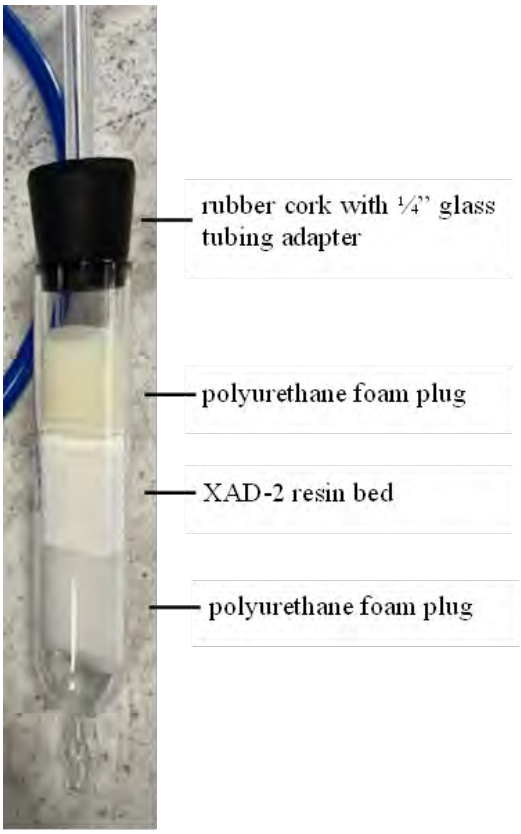

Figure 6 CORBO 1500 gas sampling apparatus used for Method TO-13 to collect vapor phase PFAS.

Table 4 Gas sampling parameters.

	Gas collection duration (min)	Gas sampling flow rate (lpm)	Total sampling gas volume (l)
Biosolids	120	1.0	120
AFFF Wastewater	60	1.0	60
RO Reject	160	1.0	160
GAC	45	1.0	45
Resin	45	1.0	45

1.1.5.7 Percent Solids

Method 1684 was used to measure percent solids in waste influent and effluent samples. A 25 g sample aliquot was dried at 103°C for 12 hours, cooled and weighed. The change in weight was used to calculate percent solids on a weight percent basis.

1.1.6 Post-Test Liquid Waste and Emissions Management

Pilot SCWO tests have been conducted at our partners' GA-EMS R&D facility in San Diego. Data from their previous tests show low gas emission levels generated from waste that contained higher concentration of PFAS than our wastes.[1,2,7] Therefore, the decision to release the gas effluent directly into the air was made by GA-EMS facility according to their R&D treatability permit.

All the collected liquid effluent was temporarily stored in designated storage tanks equipped with appropriate safety and containment measures while comprehensive analysis was conducted on the liquid samples. Our goal is to ensure that the disposal process minimizes any potential risks to human health and the environment while adhering to legal and regulatory requirements.

Once all analytical tests were completed and detailed reports were generated, we assessed both the volume of the liquid, and the specific characteristics identified in the analyses and compared them to the City of San Diego industrial wastewater discharge limits further discussed in Section 5. However, for this project, no liquid effluent that could potentially contain PFAS will be intentionally discharged to the sewer. As a precaution, any liquid effluent that could potentially contain PFAS will be handled by ACTenviro. ACTenviro is a hazardous waste management firm that has been hired to remove and handle all liquid effluents, with and without solids. The final disposal for these effluents, regardless of PFAS concentration, will be incineration.

1.2 Results

1.2.1 AFS Process to Meet SCWO Feed Boundary Conditions

The grinding process, performed utilizing the AFS as the grinding source, proved to be more effective in producing a feed that met the boundary conditions, than previously used methods at GA. GA utilized a Greerco W250 Colloid Mill for grinding solid waste mixed with water and additives into a slurry of approximately 10% solids, continually circulating through a holding tank until the solids were reduced to a desired level. In these previous tests, starting GAC with an approximate granular diameter of 4mm was only able to be reduced to between 500-700 microns over approximately two months. The waste could not be processed to SCWO because the particle size boundary condition was not met, and pumps consistently clogged. Comparatively, the waste processed through the AFS did meet the boundary condition of <250 micron, supporting a steady state SCWO operation. Moreover, the AFS was able to reduce processing time from months to hours, which leads to decreased operating costs. Lastly, the AFS proved able to process solid wastes and produce a feed for the SCWO process at a 3 GPM flow rate that supported continuous flow conditions.

For GAC and resin, the raw materials were moist solids with a certain moisture content. These solids needed to be converted into 10 wt.% GAC slurry and 15 wt.% resin slurry using the AFS. This wt% boundary condition was based on GA's recommendation that they have from previous experience running GAC and resin. A greater percentage of solids had previously clogged the pump and system lines.

Particle size analysis on both GAC and resin after being processed through the AFS results are shown in Figure 7 and Figure *8*. Nearly 90% of GAC particles were ground from 2mm to < 45 microns in a couple hours, while the remaining 10% were <186 microns, which is well below the target particle size boundary condition of 250 microns. In contrast, for the resin, only 60% particles of the total volume are <250 microns, with the remaining 40% ranging between 250 and 1,110 microns.

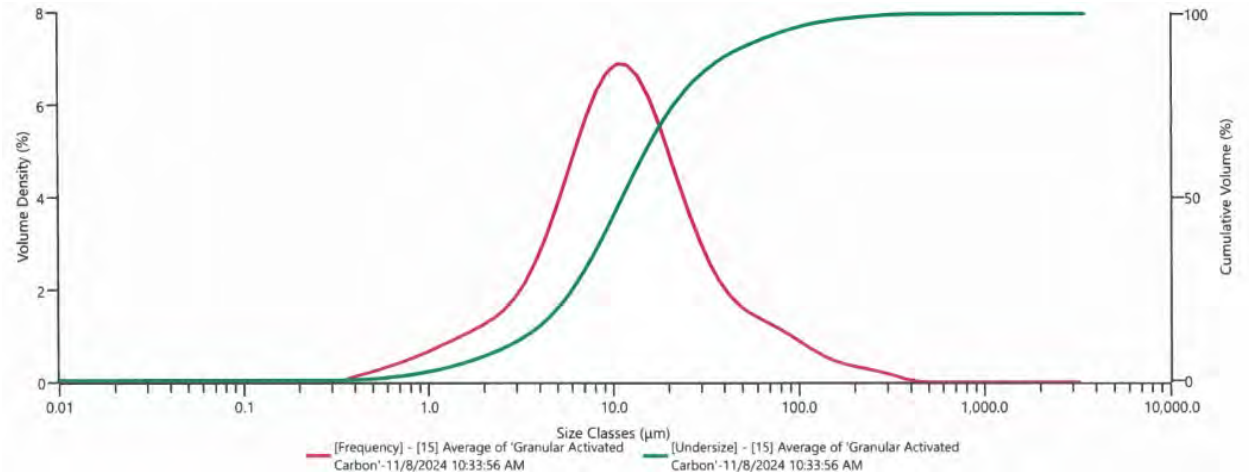

Figure 7 Frequency and undersize plots of GAC particle size analysis using Malvern 3000 liquid analysis (PTL, Downers grove, IL).

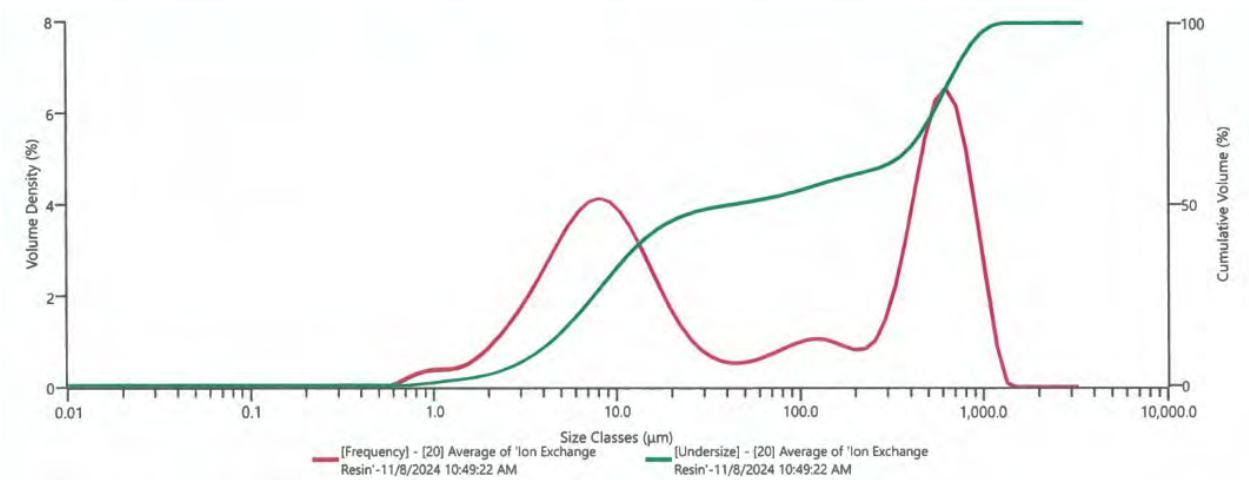

Figure 8 Frequency and undersize plots of resin particle size analysis using Malvern 3000 liquid analysis (PTL, Downers grove, IL).

Table 5. Characteristics of feed for wastes post-AFS processing.

Waste	BTU	PFAS (µg/L)	Total Salt (mg/L)	Percent Solids (wt%)	Particle Size of Solids (micron)
Biosolids	1,800	5.0	99.0	10%	< 756
AFFF wastewater	ND < 1	10.1	2.7	0%	N/A
RO reject	4,400	13.4	2,508	0%	N/A
GAC	20,000	1,115.2	257	10%	< 51.8
Resin	880	134.1	268	15%	< 859

1.2.2 SCWO reactor stability while processing different wastes

Figure 9 shows an example of the data from real-time monitoring of the SCWO process parameters that is recorded for each run. The system records flow rate of:

- the water (feed water) added in the system to prevent the reactor from overheating
- the quench water used to cool and neutralize the effluent
- the waste feed flow, and
- the reactor temperature and pressure.

Figure 10 shows an example of the real-time gas effluent monitoring available, showing the AFFF run as an example. In both Figure 9 and Figure 10, the green boxes indicate the 135-minute steady state period for each test. Typically, after initiating the SCWO process for a certain period, there is a steady state period during which the system ensures stable material feed and maintains temperature and pressure without significant fluctuations. Figure 9 shows that the temperature and pressure were stable during the run of AFFF. Instability would be indicated by significant temperature or pressure spikes or drops.

All waste materials tested in the SCWO experiments exhibited stable time series plots within the tested steady-state time range, with no unusual fluctuations observed in temperature, pressure, or other parameters. These results indicate that all wastes processed by the AFS met the boundary conditions for stable SCWO operation. Table 5 summarizes key physical and chemical characteristics of the waste feed after AFS-processing. The time series plots for the other waste materials remained stable within the steady state range.

The results of the real-time gas monitoring further indicate steady reactor conditions. Table 6 lists emissions CO, NO_x, and SO_x monitored in real-time for all wastes. Inefficient reactor conditions would be indicated by increased CO levels, indicating incomplete oxidation of organic carbon substances to CO instead of CO_2. Figure 10 is a representative schematic of real-time monitoring of these gas effluents. The levels for each gas were expected to be below the National Air Ambient Air Quality Standard. Except for O_2, all the other gas effluents were at very low levels and cannot be distinguished in the graph. Excess O_2 is introduced to the system to support complete waste oxidation.

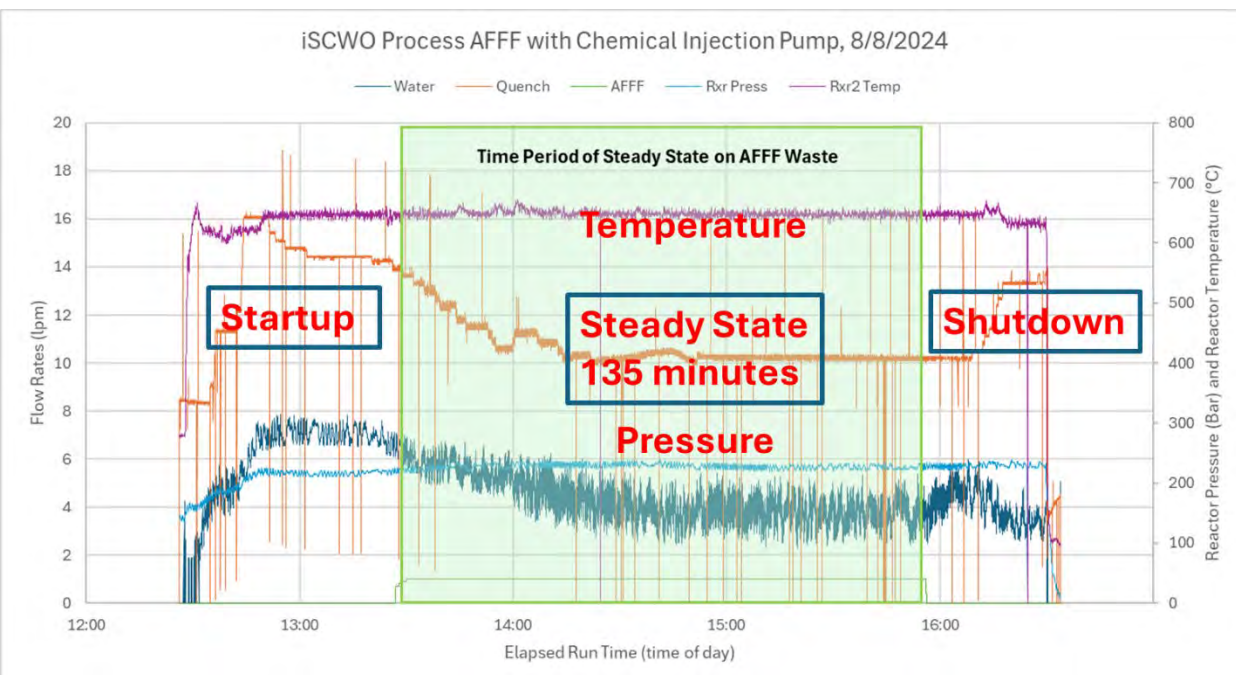

Figure 9 Time series plot of SCWO test on the AFFF wastewater (steady state marked by green box).

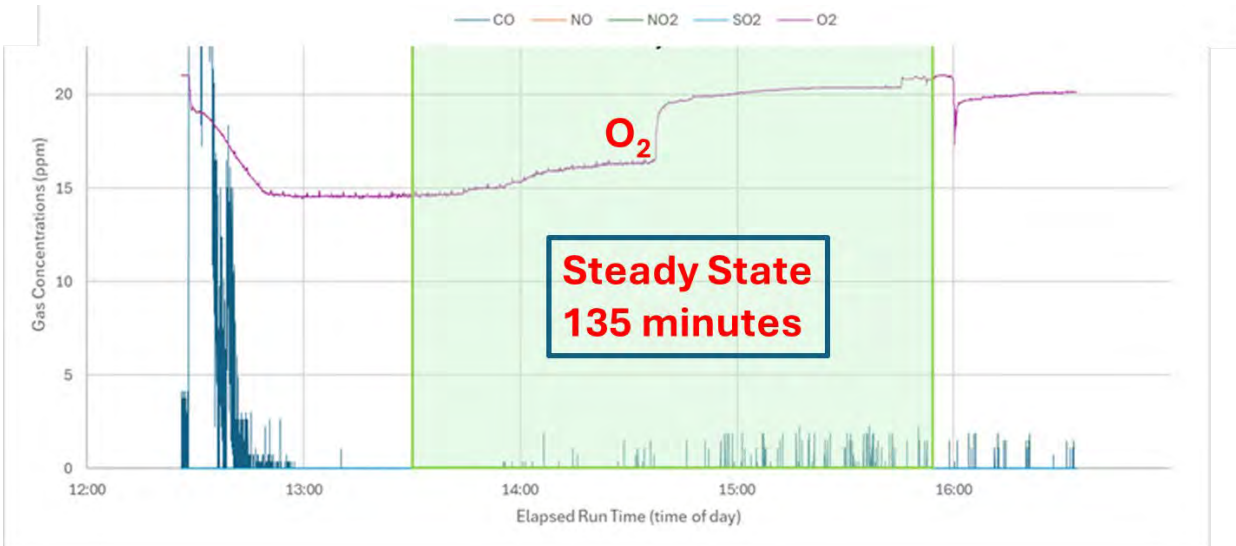

Figure 10 Time series plot of Testo gas effluent analysis during the AFFF wastewater SCWO test (steady state marked by green box).

Table 6 Average concentration of measured analytes in gas effluent.

Waste	Average Concentration (ppm)					
	HF	CO	NO	O_2	NO_2	SO_2
Biosolids	ND <0.5	0.05	0.00	12.18	1.55	0.01
AFFF	ND <0.5	1.17	0.01	18.06	0.01	0.02
RO Reject	ND <0.5	0.45	0.87	15.12	1.36	0.01
GAC	ND <0.5	1.14	0.02	14.24	0.04	0.02
Resin	ND <0.5	1.67	0.01	14.11	1.47	0.02

1.2.3 Concentrations of Compounds of Concern in Influent versus Effluent

The analytical results for compounds of concern (COCs) (i.e., PFAS, VOC, metals) are presented in the tables below. Only measurements that exceed both the Reporting Limit (RL) and the Method Detection Limit (MDL) are considered. Samples below the RL but above the MDL are appropriately noted. Samples with non-detectable (ND) values (below the MDL) are excluded from the display. For calculation purposes, non-detectable values have been substituted with the MDL. Baseline measurements were taken of the effluent from the SCWO system before any wastes for this test were run. The results are presented by waste type in the order in which they were tested: baseline, biosolids, AFFF, RO reject, GAC, then resin. Within each waste type, we represent analysis of COCs in the liquid influent versus the liquid effluent, followed by analysis of COCs in gas effluent. Percent reduction in total PFAS and TF among all the wastes tested is summarized in discussion Section 1.3.

1.2.4 Baseline liquid and gas sample analytical results

Baseline liquid and gas samples were first collected during the operation of the SCWO reactor with water only prior to any wastes running through the system. In the baseline liquid effluent sample (see Table 7), low levels (1.1–6.9 ng/L) of PFAS compounds were detected, except for PFBA, which was measured at 200 ng/L. The liquid effluent exhibited high concentrations of total fluoride, sodium, and sulfate. Specifically, fluoride concentration in the baseline was measured at 18 mg/L.

The gas samples were collected at 1 lpm for 15 minutes (i.e., total gas sample volume 15 liters) after the reactor reached steady-state conditions and before processing any waste samples. In the gas sample, two PFAS compounds were detected: PFOA at 0.27 ng/sample and PFHxS at 0.16 ng/sample (see Table 8). Additionally, media checks were done for the gas sampling apparatus to check for any background PFAS on the sorption media used in TO-13 method (see Table 8). This media check remains within the lab and no air is passed through these samples. It is packaged and prepared alongside other gas sampling materials to specifically assess the media sent for analysis. The media check gas samples showed PFAS levels below RL but above the MDL for PFOA (0.27 ng/sample) and PFHxS (0.16 ng/sample). Similar PFAS levels were observed in the baseline sample when measured in ng/sample. These two compounds are common low-level background PFAS detected in all gas samples. The sample units were reported as ng/sample but were converted to ng/L for testing to adjust for the volume passed through the resin tube.

Since the PFOA and PFHxS in baseline and media check gas samples are very similar, it can be concluded that no PFOA or PFHxS in gas effluent came from the SCWO process line. The elevated PFAS concentrations in the liquid effluent indicates that there may have been some carry over of PFAS from prior testing.

Table 7 Analytical results of the baseline liquid effluent.

Compound	Effluent		Qualifiers
PFAS			
PFBA	200	ng/L	
PFPeA	1	ng/L	J
PFHxA	1.1	ng/L	J
PFHpA	<0.45	ng/L	ND
PFOA	1	ng/L	J
PFNA	<0.45	ng/L	ND
PFDA	<0.45	ng/L	ND
PFUnA	<0.45	ng/L	ND
PFDoA	<0.45	ng/L	ND
PFTrDA	<0.45	ng/L	ND
PFTeDA	<0.45	ng/L	ND
PFBS	2.9	ng/L	
PFPeS	<0.45	ng/L	ND
PFHxS	2.2	ng/L	
PFHpS	<0.45	ng/L	ND
PFOS	6.9	ng/L	B
PFNS	<0.45	ng/L	ND
PFDS	<0.45	ng/L	ND
PFDoS	<0.55	ng/L	ND
4:2 FTS	<0.91	ng/L	ND
6:2 FTS	1.8	ng/L	J
8:2 FTS	<0.91	ng/L	ND
PFOSA	1.7	ng/L	J
NMeFOSA	<0.45	ng/L	ND
NEtFOSA	<0.45	ng/L	ND

NMeFOSAA	<0.45	ng/L	ND
NEtFOSAA	<0.45	ng/L	ND
NMeFOSE	<2.3	ng/L	ND
NEtFOSE	<2.3	ng/L	ND
HFPO-DA	<1.1	ng/L	ND
ADONA	<0.45	ng/L	ND
PFMPA	<0.45	ng/L	ND
PFMBA	<0.45	ng/L	ND
NFDHA	<0.45	ng/L	ND
9Cl-PF3ONS	<0.45	ng/L	ND
11Cl-PF3OUdS	<0.45	ng/L	ND
PFEESA	<0.45	ng/L	ND
3:3 FTCA	<0.91	ng/L	ND
5:3 FTCA	<2.5	ng/L	ND
7:3 FTCA	<2.3	ng/L	ND
Other			
TOC	5.50	mg/L	
TF	19	mg/L	Including ALL organic and inorganic fluorine
Anions			
Bromide	0.76	mg/L	J
Chloride	280	mg/L	
Fluoride	18	mg/L	
Sulfate	440	mg/L	
Metals			
Arsenic	<0.016	mg/L	ND
Barium	0.02	mg/L	
Cadmium	<0.0010	mg/L	ND
Chromium	0.03	mg/L	
Lead	<0.015	mg/L	ND
Nickel	0.13	mg/L	
Selenium	<0.016	mg/L	ND
Silver	<0.0041	mg/L	ND
Sodium	480	mg/L	
Titanium	0.11	mg/L	
Tungsten	<0.0052	mg/L	ND
VOC			
1,4-Dioxane	<0.17	µ/L	ND
Benzene	<0.30	µ/L	ND
Carbon tetrachloride	<0.30	µ/L	ND
Chlorobenzene	<0.30	µ/L	ND
Chloroform	<0.30	µ/L	ND
1,2-Dichloroethane	<0.30	µ/L	ND

1,1-Dichloroethene	<0.30	µ/L	ND
2-Butanone	<0.50	µ/L	ND
Tetrachloroethene	<0.30	µ/L	ND
Trichloroethene	<0.30	µ/L	ND
Vinyl chloride	<0.30	µ/L	ND

J, results are less than the RL but greater than or equal to MDL and the concentration is an approximate value; ND, non-detected at the reporting limit; B, compound was found in the lab blank.

Table 8 Analytical PFAS concentrations of the media check for gas effluent analysis.

Compound	Media Check Effluent	
PFAS		
PFHpA	<1.1	ng/sample
PFBA	<4.1	ng/sample
PFPeA	<0.8	ng/sample
PFHxA	<0.37	ng/sample
PFOA	0.20	ng/sample
PFNA	<0.17	ng/sample
PFDA	<0.15	ng/sample
PFUnA	<0.22	ng/sample
PFDoA	<0.091	ng/sample
PFTriA	<0.24	ng/sample
PFTeA	<0.29	ng/sample
PFHxDA	<0.18	ng/sample
PFODA	<0.16	ng/sample
PFBS	<0.17	ng/sample
PFPeS	<0.058	ng/sample
PFHxS	0.15	ng/sample
PFHpS	<0.15	ng/sample
PFOS	<0.24	ng/sample
PFNS	<0.14	ng/sample
PFDS	<0.11	ng/sample
PFDoS	<0.22	ng/sample
FOSA	<0.87	ng/sample
NEtFOSA	<0.32	ng/sample
NMeFOSA	<0.35	ng/sample
NMeFOSAA	<0.21	ng/sample
NEtFOSAA	<0.19	ng/sample
NMeFOSE	<0.3	ng/sample
NEtFOSE	<0.39	ng/sample
4:2 FTS	<0.095	ng/sample
6:2 FTS	<0.25	ng/sample
8:2 FTS	<0.14	ng/sample

10:2 FTS	<0.23	ng/sample
DONA	<0.22	ng/sample
HFPO-DA	<2.4	ng/sample
9Cl-PF3ONS	<0.12	ng/sample
11Cl-PF3OUdS	<0.14	ng/sample
6:2 FTUCA	<0.1	ng/sample
7:3 FTCA	<0.38	ng/sample
10:2 FTCA	<0.93	ng/sample
8:2 FTCA	<0.59	ng/sample
PES	<0.088	ng/sample
8:2 FTUCA	<0.13	ng/sample
PFECA F	<0.57	ng/sample
PFECA A	<0.086	ng/sample
5:3 FTCA	<0.34	ng/sample
6:2 FTCA	<0.72	ng/sample
3:3 FTCA	<0.28	ng/sample
PFECHS	<0.095	ng/sample
PFECA B	<0.19	ng/sample

J, results are less than the RL but greater than or equal to MDL and the concentration is an approximate value; ND, non-detected at the reporting limit.

Table 9 Analytical PFAS concentrations of the media check for gas effluent analysis.

Compound	Baseline Effluent		Qualifiers	Baseline Calculated Effluent	
PFAS					
PFHpA	<1.1	ng/sample	ND	<0.073	ng/L
PFBA	<4.1	ng/sample	ND	<0.273	ng/L
PFPeA	<0.8	ng/sample	ND	<0.053	ng/L
PFHxA	<0.37	ng/sample	ND	<0.025	ng/L
PFOA	0.27	ng/sample	J	0.018	ng/L
PFNA	<0.17	ng/sample	ND	<0.011	ng/L
PFDA	<0.15	ng/sample	ND	<0.010	ng/L
PFUnA	<0.22	ng/sample	ND	<0.006	ng/L
PFDoA	<0.091	ng/sample	ND	<0.016	ng/L
PFTriA	<0.24	ng/sample	ND	<0.019	ng/L
PFTeA	<0.29	ng/sample	ND	<0.012	ng/L
PFHxDA	<0.18	ng/sample	ND	<0.011	ng/L
PFODA	<0.16	ng/sample	ND	<0.011	ng/L
PFBS	<0.17	ng/sample	ND	<0.004	ng/L
PFPeS	<0.058	ng/sample	ND	<0.010	ng/L
PFHxS	0.16	ng/sample	J	0.011	ng/L
PFHpS	<0.15	ng/sample	ND	<0.010	ng/L
PFOS	<0.24	ng/sample	ND	<0.016	ng/L

PFNS	<0.14	ng/sample	ND	<0.007	ng/L
PFDS	<0.11	ng/sample	ND	<0.015	ng/L
PFDoS	<0.22	ng/sample	ND	<0.058	ng/L
FOSA	<0.87	ng/sample	ND	<0.021	ng/L
NEtFOSA	<0.32	ng/sample	ND	<0.023	ng/L
NMeFOSA	<0.35	ng/sample	ND	<0.026	ng/L
NMeFOSAA	<0.21	ng/sample	ND	<0.014	ng/L
NEtFOSAA	<0.19	ng/sample	ND	<0.013	ng/L
NMeFOSE	<0.3	ng/sample	ND	<0.020	ng/L
NEtFOSE	<0.39	ng/sample	ND	<0.006	ng/L
4:2 FTS	<0.095	ng/sample	ND	<0.017	ng/L
6:2 FTS	<0.25	ng/sample	ND	<0.027	ng/L
8:2 FTS	<0.14	ng/sample	ND	<0.015	ng/L
10:2 FTS	<0.23	ng/sample	ND	<0.015	ng/L
DONA	<0.22	ng/sample	ND	<0.160	ng/L
HFPO-DA	<2.4	ng/sample	ND	<0.008	ng/L
9Cl-PF3ONS	<0.12	ng/sample	ND	<0.009	ng/L
11Cl-PF3OUdS	<0.14	ng/sample	ND	<0.007	ng/L
6:2 FTUCA	<0.1	ng/sample	ND	<0.007	ng/L
7:3 FTCA	<0.38	ng/sample	ND	<0.025	ng/L
10:2 FTCA	<0.93	ng/sample	ND	<0.062	ng/L
8:2 FTCA	<0.59	ng/sample	ND	<0.039	ng/L
PES	<0.088	ng/sample	ND	<0.006	ng/L
8:2 FTUCA	<0.13	ng/sample	ND	<0.009	ng/L
PFECA F	<0.57	ng/sample	ND	<0.038	ng/L
PFECA A	<0.086	ng/sample	ND	<0.006	ng/L
5:3 FTCA	<0.34	ng/sample	ND	<0.023	ng/L
6:2 FTCA	<0.72	ng/sample	ND	<0.048	ng/L
3:3 FTCA	<0.28	ng/sample	ND	<0.019	ng/L
PFECHS	<0.095	ng/sample	ND	<0.006	ng/L
PFECA B	<0.19	ng/sample	ND	<0.013	ng/L

J, results are less than the RL but greater than or equal to MDL and the concentration is an approximate value; ND, non-detected at the reporting limit.

1.2.5 Liquid and gas sample analytical results for biosolids waste

Table 10 lists the influent and effluent PFAS concentrations in testing biosolids waste. All PFAS were reduced following SCWO treatment. Total PFAS compounds in biosolids were reduced by 98.75% by the SCWO process. Specifically, PFBA was reduced by 95.9%, PFBS by 96.0%, PFOS by 99.4%, PFOA by 96.2%, PFHxS by 97.3% and 8:2 FTS by 87.9%. TF levels significantly increased 70% after SCWO treatment. We believe this is due to the inability of method 1633 to accurately quantify PFAS on solids in the influent, whereas oxidation that occurs in SCWO makes TF more available for measurement. This is further discussed in Section 1.3.4.

COD was also significantly reduced by 99.7%, indicating that the SCWO reactor effectively converted most oxidizable compounds. TOC was reduced from 36 mg/L in the influent to 9.4 mg/L in the effluent, showing the value of SCWO in treating this type of wastewater. Bromide, chloride, fluoride and sulfate ions significantly increased in the effluent samples likely due to mineralization of PFAS and other organic compounds in the waste. The sodium concentration increased due to the addition of NaOH used to neutralize acids formed in the SCWO reactor. Some metals increased in the effluent (chromium, lead, nickel, titanium, tungsten) likely due to slight corrosion of the reactor liner. However, these levels were typical of normal SCWO operation and do not indicate significant corrosion. No VOCs were detected in the influent or effluent of the biosolids.

Table 10 Analytical results of the biosolids liquid influent and effluent.

Compound	Influent		Qualifiers	Effluent		Qualifiers
PFAS						
PFBA	860	ng/L		35	ng/L	
PFPeA	37	ng/L		2.8	ng/L	J
PFHxA	100	ng/L		2.4	ng/L	J
PFHpA	<8	ng/L	ND	<1.4	ng/L	ND
PFOA	66	ng/L		2.5	ng/L	J
PFNA	6.2	ng/L	J	<0.86	ng/L	ND
PFDA	<5	ng/L	ND	<0.86	ng/L	ND
PFUnA	<5	ng/L	ND	<0.86	ng/L	ND
PFDoA	<5	ng/L	ND	<0.86	ng/L	ND
PFTrDA	<5	ng/L	ND	<0.86	ng/L	ND
PFTeDA	<5	ng/L	ND	<0.86	ng/L	ND
PFBS	130	ng/L	I	5.2	ng/L	
PFPeS	12	ng/L	J	0.9	ng/L	J
PFHxS	160	ng/L		4.4	ng/L	
PFHpS	11	ng/L	J	<0.86	ng/L	ND
PFOS	1,200	ng/L		7.3	ng/L	I
PFNS	<5	ng/L	ND	<0.86	ng/L	ND
PFDS	<5	ng/L	ND	<0.86	ng/L	ND
PFDoS	<6	ng/L	ND	1	ng/L	ND
4:2 FTS	<10	ng/L	ND	1.7	ng/L	ND
6:2 FTS	90	ng/L		5.7	ng/L	J
8:2 FTS	14	ng/L	J	1.7	ng/L	ND
PFOSA	76	ng/L		<0.86	ng/L	ND
NMeFOSA	<5	ng/L	ND	<0.86	ng/L	ND
NEtFOSA	<5	ng/L	ND	<0.86	ng/L	ND
NMeFOSAA	<5	ng/L	ND	<0.86	ng/L	ND
NEtFOSAA	<5	ng/L	ND	<0.86	ng/L	ND
NMeFOSE	<25	ng/L	ND	<4.3	ng/L	ND
NEtFOSE	<25	ng/L	ND	<4.3	ng/L	ND
HFPO-DA	<12	ng/L	ND	4.9	ng/L	J
ADONA	<5	ng/L	ND	<0.86	ng/L	ND
PFMPA	7.5	ng/L	J	<0.86	ng/L	ND

PFMBA	<5	ng/L	ND	<0.86	ng/L	ND
NFDHA	<5	ng/L	ND	<0.86	ng/L	ND
9Cl-PF3ONS	<5	ng/L	ND	<0.86	ng/L	ND
11Cl-PF3OUdS	<5	ng/L	ND	<0.86	ng/L	ND
PFEESA	<5	ng/L	ND	<0.86	ng/L	ND
3:3 FTCA	<10	ng/L		<1.7	ng/L	ND
5:3 FTCA	2,000	ng/L		<4.8	ng/L	ND
7:3 FTCA	<25	ng/L		<1.3	ng/L	ND
Other						
TOC	36	mg/L		9.40	mg/L	
COD	11,060	mg/L		33.8	mg/L	
TF	2,800	mg/L		9,600	mg/L	
Anions						
Bromide	<2.8	mg/L	ND	3	mg/L	J
Chloride	63	mg/L		540	mg/L	
Fluoride	<2.0	mg/L	ND	4.2	mg/L	
Sulfate	34	mg/L		1,200	mg/L	
Metals						
Arsenic	<0.41	mg/L	ND	<0.016	mg/L	ND
Barium	2.2	mg/L		3.20	mg/L	
Cadmium	0.036	mg/L	J	0.07	mg/L	
Chromium	<0.077	mg/L	ND	0.40	mg/L	
Lead	<0.18	mg/L	ND	0.25	mg/L	
Nickel	0.12	mg/L	J	0.24	mg/L	
Selenium	<0.41	mg/L	ND	<0.016	mg/L	ND
Silver	<0.10	mg/L	ND	<0.0041	mg/L	ND
Sodium	51	mg/L		1,400	mg/L	
Titanium	0.17	mg/L	J	1.10	mg/L	
Tungsten	<0.13	mg/L	ND	0.13	mg/L	J
VOC						
1,4-Dioxane	<3.0	µ/L	ND	<0.17	µ/L	ND
Benzene	<3.0	µ/L	ND	<0.30	µ/L	ND
Carbon tetrachloride	<3.0	µ/L	ND	<0.30	µ/L	ND
Chlorobenzene	<3.0	µ/L	ND	<0.30	µ/L	ND
Chlororform	<3.0	µ/L	ND	<0.30	µ/L	ND
1,2-Dichloroethane	<3.0	µ/L	ND	<0.30	µ/L	ND
1,1-Dichloroethene	<3.0	µ/L	ND	<0.30	µ/L	ND
2-Butanone	<5.0	µ/L	ND	<0.50	µ/L	ND
Tetrachloroethene	<3.0	µ/L	ND	<0.30	µ/L	ND
Trichloroethene	<3.0	µ/L	ND	<0.30	µ/L	ND
Vinyl chloride	<3.0	µ/L	ND	<0.30	µ/L	ND

J, results are less than the RL but greater than or equal to MDL and the concentration is an approximate value; ND, non-detected at the reporting limit; B, compound was found in the lab blank; I, value is estimated maximum possible concentration.

Eleven identified PFAS compounds were identified in the gas effluent of treated biosolids over a two hour collection period (per sample data listed in Table 11). Effluent data was converted from a per sample concentration to a per volume concentration by accounting for the gas emission flow rate over time (ng/L data listed in Table 11). Six compounds were detected above the reporting limit and five other compounds were detected above the MDL but below the RL, resulting in estimated concentration values. Two compounds, PFOA and PFHxS, detected in biosolids gas effluent were also found in the media check, so the actual presence of these in the gas effluent is non-conclusive. Repeat measurements would be required in further testing to determine if PFOA and PFHxS are present in the gas effluent from treating biosolids in a significant level above what was observed on the media check. Other PFAS compounds measured in the gas effluent include PFPeA (1.2 ng/sample), PFHxA (3.9 ng/sample), PFOS (1.4 ng/sample), 6:2 FTS (2.9 ng/sample), 8:2 FTS (1.7 ng/sample) were detected above the reporting limit. However, we note that these results are total collection over 2 hours of running the waste. When converting these results to a concentration of ng/L, the concentrations are all very low, less than 0.043 ng/L. Future testing will include more detailed sampling and analysis to support mass balance calculations and optimizing SCWO reaction parameters to achieve total non-detect in the gas emissions. We note that mass balance calculations was determined to be out of scope of this program, as described in the Sampling Design and Rationale plan.

Table 11 Analytical PFAS concentrations of the biosolids gas effluent.

Compound	Concentration in Effluent per sample		Qualifiers	Calculated Concentration in Effluent	
PFAS					
PFHpA	<3	ng/sample	ND	<0.025	ng/L
PFBA	5.1	ng/sample	J	0.043	ng/L
PFPeA	1.2	ng/sample		0.010	ng/L
PFHxA	3.9	ng/sample		0.033	ng/L
PFOA	3.4	ng/sample		0.028	ng/L
PFNA	0.2	ng/sample	J	0.002	ng/L
PFDA	0.54	ng/sample	J	0.005	ng/L
PFUnA	<0.22	ng/sample	ND	<0.002	ng/L
PFDoA	0.3	ng/sample	J	0.003	ng/L
PFTriA	<0.24	ng/sample	ND	<0.002	ng/L
PFTeA	<0.29	ng/sample	ND	<0.002	ng/L
PFHxDA	<0.18	ng/sample	ND	<0.002	ng/L
PFODA	<0.16	ng/sample	ND	<0.001	ng/L
PFBS	<0.17	ng/sample	ND	<0.001	ng/L
PFPeS	<0.058	ng/sample	ND	<0.000	ng/L
PFHxS	0.47	ng/sample	J	0.001	ng/L
PFHpS	<0.15	ng/sample	ND	<0.001	ng/L
PFOS	1.4	ng/sample		0.002	ng/L
PFNS	<0.14	ng/sample	ND	<0.001	ng/L
PFDS	<0.11	ng/sample	ND	<0.001	ng/L
PFDoS	<0.22	ng/sample	ND	<0.002	ng/L

FOSA	<0.87	ng/sample	ND	<0.007	ng/L
NEtFOSA	<0.32	ng/sample	ND	<0.003	ng/L
NMeFOSA	<0.35	ng/sample	ND	<0.003	ng/L
NMeFOSAA	<0.21	ng/sample	ND	<0.002	ng/L
NEtFOSAA	<0.19	ng/sample	ND	<0.002	ng/L
NMeFOSE	<0.3	ng/sample	ND	<0.003	ng/L
NEtFOSE	<0.39	ng/sample	ND	<0.003	ng/L
4:2 FTS	<0.095	ng/sample	ND	<0.001	ng/L
6:2 FTS	2.9	ng/sample		0.002	ng/L
8:2 FTS	1.7	ng/sample		0.001	ng/L
10:2 FTS	<0.23	ng/sample	ND	<0.002	ng/L
DONA	<0.22	ng/sample	ND	<0.002	ng/L
HFPO-DA	<2.4	ng/sample	ND	<0.020	ng/L
9Cl-PF3ONS	<0.12	ng/sample	ND	<0.001	ng/L
11Cl-PF3OUdS	<0.14	ng/sample	ND	<0.001	ng/L
6:2 FTUCA	<0.1	ng/sample	ND	<0.001	ng/L
7:3 FTCA	<0.38	ng/sample	ND	<0.003	ng/L
10:2 FTCA	<0.93	ng/sample	ND	<0.008	ng/L
8:2 FTCA	<0.59	ng/sample	ND	<0.005	ng/L
PES	<0.088	ng/sample	ND	<0.001	ng/L
8:2 FTUCA	<0.13	ng/sample	ND	<0.001	ng/L
PFECA F	<0.57	ng/sample	ND	<0.005	ng/L
PFECA A	<0.086	ng/sample	ND	<0.001	ng/L
5:3 FTCA	<0.34	ng/sample	ND	<0.003	ng/L
6:2 FTCA	<0.72	ng/sample	ND	<0.006	ng/L
3:3 FTCA	<0.28	ng/sample	ND	<0.002	ng/L
PFECHS	<0.095	ng/sample	ND	<0.001	ng/L
PFECA B	<0.19	ng/sample	ND	<0.002	ng/L

Italicized PFAS were proposed as hazardous substances by the EPA under Title 40 of the Code of the Federal Regulations Part 261 Appendix VIII. J, results are less than the RL but greater than or equal to MDL and the concentration is an approximate value; ND, non-detected at the reporting limit.

1.2.6 Liquid and gas sample analytical results for AFFF wastewater

PFAS compounds in AFFF wastewater were reduced by 95.91% by the SCWO process, with PFBA exhibiting the highest effluent concentration at 120 ng/L. COD was significantly reduced by 94.7%. TOC was low in the influent and reduced from 46 mg/L to 34 mg/L. TF levels increased by 70% after SCWO treatment, which is further explained in Section 1.3.4. Chloride, fluoride and sulfate ions significantly increased in the effluent samples which could be due to mineralization of PFAS and other organic compounds in the waste. The sodium concentration also increased due to the addition of NaOH added into the quench water to neutralize HF in the effluent. Some metals increased in the effluent (chromium, lead, nickel, titanium, tungsten). However, these levels were typical of normal (levels below 2 mg/L) SCWO operation and do not indicate significant corrosion. Only other COC detected in the influent was 2-butanone, which was reduced to non-detect levels in the effluent.

Table 12 Analytical results for the AFFF wastewater liquid influent and effluent.

Compound	Influent		Qualifiers	Effluent		Qualifiers
PFAS						
PFBA	2,300	ng/L		120	ng/L	
PFPeA	220	ng/L		2.2	ng/L	
PFHxA	1,000	ng/L		1.7	ng/L	J
PFHpA	150	ng/L	J	0.8	ng/L	ND
PFOA	120	ng/L	J	1.6	ng/L	J
PFNA	<48	ng/L	ND	0.5	ng/L	ND
PFDA	<48	ng/L	ND	0.5	ng/L	ND
PFUnA	<48	ng/L	ND	0.5	ng/L	ND
PFDoA	<48	ng/L	ND	0.5	ng/L	ND
PFTrDA	<48	ng/L	ND	0.5	ng/L	ND
PFTeDA	<48	ng/L	ND	0.5	ng/L	ND
PFBS	170	ng/L	J	7.7	ng/L	
PFPeS	110	ng/L	J	0.97	ng/L	J
PFHxS	490	ng/L		6	ng/L	
PFHpS	<48	ng/L	ND	0.57	ng/L	J
PFOS	1,400	ng/L	B	20	ng/L	B
PFNS	<48	ng/L	ND	0.5	ng/L	ND
PFDS	<48	ng/L	ND	0.5	ng/L	ND
PFDoS	<58	ng/L	ND	0.6	ng/L	ND
4:2 FTS	<96	ng/L	ND	0.99	ng/L	ND
6:2 FTS	1,700	ng/L		8.7	ng/L	
8:2 FTS	<96	ng/L	J	3.1	ng/L	J
PFOSA	<48	ng/L		0.5	ng/L	ND
NMeFOSA	<48	ng/L	ND	0.5	ng/L	ND
NEtFOSA	<48	ng/L	ND	0.5	ng/L	ND
NMeFOSAA	<48	ng/L	ND	0.5	ng/L	ND
NEtFOSAA	<48	ng/L	ND	0.5	ng/L	ND
NMeFOSE	<240	ng/L	ND	2.5	ng/L	ND
NEtFOSE	<240	ng/L	ND	2.5	ng/L	ND
HFPO-DA	<120	ng/L	ND	1.5	ng/L	ND
ADONA	<48	ng/L	ND	10	ng/L	
PFMPA	<48	ng/L	J	0.5	ng/L	ND
PFMBA	<48	ng/L	ND	0.5	ng/L	ND
NFDHA	<48	ng/L	ND	0.5	ng/L	ND
9Cl-PF3ONS	<48	ng/L	ND	0.5	ng/L	ND
11Cl-PF3OUdS	<48	ng/L	ND	0.5	ng/L	ND
PFEESA	<48	ng/L	ND	0.5	ng/L	ND
3:3 FTCA	<96	ng/L		0.99	ng/L	ND
5:3 FTCA	<270	ng/L		2.8	ng/L	ND

7:3 FTCA	<240	ng/L		2.5	ng/L	ND
Other						
TOC	46	mg/L		34	mg/L	
COD	720	mg/L		38.2	mg/L	
TF	2,800	mg/L		9,200	mg/L	
Anions						
Bromide	<2.8	mg/L	ND	<1.4	mg/L	J
Chloride	0.63	mg/L	J	640	mg/L	
Fluoride	1.1	mg/L		4.4	mg/L	
Sulfate	1.0	mg/L		880	mg/L	
Metals						
Arsenic	<0.016	mg/L	ND	<0.016	mg/L	ND
Barium	0.015	mg/L		0.10	mg/L	
Cadmium	<0.0010	mg/L	ND	<0.0010	mg/L	ND
Chromium	<0.0031	mg/L	ND	0.42	mg/L	
Lead	<0.0073	mg/L	ND	<0.0073	mg/L	ND
Nickel	<0.0022	mg/L	ND	0.15	mg/L	
Selenium	<0.016	mg/L	ND	0.024	mg/L	J
Silver	<0.0041	mg/L	ND	<0.0041	mg/L	ND
Sodium	23	mg/L		4,000	mg/L	
Titanium	0.014	mg/L		0.16	mg/L	
Tungsten	<0.0052	mg/L	ND	0.012	mg/L	J
VOC						
1,4-Dioxane	<3.0	µ/L	ND	<1.7	µ/L	ND
Benzene	<3.0	µ/L	ND	<3.0	µ/L	ND
Carbon tetrachloride	<3.0	µ/L	ND	<3.0	µ/L	ND
Chlorobenzene	<3.0	µ/L	ND	<3.0	µ/L	ND
Chloroform	<3.0	µ/L	ND	<3.0	µ/L	ND
1,2-Dichloroethane	<3.0	µ/L	ND	<3.0	µ/L	ND
1,1-Dichloroethene	<3.0	µ/L	ND	<3.0	µ/L	ND
2-Butanone	890	µ/L		<5.0	µ/L	ND
Tetrachloroethene	<3.0	µ/L	ND	<3.0	µ/L	ND
Trichloroethene	<3.0	µ/L	ND	<3.0	µ/L	ND
Vinyl chloride	<3.0	µ/L	ND	<3.0	µ/L	ND

J, results are less than the RL but greater than or equal to MDL and the concentration is an approximate value; ND, non-detected at the reporting limit; B, compound was found in the lab blank.

Two PFAS compounds, PFOA (0.34 ng/sample) and PFHxS (0.16 ng/sample) were detected in the gas effluent samples of AFFF wastewater. However, we note that both are above the method detection limit but below the reporting limit, so these are only approximate values. These are similar concentrations found in the media check, so it is not conclusive if these were actually emitted from the process. As noted above, additional testing with replicate samples is required for future testing.

Table 13 Analytical PFAS concentrations of the AFFF wastewater gas effluent.

Compound	Effluent		Qualifiers	Calculated Effluent	
PFAS					
PFHpA	<1.1	ng/sample	ND	<0.018	ng/L
PFBA	<4.1	ng/sample	ND	<0.068	ng/L
PFPeA	<0.8	ng/sample	ND	<0.013	ng/L
PFHxA	<0.37	ng/sample	ND	<0.006	ng/L
PFOA	0.34	ng/sample	J	0.006	ng/L
PFNA	<0.17	ng/sample	ND	<0.003	ng/L
PFDA	<0.15	ng/sample	ND	<0.003	ng/L
PFUnA	<0.22	ng/sample	ND	<0.004	ng/L
PFDoA	<0.091	ng/sample	ND	<0.002	ng/L
PFTriA	<0.24	ng/sample	ND	<0.004	ng/L
PFTeA	<0.29	ng/sample	ND	<0.005	ng/L
PFHxDA	<0.18	ng/sample	ND	<0.003	ng/L
PFODA	<0.16	ng/sample	ND	<0.003	ng/L
PFBS	<0.17	ng/sample	ND	<0.003	ng/L
PFPeS	<0.058	ng/sample	ND	<0.001	ng/L
PFHxS	0.16	ng/sample	J	0.003	ng/L
PFHpS	<0.15	ng/sample	ND	<0.003	ng/L
PFOS	<0.24	ng/sample	ND	<0.004	ng/L
PFNS	<0.14	ng/sample	ND	<0.002	ng/L
PFDS	<0.11	ng/sample	ND	<0.002	ng/L
PFDoS	<0.22	ng/sample	ND	<0.004	ng/L
FOSA	<0.87	ng/sample	ND	<0.015	ng/L
NEtFOSA	<0.32	ng/sample	ND	<0.005	ng/L
NMeFOSA	<0.35	ng/sample	ND	<0.006	ng/L
NMeFOSAA	<0.21	ng/sample	ND	<0.004	ng/L
NEtFOSAA	<0.19	ng/sample	ND	<0.003	ng/L
NMeFOSE	<0.3	ng/sample	ND	<0.005	ng/L
NEtFOSE	<0.39	ng/sample	ND	<0.007	ng/L
4:2 FTS	<0.095	ng/sample	ND	<0.002	ng/L
6:2 FTS	<0.25	ng/sample	ND	<0.004	ng/L
8:2 FTS	<0.14	ng/sample	ND	<0.002	ng/L
10:2 FTS	<0.23	ng/sample	ND	<0.004	ng/L
DONA	<0.22	ng/sample	ND	<0.004	ng/L
HFPO-DA	<2.4	ng/sample	ND	<0.040	ng/L
9Cl-PF3ONS	<0.12	ng/sample	ND	<0.002	ng/L
11Cl-PF3OUdS	<0.14	ng/sample	ND	<0.002	ng/L
6:2 FTUCA	<0.1	ng/sample	ND	<0.002	ng/L
7:3 FTCA	<0.38	ng/sample	ND	<0.006	ng/L
10:2 FTCA	<0.93	ng/sample	ND	<0.016	ng/L

8:2 FTCA	<0.59	ng/sample	ND	<0.010	ng/L
PES	<0.088	ng/sample	ND	<0.001	ng/L
8:2 FTUCA	<0.13	ng/sample	ND	<0.002	ng/L
PFECA F	<0.57	ng/sample	ND	<0.010	ng/L
PFECA A	<0.086	ng/sample	ND	<0.001	ng/L
5:3 FTCA	<0.34	ng/sample	ND	<0.006	ng/L
6:2 FTCA	<0.72	ng/sample	ND	<0.012	ng/L
3:3 FTCA	<0.28	ng/sample	ND	<0.005	ng/L
PFECHS	<0.095	ng/sample	ND	<0.002	ng/L
PFECA B	<0.19	ng/sample	ND	<0.003	ng/L

J, results is less than the RL but greater than or equal to MDL and the concentration is an approximate value; ND, non-detected at the reporting limit.

1.2.7 Liquid and gas sample analytical results for RO reject waste

The results of RO reject testing in Table 14 indicate that PFAS compounds were reduced by 99.66% by the SCWO process. Most metals in the influent maintained similar concentrations, however nickel decreased from 0.65 mg/L to 0.022 mg/L. This decrease is discussed in Section 5. Bromide concentrations also increased, while TOC and COD levels were significantly reduced by 99.67%. TF concentration increased by 74%. In addition to 1,4-dioxane, benzene and 2-butanone were detected in the influent and reduced to non-detect levels in the effluent.

No PFAS was detected in gas effluent samples above RL (see Table 15). Again PFOA (0.18 ng/sample) and PFHxS (0.17 ng/sample) were detected above the MDL at approximate values, mimicking the presence of those in the media checks. PFHxA (0.37 ng/sample) was detected though right at the method detection limit (< 0.37 ng/sample), so this is not considered conclusive.

Table 14 Analytical results of the RO reject liquid influent and effluent.

Compound	Influent		Qualifiers	Effluent		Qualifiers
PFAS						
PFBA	9,300	ng/L		3.5	ng/L	J
PFPeA	190	ng/L	J	0.086	ng/L	J
PFHxA	330	ng/L		0.99	ng/L	J
PFHpA	31	ng/L	I	<0.86	ng/L	ND
PFOA	68	ng/L		<0.96	ng/L	ND
PFNA	<5	ng/L	ND	<0.54	ng/L	ND
PFDA	<5	ng/L	ND	<0.54	ng/L	ND
PFUnA	<5	ng/L	ND	<0.54	ng/L	ND
PFDoA	<5	ng/L	ND	<0.54	ng/L	ND
PFTrDA	<5	ng/L	ND	<0.54	ng/L	ND
PFTeDA	<5	ng/L	ND	<0.54	ng/L	ND
PFBS	2,200	ng/L	J	2.4	ng/L	
PFPeS	<5	ng/L		<0.54	ng/L	ND
PFHxS	69	ng/L		1.9	ng/L	J
PFHpS	<5	ng/L	ND	<0.54	ng/L	ND
PFOS	200	ng/L		4.3	ng/L	
PFNS	<5	ng/L	ND	<0.54	ng/L	ND

PFDS	<5	ng/L	ND	<0.54	ng/L	ND
PFDoS	<6	ng/L	ND	<0.64	ng/L	ND
4:2 FTS	<10	ng/L	ND	<1.1	ng/L	ND
6:2 FTS	500	ng/L		<1.1	ng/L	ND
8:2 FTS	<10	ng/L	ND	<1.1	ng/L	ND
PFOSA	<5	ng/L	ND	<0.54	ng/L	ND
NMeFOSA	<5	ng/L	ND	<0.54	ng/L	ND
NEtFOSA	<5	ng/L	ND	<0.54	ng/L	ND
NMeFOSAA	<5	ng/L	ND	<0.54	ng/L	ND
NEtFOSAA	<5	ng/L	ND	<0.54	ng/L	ND
NMeFOSE	<25	ng/L	ND	<2.7	ng/L	ND
NEtFOSE	<25	ng/L	ND	<2.7	ng/L	ND
HFPO-DA	<12	ng/L	ND	<1.3	ng/L	ND
ADONA	12	ng/L	J	<0.54	ng/L	ND
PFMPA	210	ng/L		<0.54	ng/L	ND
PFMBA	<5	ng/L	ND	<0.54	ng/L	ND
NFDHA	<5	ng/L	ND	<0.54	ng/L	ND
9Cl-PF3ONS	<5	ng/L	ND	<0.54	ng/L	ND
11Cl-PF3OUdS	<5	ng/L	ND	<0.54	ng/L	ND
PFEESA	<5	ng/L	ND	<0.54	ng/L	ND
3:3 FTCA	<10	ng/L		<1.1	ng/L	ND
5:3 FTCA	110	ng/L		<3	ng/L	ND
7:3 FTCA	<25	ng/L		<2.7	ng/L	ND
Other						
TOC	46	mg/L		8.00	mg/L	
COD	720	mg/L		35.4	mg/L	
TF	2,800	mg/L		11,000	mg/L	
Anions						
Bromide	6.5	mg/L	J	41	mg/L	
Chloride	900	mg/L		910	mg/L	
Fluoride	<1.0	mg/L	ND	5.1	mg/L	J
Sulfate	1,600	mg/L		1,800	mg/L	
Metals						
Arsenic	<0.082	mg/L	ND	<0.016	mg/L	ND
Barium	0.11	mg/L		0.084	mg/L	
Cadmium	0.0054	mg/L	J	0.0030	mg/L	J
Chromium	0.23	mg/L		0.18	mg/L	
Lead	0.082	mg/L		0.043	mg/L	
Nickel	0.65	mg/L		0.022	mg/L	J
Selenium	<0.082	mg/L	ND	0.024	mg/L	J
Silver	<0.021	mg/L	ND	<0.0041	mg/L	ND
Sodium	2,000	mg/L		1,700	mg/L	

Titanium	0.018	mg/L	J	0.041	mg/L	
Tungsten	1.0	mg/L		0.067	mg/L	
VOC						
1,4-Dioxane	6,500	µ/L		<1.7	µ/L	ND
Benzene	220	µ/L	J	<3.0	µ/L	ND
Carbon tetrachloride	<150	µ/L	ND	<3.0	µ/L	ND
Chlorobenzene	<150	µ/L	ND	<3.0	µ/L	ND
Chloroform	<150	µ/L	ND	<3.0	µ/L	ND
1,2-Dichloroethane	<150	µ/L	ND	<3.0	µ/L	ND
1,1-Dichloroethene	<150	µ/L	ND	<3.0	µ/L	ND
2-Butanone	21,000	µ/L		<5.0	µ/L	ND
Tetrachloroethene	<150	µ/L	ND	<3.0	µ/L	ND
Trichloroethene	<150	µ/L	ND	<3.0	µ/L	ND
Vinyl chloride	<150	µ/L	ND	<3.0	µ/L	ND

J, results are less than the RL but greater than or equal to MDL and the concentration is an approximate value; ND, Non detected at the reporting limit.

Table 15 Analytical PFAS concentrations of the RO reject gas effluent.

Compound	Effluent		Qualifiers	Calculated Effluent	
PFAS					
PFHpA	<1.1	ng/sample	ND	<0.007	ng/L
PFBA	<4.1	ng/sample	ND	<0.026	ng/L
PFPeA	<0.8	ng/sample	ND	<0.005	ng/L
PFHxA	0.37	ng/sample	J	0.004	ng/L
PFOA	0.18	ng/sample	J	0.002	ng/L
PFNA	<0.17	ng/sample	ND	<0.001	ng/L
PFDA	<0.15	ng/sample	ND	<0.001	ng/L
PFUnA	<0.22	ng/sample	ND	<0.001	ng/L
PFDoA	<0.091	ng/sample	ND	<0.001	ng/L
PFTriA	<0.24	ng/sample	ND	<0.002	ng/L
PFTeA	<0.29	ng/sample	ND	<0.002	ng/L
PFHxDA	<0.18	ng/sample	ND	<0.001	ng/L
PFODA	<0.16	ng/sample	ND	<0.001	ng/L
PFBS	<0.17	ng/sample	ND	<0.001	ng/L
PFPeS	<0.058	ng/sample	ND	<0.000	ng/L
PFHxS	0.17	ng/sample	J	0.001	ng/L
PFHpS	<0.15	ng/sample	ND	<0.001	ng/L
PFOS	<0.24	ng/sample	ND	<0.002	ng/L
PFNS	<0.14	ng/sample	ND	<0.001	ng/L
PFDS	<0.11	ng/sample	ND	<0.001	ng/L
PFDoS	<0.22	ng/sample	ND	<0.001	ng/L
FOSA	<0.87	ng/sample	ND	<0.005	ng/L

NEtFOSA	<0.32	ng/sample	ND	<0.002	ng/L
NMeFOSA	<0.35	ng/sample	ND	<0.002	ng/L
NMeFOSAA	<0.21	ng/sample	ND	<0.002	ng/L
NEtFOSAA	<0.19	ng/sample	ND	<0.001	ng/L
NMeFOSE	<0.3	ng/sample	ND	<0.001	ng/L
NEtFOSE	<0.39	ng/sample	ND	<0.002	ng/L
4:2 FTS	<0.095	ng/sample	ND	<0.002	ng/L
6:2 FTS	<0.25	ng/sample	ND	<0.001	ng/L
8:2 FTS	<0.14	ng/sample	ND	<0.002	ng/L
10:2 FTS	<0.23	ng/sample	ND	<0.001	ng/L
DONA	<0.22	ng/sample	ND	<0.001	ng/L
HFPO-DA	<2.4	ng/sample	ND	<0.015	ng/L
9Cl-PF3ONS	<0.12	ng/sample	ND	<0.001	ng/L
11Cl-PF3OUdS	<0.14	ng/sample	ND	<0.001	ng/L
6:2 FTUCA	<0.1	ng/sample	ND	<0.001	ng/L
7:3 FTCA	<0.38	ng/sample	ND	<0.002	ng/L
10:2 FTCA	<0.93	ng/sample	ND	<0.006	ng/L
8:2 FTCA	<0.59	ng/sample	ND	<0.004	ng/L
PES	<0.088	ng/sample	ND	<0.001	ng/L
8:2 FTUCA	<0.13	ng/sample	ND	<0.001	ng/L
PFECA F	<0.57	ng/sample	ND	<0.004	ng/L
PFECA A	<0.086	ng/sample	ND	<0.001	ng/L
5:3 FTCA	<0.34	ng/sample	ND	<0.002	ng/L
6:2 FTCA	<0.72	ng/sample	ND	<0.005	ng/L
3:3 FTCA	<0.28	ng/sample	ND	<0.002	ng/L
PFECHS	<0.095	ng/sample	ND	<0.001	ng/L
PFECA B	<0.19	ng/sample	ND	<0.001	ng/L

J, results are less than the RL but greater than or equal to MDL and the concentration is an approximate value; ND, non-detected at the reporting limit.

1.2.8 Liquid and gas sample analytical results for GAC waste

The prototype AFS was built and employed for grinding the GAC particle size from about 2 mm to <250 microns as specified by the SCWO feed specification. As a result, the system effectively reduced the particle size of GAC to below 250 µm, ensuring a consistent feed that supports the continuous operation of the SCWO reactor.

PFAS compounds in GAC were reduced by 99.995% by the SCWO process (see Table 20). Specifically, PFOA, PFHxS, and PFOS were reduced by 99.99%, although PFOS in the effluent was still at 5.2 ng/L (Table 16). Additionally, the fluoride concentration in the effluent increased to five times that of the influent. Both TOC and COD were significantly reduced by over 90%, and metal concentrations in the effluent increased. Furthermore, the same two PFAS compounds (PFOA at 0.26 ng/sample and PFHxS at 0.17 ng/sample, Table 17) observed in the media check were again observed in the gas emission samples of GAC waste.

Table 16 Analytical results for GAC slurry liquid influent and effluent.

Compound	Influent		Qualifiers	Effluent		Qualifiers
PFAS						
PFBA	3,600	ng/L		1.8	ng/L	J
PFPeA	12,000	ng/L		<0.6	ng/L	ND
PFHxA	31,000	ng/L		0.58	ng/L	J
PFHpA	7,500	ng/L		<0.8	ng/L	ND
PFOA	69,000	ng/L		0.9	ng/L	J
PFNA	1,400	ng/L		<0.5	ng/L	ND
PFDA	<50	ng/L	ND	<0.5	ng/L	ND
PFUnA	<50	ng/L	ND	<0.5	ng/L	ND
PFDoA	<50	ng/L	ND	<0.5	ng/L	ND
PFTrDA	<50	ng/L	ND	<0.5	ng/L	ND
PFTeDA	<50	ng/L	ND	<0.5	ng/L	ND
PFBS	39,000	ng/L		2.1	ng/L	
PFPeS	50,000	ng/L		<0.5	ng/L	ND
PFHxS	400,000	ng/L		<0.8	ng/L	ND
PFHpS	16,000	ng/L		3.3	ng/L	
PFOS	450,000	ng/L		5.2	ng/L	
PFNS	<50	ng/L	ND	<0.5	ng/L	ND
PFDS	<50	ng/L	ND	<0.5	ng/L	ND
PFDoS	<50	ng/L	ND	<0.6	ng/L	ND
4:2 FTS	<100	ng/L	ND	<1	ng/L	ND
6:2 FTS	32,000	ng/L		<1.1	ng/L	J
8:2 FTS	1,300	ng/L	J	<1	ng/L	ND
PFOSA	<50	ng/L	ND	<1	ng/L	ND
NMeFOSA	<50	ng/L	ND	<0.5	ng/L	ND
NEtFOSA	<48	ng/L	ND	<0.5	ng/L	ND
NMeFOSAA	<50	ng/L	ND	<0.5	ng/L	ND
NEtFOSAA	<50	ng/L	ND	<0.5	ng/L	ND
NMeFOSE	<250	ng/L	ND	<2.5	ng/L	ND
NEtFOSE	400	ng/L	J	<2.5	ng/L	ND
HFPO-DA	<50	ng/L	ND	<1.2	ng/L	ND
ADONA	<50	ng/L	ND	<0.5	ng/L	ND
PFMPA	<50	ng/L	ND	<0.5	ng/L	ND
PFMBA	<50	ng/L	ND	<0.5	ng/L	ND
NFDHA	<50	ng/L	ND	<0.5	ng/L	ND
9Cl-PF3ONS	<50	ng/L	ND	<0.5	ng/L	ND
11Cl-PF3OUdS	<50	ng/L	ND	<0.5	ng/L	ND
PFEESA	<50	ng/L	ND	<0.5	ng/L	ND
3:3 FTCA	<96	ng/L	ND	<1	ng/L	ND
5:3 FTCA	<280	ng/L	ND	<2.8	ng/L	ND

7:3 FTCA	<250	ng/L	ND	<2.5	ng/L	ND
Other						
TOC	710	mg/L		3.8	mg/L	
COD	14,980	mg/L		1,077	mg/L	
TF	3,000	mg/L		8,200	mg/L	
Anions						
Bromide	0.14	mg/L	J	0.35	mg/L	J
Chloride	160	mg/L		310	mg/L	
Fluoride	0.59	mg/L	J	4	mg/L	
Sulfate	96	mg/L		540	mg/L	
Metals						
Arsenic	<0.082	mg/L	ND	<0.082	mg/L	ND
Barium	1.5	mg/L		1.3	mg/L	
Cadmium	<0.0052	mg/L	ND	<0.0052	mg/L	ND
Chromium	0.094	mg/L		0.27	mg/L	
Lead	<0.037	mg/L	J	<0.037	mg/L	ND
Nickel	0.11	mg/L		0.23	mg/L	
Selenium	<0.082	mg/L	ND	<0.082	mg/L	ND
Silver	<0.021	mg/L	ND	<0.0021	mg/L	ND
Sodium	100	mg/L		840	mg/L	
Titanium	0.020	mg/L	J	0.093	mg/L	
Tungsten	<0.026	mg/L	ND	0.032	mg/L	J
VOC						
1,4-Dioxane	<8.5	µ/L	ND	<0.17	µ/L	ND
Benzene	<150	µ/L	ND	<0.30	µ/L	ND
Carbon tetrachloride	<150	µ/L	ND	<0.30	µ/L	ND
Chlorobenzene	<150	µ/L	ND	<0.30	µ/L	ND
Chloroform	170	µ/L	J	<0.30	µ/L	ND
1,2-Dichloroethane	<150	µ/L	ND	<0.30	µ/L	ND
1,1-Dichloroethene	<150	µ/L	ND	<0.30	µ/L	ND
2-Butanone	<250	µ/L	ND	<0.50	µ/L	ND
Tetrachloroethene	<150	µ/L	ND	<0.30	µ/L	ND
Trichloroethene	1,300	µ/L		<0.30	µ/L	ND
Vinyl chloride	<150	µ/L	ND	<0.30	µ/L	ND

J, results are less than the RL but greater than or equal to MDL and the concentration is an approximate value; ND, non-detected at the reporting limit.

Table 17 Analytical PFAS concentrations of the GAC gas effluent.

Compound	Effluent		Qualifiers	Calculated Effluent	
PFAS					
PFHpA	<1.1	ng/sample	ND	<0.024	ng/L
PFBA	<4.1	ng/sample	ND	<0.091	ng/L
PFPeA	<0.8	ng/sample	ND	<0.018	ng/L

PFHxA	<0.37	ng/sample	ND	<0.008	ng/L
PFOA	0.26	ng/sample	J	0.006	ng/L
PFNA	<0.17	ng/sample	ND	<0.004	ng/L
PFDA	<0.15	ng/sample	ND	<0.003	ng/L
PFUnA	<0.22	ng/sample	ND	<0.005	ng/L
PFDoA	<0.091	ng/sample	ND	<0.002	ng/L
PFTriA	<0.24	ng/sample	ND	<0.005	ng/L
PFTeA	<0.29	ng/sample	ND	<0.006	ng/L
PFHxDA	<0.18	ng/sample	ND	<0.004	ng/L
PFODA	<0.16	ng/sample	ND	<0.004	ng/L
PFBS	<0.17	ng/sample	ND	<0.004	ng/L
PFPeS	<0.058	ng/sample	ND	<0.001	ng/L
PFHxS	0.17	ng/sample	J	0.004	ng/L
PFHpS	<0.15	ng/sample	ND	<0.003	ng/L
PFOS	<0.24	ng/sample	ND	<0.005	ng/L
PFNS	<0.14	ng/sample	ND	<0.003	ng/L
PFDS	<0.11	ng/sample	ND	<0.002	ng/L
PFDoS	<0.22	ng/sample	ND	<0.005	ng/L
FOSA	<0.87	ng/sample	ND	<0.019	ng/L
NEtFOSA	<0.32	ng/sample	ND	<0.007	ng/L
NMeFOSA	<0.35	ng/sample	ND	<0.008	ng/L
NMeFOSAA	<0.21	ng/sample	ND	<0.008	ng/L
NEtFOSAA	<0.19	ng/sample	ND	<0.005	ng/L
NMeFOSE	<0.3	ng/sample	ND	<0.004	ng/L
NEtFOSE	<0.39	ng/sample	ND	<0.007	ng/L
4:2 FTS	<0.095	ng/sample	ND	<0.009	ng/L
6:2 FTS	<0.25	ng/sample	ND	<0.002	ng/L
8:2 FTS	<0.14	ng/sample	ND	<0.006	ng/L
10:2 FTS	<0.23	ng/sample	ND	<0.003	ng/L
DONA	<0.22	ng/sample	ND	<0.005	ng/L
HFPO-DA	<2.4	ng/sample	ND	<0.053	ng/L
9Cl-PF3ONS	<0.12	ng/sample	ND	<0.003	ng/L
11Cl-PF3OUdS	<0.14	ng/sample	ND	<0.003	ng/L
6:2 FTUCA	<0.1	ng/sample	ND	<0.002	ng/L
7:3 FTCA	<0.38	ng/sample	ND	<0.008	ng/L
10:2 FTCA	<0.93	ng/sample	ND	<0.021	ng/L
8:2 FTCA	<0.59	ng/sample	ND	<0.013	ng/L
PES	<0.088	ng/sample	ND	<0.002	ng/L
8:2 FTUCA	<0.13	ng/sample	ND	<0.003	ng/L
PFECA F	<0.57	ng/sample	ND	<0.013	ng/L
PFECA A	<0.086	ng/sample	ND	<0.002	ng/L
5:3 FTCA	<0.34	ng/sample	ND	<0.008	ng/L

6:2 FTCA	<0.72	ng/sample	ND	<0.016	ng/L
3:3 FTCA	<0.28	ng/sample	ND	<0.006	ng/L
PFECHS	<0.095	ng/sample	ND	<0.002	ng/L
PFECA B	<0.19	ng/sample	ND	<0.004	ng/L

J, results are less than the RL but greater than or equal to MDL and the concentration is an approximate value; ND, non-detected at the reporting limit.

1.2.9 Liquid and gas sample analytical results for resin waste

PFAS compounds in resin waste were reduced by 98.52% by the SCWO process (see Table 21). PFBA demonstrating the lowest removal efficiency at 17%, however the presence of PFBA in the baselines indicates that there may be some source of PFBA outside of the treated waste. Trichloromethane and trichloroethene was detected in the influent but was non-detectable in the effluent. Additionally, both TOC and COD levels decreased in the effluent, while TF concentrations increased by 79%. Chloride and sulfate concentrations increased in the effluent, suggesting COC degradation. Lead and nickel increased in the effluent, again showing low levels of reactor lining degradation as expected. Furthermore, two PFAS compounds (PFOA at 0.27 ng/sample and PFHxS at 0.16 ng/sample) were detected in the gas phase, again mimicking that of the media check so their actual present in gas emission is not conclusive. Section 1.3 summarize these results with the other wastes tested.

Table 18 Analytical results of the resin liquid influent and effluent.

Compound	Influent		Qualifiers	Effluent		Qualifiers
PFAS						
PFBA	850	ng/L		700	ng/L	
PFPeA	4,200	ng/L		470	ng/L	
PFHxA	5,800	ng/L		290	ng/L	
PFHpA	1,200	ng/L		93	ng/L	
PFOA	7,100	ng/L		62	ng/L	
PFNA	230	ng/L		0.62	ng/L	J
PFDA	<48	ng/L	ND	<0.61	ng/L	ND
PFUnA	<48	ng/L	ND	<0.61	ng/L	ND
PFDoA	<48	ng/L	ND	<0.61	ng/L	ND
PFTrDA	<48	ng/L	ND	<0.61	ng/L	ND
PFTeDA	<48	ng/L	ND	<0.61	ng/L	ND
PFBS	2,600	ng/L		3.9	ng/L	
PFPeS	3,200	ng/L		1.2	ng/L	J
PFHxS	29,000	ng/L		8.7	ng/L	
PFHpS	1,400	ng/L		<0.61	ng/L	ND
PFOS	69,000	ng/L	B	16	ng/L	
PFNS	<48	ng/L	ND	<0.61	ng/L	ND
PFDS	<48	ng/L	ND	<0.61	ng/L	ND
PFDoS	<58	ng/L	ND	<0.74	ng/L	ND
4:2 FTS	<96	ng/L	ND	<1.2	ng/L	ND
6:2 FTS	7,000	ng/L		1.8	ng/L	J
8:2 FTS	260	ng/L	J	<1.2	ng/L	ND

PFOSA	<48	ng/L	ND	2	ng/L	J
NMeFOSA	<48	ng/L	ND	<0.61	ng/L	ND
NEtFOSA	<48	ng/L	ND	<0.61	ng/L	ND
NMeFOSAA	<48	ng/L	ND	<0.61	ng/L	ND
NEtFOSAA	<48	ng/L	ND	<0.61	ng/L	ND
NMeFOSE	<240	ng/L	ND	<3.1	ng/L	ND
NEtFOSE	<240	ng/L	ND	<3.1	ng/L	ND
HFPO-DA	<120	ng/L	ND	<1.5	ng/L	ND
ADONA	<48	ng/L	ND	<0.61	ng/L	ND
PFMPA	<48	ng/L	ND	<0.61	ng/L	ND
PFMBA	<48	ng/L	ND	<0.61	ng/L	ND
NFDHA	<48	ng/L	ND	<0.61	ng/L	ND
9Cl-PF3ONS	<48	ng/L	ND	<0.61	ng/L	ND
11Cl-PF3OUdS	<48	ng/L	ND	<0.61	ng/L	ND
PFEESA	<48	ng/L	ND	<0.61	ng/L	ND
3:3 FTCA	<96	ng/L		<1.2	ng/L	ND
5:3 FTCA	<270	ng/L		<3.4	ng/L	ND
7:3 FTCA	<240	ng/L		<3.1	ng/L	ND
Other						
TOC	590	mg/L		3.30	mg/L	
COD	10,290	mg/L		57.8	mg/L	
TF	1,300	mg/L		6,300	mg/L	
Anions						
Bromide	<0.14	mg/L	ND	2.3	mg/L	
Chloride	98	mg/L		380	mg/L	
Fluoride	0.27	mg/L	J	3.8	mg/L	
Sulfate	170	mg/L		660	mg/L	
Metals						
Arsenic	<0.082	mg/L	ND	<0.082	mg/L	ND
Barium	0.43	mg/L		0.40	mg/L	
Cadmium	<0.0052	mg/L	ND	<0.0052	mg/L	ND
Chromium	0.21	mg/L		<0.015	mg/L	ND
Lead	0.044	mg/L	J	0.11	mg/L	
Nickel	0.14	mg/L		0.31	mg/L	
Selenium	<0.082	mg/L	ND	<0.082	mg/L	ND
Silver	<0.021	mg/L	ND	<0.0041	mg/L	ND
Sodium	180	mg/L		470	mg/L	
Titanium	0.087	mg/L		0.052	mg/L	
Tungsten	<0.026	mg/L	ND	<0.026	mg/L	ND
VOC						
1,4-Dioxane	<8.5	µ/L	ND	<0.17	µ/L	ND
Benzene	<15	µ/L	ND	<0.30	µ/L	ND

Carbon tetrachloride	<15	µ/L	ND	<0.30	µ/L	ND
Chlorobenzene	<15	µ/L	ND	<0.30	µ/L	ND
Chlororform	<15	µ/L	ND	<0.30	µ/L	ND
1,2-Dichloroethane	<15	µ/L	ND	<0.30	µ/L	ND
1,1-Dichloroethene	<15	µ/L	ND	<0.30	µ/L	ND
2-Butanone	<25	µ/L	ND	<0.50	µ/L	ND
Tetrachloroethene	<15	µ/L	ND	<0.30	µ/L	ND
Trichloroethene	130	µ/L		<0.30	µ/L	ND
Vinyl chloride	<15	µ/L	ND	<0.30	µ/L	ND

J, results are less than the RL but greater than or equal to MDL and the concentration is an approximate value; ND, non-detected at the reporting limit.

Table 19 Analytical PFAS concentrations of the resin gas effluent.

Compound	Effluent		Qualifiers	Calculated Effluent	
PFAS					
PFHpA	<1.1	ng/sample	ND	<0.024	ng/L
PFBA	<4.1	ng/sample	ND	<0.089	ng/L
PFPeA	<0.8	ng/sample	ND	<0.017	ng/L
PFHxA	<0.37	ng/sample	ND	<0.008	ng/L
PFOA	0.27	ng/sample	J	0.006	ng/L
PFNA	<0.17	ng/sample	ND	<0.004	ng/L
PFDA	<0.15	ng/sample	ND	<0.003	ng/L
PFUnA	<0.22	ng/sample	ND	<0.005	ng/L
PFDoA	<0.091	ng/sample	ND	<0.002	ng/L
PFTriA	<0.24	ng/sample	ND	<0.005	ng/L
PFTeA	<0.29	ng/sample	ND	<0.006	ng/L
PFHxDA	<0.18	ng/sample	ND	<0.004	ng/L
PFODA	<0.16	ng/sample	ND	<0.003	ng/L
PFBS	<0.17	ng/sample	ND	<0.004	ng/L
PFPeS	<0.058	ng/sample	ND	<0.001	ng/L
PFHxS	0.16	ng/sample	J	0.003	ng/L
PFHpS	<0.15	ng/sample	ND	<0.003	ng/L
PFOS	<0.24	ng/sample	ND	<0.005	ng/L
PFNS	<0.14	ng/sample	ND	<0.003	ng/L
PFDS	<0.11	ng/sample	ND	<0.002	ng/L
PFDoS	<0.22	ng/sample	ND	<0.005	ng/L
FOSA	<0.87	ng/sample	ND	<0.019	ng/L
NEtFOSA	<0.32	ng/sample	ND	<0.007	ng/L
NMeFOSA	<0.35	ng/sample	ND	<0.008	ng/L
NMeFOSAA	<0.21	ng/sample	ND	<0.005	ng/L
NEtFOSAA	<0.19	ng/sample	ND	<0.004	ng/L
NMeFOSE	<0.3	ng/sample	ND	<0.007	ng/L
NEtFOSE	<0.39	ng/sample	ND	<0.008	ng/L

4:2 FTS	<0.095	ng/sample	ND	<0.002	ng/L
6:2 FTS	<0.25	ng/sample	ND	<0.005	ng/L
8:2 FTS	<0.14	ng/sample	ND	<0.003	ng/L
10:2 FTS	<0.23	ng/sample	ND	<0.005	ng/L
DONA	<0.22	ng/sample	ND	<0.005	ng/L
HFPO-DA	<2.4	ng/sample	ND	<0.052	ng/L
9Cl-PF3ONS	<0.12	ng/sample	ND	<0.003	ng/L
11Cl-PF3OUdS	<0.14	ng/sample	ND	<0.003	ng/L
6:2 FTUCA	<0.1	ng/sample	ND	<0.002	ng/L
7:3 FTCA	<0.38	ng/sample	ND	<0.008	ng/L
10:2 FTCA	<0.93	ng/sample	ND	<0.020	ng/L
8:2 FTCA	<0.59	ng/sample	ND	<0.013	ng/L
PES	<0.088	ng/sample	ND	<0.002	ng/L
8:2 FTUCA	<0.13	ng/sample	ND	<0.003	ng/L
PFECA F	<0.57	ng/sample	ND	<0.012	ng/L
PFECA A	<0.086	ng/sample	ND	<0.002	ng/L
5:3 FTCA	<0.34	ng/sample	ND	<0.007	ng/L
6:2 FTCA	<0.72	ng/sample	ND	<0.016	ng/L
3:3 FTCA	<0.28	ng/sample	ND	<0.006	ng/L
PFECHS	<0.095	ng/sample	ND	<0.002	ng/L
PFECA B	<0.19	ng/sample	ND	<0.004	ng/L

J, results are less than the RL but greater than or equal to MDL and the concentration is an approximate value; ND, non-detected at the reporting limit.

1.2.10 Solid percentage in liquid effluents

All liquid effluent samples from the five waste materials tested were analyzed in house for total solids to determine solid percentage. The results in Table 18 indicated that all liquid effluent samples contained 0.2–1.0 wt.% total solids even though the influent solids content ranged from near zero to 10wt%. There were not enough solids in the effluent samples collected to perform separate PFAS analytical methods for the individual solids themselves, which requires at least 10 grams.

Table 20 Total solids percentage of influent versus effluent per EPA method 1684.

Waste	Percent Solids Influent	Percent Solids effluent	Solids Reduction	Observations
Biosolids	10%	0.6%	94%	Brownish particles
AFFF wastewater	0.01%	1.0%	NA	White salt-like particles
RO reject	2%	0.5%	15%	Yellow green particles
GAC	10%	0.4%	96%	White salt-like particles
Resin	10%	0.2%	98%	Brownish particles

1.3 Discussion

1.3.1 AFS Performance

This pilot test of the AFS during this project provided valuable insights into the mechanical requirements necessary to achieve the desired particle size and timing needed for continuous SCWO system operation. Specifically, this involved determining the required time to create a 10-15% solid slurry with particle sizes below 250 microns, ensuring sufficient volume for processing through the AFS.

Key learning from this test included temperature control, regulating the speed at which the solids are introduced into the mill, maintaining the appropriate water ratio, managing spills, and optimizing system plumbing. These insights informed necessary improvements to the AFS-SCWO system. Details on these parameters will remain proprietary. The ability to reduce solid waste particle size for this application has not been previously demonstrated, thus this work has significant impact for the industry on processing solid PFAS wastes. Given the current DoD's moratorium on incineration of AFFF/PFAS-related wastes,[4] the AFS-SCWO system may offer a viable alternative for their disposal.

1.3.2 PFAS and Related Analysis Overview

To aid in the discussion, we first provide a general overview of the analytical methods used in this test related to PFAS.

1.3.2.1 Targeted PFAS Analysis:

This method focuses on identifying and quantifying specific PFAS compounds in a sample. It typically uses techniques like Liquid Chromatography-Mass Spectrometry (LC-MS/MS) to detect and measure individual PFAS molecules. This method is highly accurate and sensitive, allowing for the detection of low concentrations of specific PFAS compounds. It is commonly used for regulatory compliance and monitoring purposes.

1.3.2.2 Total Fluorine Analysis:

TF analysis measures the total amount of fluorine present in a sample, including both organic and inorganic fluorine compounds. This method does not differentiate between individual PFAS compounds, but provides an overall measure of fluorine content. Techniques like CIC are used, where the sample is combusted to convert all fluorine-containing compounds to fluoride ions, which are then measured. This method is useful for assessing a non-targeted PFAS content.

1.3.2.3 Fluoride Analysis:

Fluoride analysis specifically measures the concentration of fluoride ions in a sample. This method is different from TF analysis as it only quantifies the inorganic fluoride ions and does not account for organic fluorine compounds. Techniques like Ion Chromatography (IC) are commonly used for fluoride analysis. This method is useful for monitoring fluoride generated from PFAS mineralization.

Each of these methods provides different insights into the presence and concentration of PFAS and fluorine compounds in samples. Targeted PFAS analysis is ideal for identifying specific PFAS compounds, TF analysis gives a comprehensive measure of all fluorine content, and fluoride analysis focuses on inorganic fluoride ions. Herein we discuss the results obtained from each of these methods, a discussion on any inaccuracies, conclusions, and future research directors.

1.3.3 PFAS Removal Efficiency in Liquids Post-SCWO Treatment

Figure 11 summarizes the measured targeted PFAS in influent and effluent across all wastes tested. Table 21 provides total PFAS in influent and effluent as well as percent reduction of total PFAS in each waste stream. This was calculated after the effluent total PFAS concentrations were normalized to eliminate the difference between waste feed rate and

liquid effluent rate of each waste test. There is some dilution that occurs after the SCWO reactor due to the addition of quench water used to neutralize acids formed in the reactor. This dilution has been accounted for in the data presented in Table *21*.

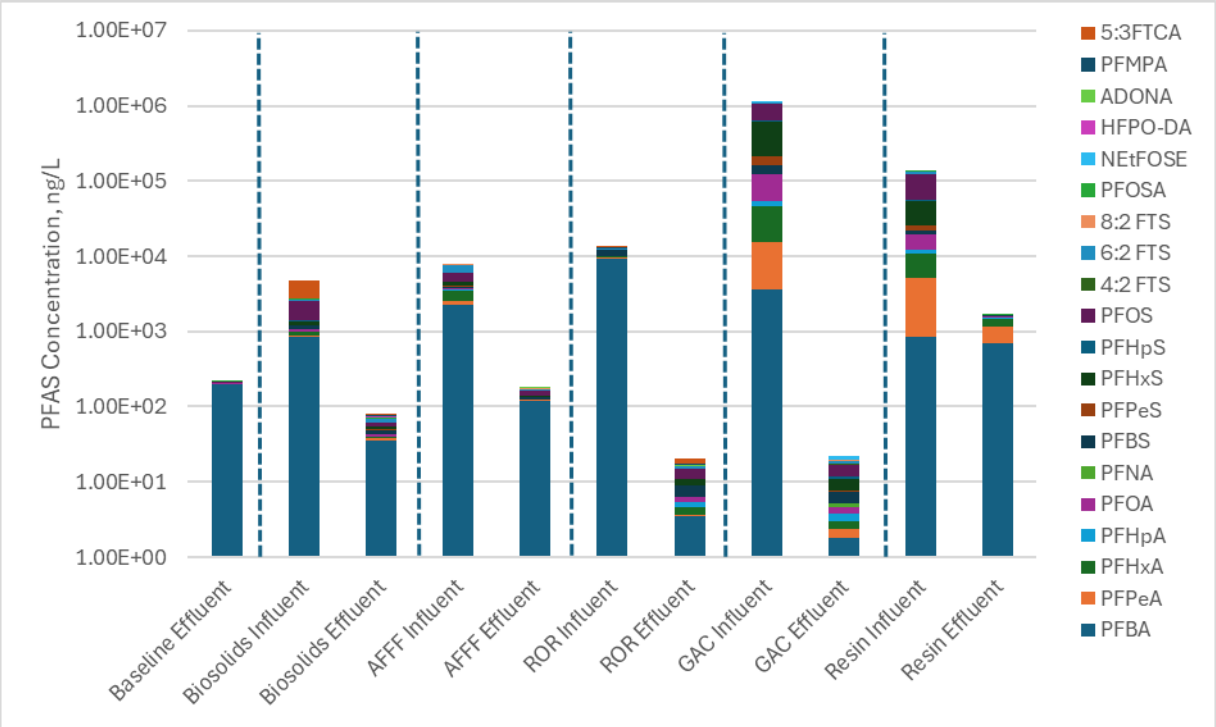

Figure 11. Compiled results of PFAS concentrations in influent and effluent of all wastes tested.

Taken together, these results show that significant (95.91% - 99.995%) reduction in PFAS across all waste types. Even with the high concentration, solid wastes of GAC and resin, 99.995% and 98.52% reductions were seen. These results, combined with the fact that all waste materials tested in the SCWO experiments exhibited stable time series plots, indicate that the AFS succeeded in preparing feeds to meet the boundary conditions for stable SCWO operation. These results support the utility of the AFS to process waste into a form that both supports continuous SCWO reactions as well as target COC reduction.

Although PFAS reduced significantly, some PFAS were detected in all effluents at very low concentrations. PFAS in the effluent could be present for a number of reasons. First, untreated tap water that is used in the quench water could have PFAS because most municipally supplied tap water has PFAS. San Diego city water report from 2023 indicated 10 ppt of PFHxS and PFNA, PFBA was not included in the analyte list at that time.[5] A water softening system is installed at the pilot facility. Water used in the process needs to be analyzed to determine if this is a factor (results pending). The second cause could be residuals from prior testing of high concentrations wastes also known as “bleed-over” or “increasing baseline”. The start-up/shut-down procedure for the SCWO system involves running the system with just water and diesel in the influent for 30-60 mins. Baseline sample was only taken at the beginning of the test, not between each sample. Therefore we cannot be sure that increasing baseline was not an issue, but we hypothesize that it is not a major contributor considering the significant percent reductions seen between the influent and the effluent PFAS concentrations even for very high starting concentrations. Also the fact that effluent concentrations are at or below the baseline measurement suggests that increasing baseline was not occurring. The third cause of PFAS in the effluent would be incomplete destruction of PFAS in the reactor zone. GAC PFAS effluent concentrations were lower than that of AFFF effluent, which had lower starting PFAS and contained no solids, so AFFF would presumably undergo more

complete oxidation but this did not occur. In contrast, resin PFAS effluent concentrations were higher than that of AFFF effluent, so AFFF would presumably undergo more complete oxidation, which did occur. Based on these conflicting results, further testing is needed to determine if background PFAS levels are impacting effluent concentrations. Once background PFAS can be eliminated, we can better understand the role of particle size and waste composition on PFAS destruction efficacy.

PFBA is the most dominant PFAS across all samples, including the baseline, and at similar concentrations. PFOS was higher than 6.5 ng/L in every waste liquid effluent and PFOA was higher than 7 ng/L in the resin liquid. These results support the hypothesis of background contamination from tap water. Untreated tap water mixes with the effluent in the quench tank. As a precaution, any liquid effluent that could potentially contain PFAS will be handled by ACTenviro.

Nearly 90% of GAC particles were ground from 2mm to < 45 microns in a couple hours, while the remaining 10% were <186 microns, which is well below the target particle size boundary condition of 250 microns. In contrast, for the resin, only 60% particles of the total volume are <250 microns, with the remaining 40% ranging between 250 and 1,110 microns. Based on these results, our particle size reduction is more effective against GAC than resin. However, we note that the resin slurry did still support stable SCWO operation. This test suggests that particle size alone does not determine processability. Particle composition, elasticity, and aspect ratio may also impact processability. The resin beads are made of a modified polystyrene, which can deform under pressure. We hypothesize that resin particles larger than 200 microns may be able to deform and flow through pumps without bridging and clogging that occurs with the harder GAC. This is corroborated with the particle size measurements shown in Table 5.

Table 21 Summary of total PFAS compound removal efficiency by SCWO process.

Waste	Influent Total PFAS (ng/L)	Effluent Total PFAS (ng/L)	*Normalized Effluent Total PFAS (ng/L)	*Removal Efficiency (%)
Biosolids	4,969.7	111.4	62.1	98.75%
AFFF wastewater	10,124	207.2	414.4	95.91%
RO reject	13,443	44.3	45.4	99.66%
GAC	1,115,224	43	50.4	99.995%
Resin	134,112	1,679.4	1,991.3	98.52%

*Values normalized by elimination of difference on waste feed rate and liquid effluent rate (Table 2).

The effect of particle size on PFAS destruction showed a loose correlation in this test, though we caveat any conclusions with the fact that our starting concentration of PFAS on resins may be inaccurately low. The AFS achieved significant particle size reduction of GAC, decreasing from over 2,000 µm to below 250 µm, followed SCWO destruction of 99.995% reduction in total PFAS. In contrast, the resin material achieved a slightly lower particle size reduction, though still successful, 98.52% PFAS reduction. We hypothesize that this lower % PFAS reduction seen from resin compared to GAC could result from two effects. First, the resin particles post-AFS treatment were larger than GAC, which may have led to lower total material destruction in the reactor. PFAS in the interior of the particles may not have been fully oxidized. However, we note that low solid wt% in effluent resulted from resins so this reasoning is not highly likely. Second, the total PFAS concentration reported for the resin influent may be inaccurately low, which would in turn result in an overall lower reported % reduction. Others have reported on inaccurately low PFAS concentrations when using these standard methods 1633 and 537.1 on resins.[3] Resin has a higher loading capacity for PFAS than GAC, yet the measured PFAS concentration on the resin was lower than that of GAC. The PFAS analysis of solids in this

study used a method where the PFAS was first extracted from the solids, then measured in solution. PFAS has a very high binding affinity for resin, thus is very difficult to desorb. The binding affinity of PFAS to GAC is lower than resin, so an extraction-based analysis may pull more PFAS from GAC than resin. We suspect this second reasoning is more likely driving the results of this test. Future work will aim to get better starting concentrations of resins. The high PFAS concentration in the resin effluent may result from incomplete destruction if the starting concentration is very high. Future testing will also look to increase resonance time of resins in the SCWO reactor and different approaches for particle size reduction of low modulus materials.

Table 22 Summary of PFAS compound concentrations for all the gas effluents.

Analyte	Baseline ng/sample	Media Check ng/sample	Biosolids ng/sample	AFFF wastewater ng/sample	RO reject ng/sample	GAC ng/sample	Resin ng/sample
PFBA			5.1				
PFPeA			1.2				
PFHxA			3.9		0.62		
PFHpA			3.4				
PFOA	0.27	0.20	0.2	0.34	0.39	0.26	0.27
PFNA			0.54				
PFDoA			0.3				
PFHxS	0.16	0.15	0.47	0.16	0.17	0.17	0.16
PFOS			1.4				
6:2 FTS			2.9				
8:2 FTS			1.7				

1.3.4 PFAS in Gas Phase Post-SCWO Treatment

PFOA and PFHxS were present in higher concentrations in all the gas effluents including baseline and media check samples (Table *22*). The lab indicated that these compounds are contaminants from the XAD resin prior to gas sampling and are common low-level background PFAS. Eurofins Scientific implemented a three-day cleaning process for the resin prior to use, but these contaminants remained. Biosolids gas samples revealed the presence of very low levels of other PFAS compounds and RO reject sample showed a concentration of 0.62 ng/sample PFHxA. Future testing aims to determine whether these compounds are truly present in SCWO gas effluent or are residual contaminants from the system by pretesting of the resin used in TO-13 for sampling. Biosolids gas analysis showed the most PFAS in gas sampling over any other wastes. Considering biosolids were first tested after the baseline sample was collected, it is not likely that this is a bleed-over effect. Considering the PFAS concentrations were similar for biosolids and RO reject, it is unexpected that these wastes did not exhibit similar PFAS gas effluent concentrations if these PFAS are in fact from incomplete destruction. Replicate measurements would likely resolve this issue, but was out of scope and out of budget of the program. Future testing will continue to measure for targeted and non-targeted PFAS in the gas effluent.

1.3.5 Total fluorine and fluoride in liquid influent and effluents

The previous section described changes in target PFAS concentrations over time. When coupled with TF and total fluoride concentrations, further demonstration of PFAS destruction (e.g., breaking of carbon-fluorine bonds) could theoretically be assessed. Table 23 and Table 24 summarize TF and fluoride, respectively, across all the liquid/slurry samples. Methods for these analyses are described in Section 2.5. Δa and Δb numbers are the TF and fluoride difference between influent and effluent samples, respectively. TF measures all fluorine present in the sample regardless of its form in the sample. F- alternatively measures only the anionic form of fluorine present in the sample. As PFAS are destroyed, the carbon-fluorine bonds are broken and F- is released or smaller chain defluorinated

products are produced. The smaller defluorinated compounds would not be detected in the F- measurement, but would be measured in TF.

TF concentrations were higher than F^- in the influent samples, as expected because the PFAS compounds are intact. In all effluent samples, F- concentrations significantly increased by 17% to 2,152.9% (Table *24*), possibly indicating defluorination of PFAS compounds is occurring. Higher F- concentrations correlated with higher initial starting concentrations of PFAS, which aligns with mineralization of PFAS. However, F- concentration did not correlate strongly with targeted PFAS removal efficiency (Table *21*), for example biosolids demonstrated a higher RE than AFFF, but lower F-. GAC demonstrated a higher removal efficiency than resins, but lower F-. These results further support our hypotheses that the wastes with solids may have inaccurate starting targeted PFAS concentrations due to inaccurate extraction methods.

Table 23 Summary of total fluorine in all liquid/slurry waste influents/effluents.

Waste	TF Inf. (mg/L)	TF Eff. (mg/L)	*Normalized TF Eff. (mg/L)	*Eff.-Inf., Δa, (mg/L)
Baseline	-	19	-	-
Biosolids	2.8	9.6	5.3	2.5
AFFF wastewater	2.8	9.2	18.4	15.6
RO reject	14	11	11.3	-2.7
GAC	3	8.2	9.6	6.6
Resin	1.3	6.3	7.5	6.2

*Values normalized by elimination of difference on waste feed rate and liquid effluent rate (see Table 6).

Table 24 Summary of total fluoride in all liquid/slurry waste influents/effluents.

Waste	F^- Inf. (mg/L)	F^- Eff. (mg/L)	*Normalized F^- Eff. (mg/L)	*Eff.-Inf., Δb, (mg/L)	*Δb/F^- Inf. (%)
Baseline	-	18	-	-	-
Biosolids	2	4.2	2.3	0.3	17.0
AFFF wastewater	1.1	4.4	8.8	7.7	700.0
RO reject	1	5.1	5.2	4.2	422.4
GAC	0.5	4	4.7	4.2	837.1
Resin	0.2	3.8	4.5	4.3	2,152.9

*Values normalized by elimination of difference on waste feed rate and liquid effluent rate (see Table 5).

If all PFAS compounds going into SCWO were completely mineralized, then TF in effluent would equal F- in the effluent and F- in the effluent would be higher than F- in the influent. Further, if the TF method truly measures all fluorine-containing compounds, then TF in the influent would equal TF in the effluent. But this was not the case in the results from this test, which makes it difficult to draw conclusions on complete mineralization from this test. TF in the effluent samples was higher than in the influent for all wastes, except RO reject. F- concentrations in the effluent were higher than that of the influent, which may suggest mineralization. However, definitive conclusions cannot be drawn from the data resulting from this test because the baseline TF and F- concentrations were orders of magnitude higher compared to PFAS concentrations. Considering that most PFAS are in the 0.001 mg/L concentrations, it is unlikely that mineralization would make a significant impact on either TF or F-, which have concentrations in the 2-10 mg/L range. For example, the total possible F- (mg/L) generated from complete defluorination of all targeted PFAS detected above MDL is: 0.0032 mg/L for biosolids, 0.0089 for AFF, 0.147 for RO reject, 1.312 in GAC and 0.157 in resin. Results are further complicated because the baseline effluent sample (i.e., no waste running through the system), the TF concentration was 19 mg/L and F^- concentration was 18 mg/L. Typical tap water has 2 mg/L F-, so there may be some bleedover from previous testing on the SCWO reactor. Future studies would need to remove background fluoride from

tap water and ensure post-reactor components are completely washed between runs in order to be able to accurately monitor fluoride generated from PFAS mineralization. This is the focus for future work.

Table 25 Summary of total fluoride in all liquid/slurry waste influents/effluents.

Waste	Normalized [a] TF Eff. (mg/L)	*Normalized F^- Eff. (mg/L)	% Diff $= \frac{TF - F}{(TF - F)/2} X\ 100$
Baseline	-	-	-
Biosolids	5.3	2.3	78%
AFFF wastewater	18.4	8.8	70%
RO reject	11.3	5.2	73%
GAC	9.6	4.7	68.5
Resin	7.5	4.5	50%

[a] Values normalized by elimination of difference on waste feed rate and liquid effluent rate (see Table 5 and 6).

There is further uncertainly in the TF measurements because if the method was truly accounting for all fluorine, then the TF in influent should be equal to TF in effluent. TF in influent is less than TF in effluent in our test, which could be due to a number of reasons. First, the hydrolytic pyrolysis used in the TF method to mineralize organic fluorines to fluoride for measurement must not be completely mineralizing the organic fluorines. Second, bleedover of fluoride in the post-reactor components run-to-run would increase TF in the effluent. Additional error can be introduced when the wastes tested are slurries (e.g., mixtures of liquids and solids). Eurofins Scientific method (ELLE SOP in Attachment 3) involves shaking a 125 ml sample bottle, pipetting a 200 µl aliquot that includes both liquid and solid components, placing this aliquot in quartz tube for direct combustion. When the sample has solid particles or is an unstable slurry, the aliquot may not accurately represent the entire sample due to inhomogeneity. If the samples are not properly homogenized, and less solids are taken up in the pipette used to transfer sample from the bottle to the combustion tube, then reported concentrations would be inaccurately low. Lastly, solid-phase PFAS (e.g, sorbed onto solids) may not fully pyrolyze and therefore not be measured. If any of these occur, then the TF concentrations for the influent would be under reported. This would also apply to the F^- concentrations.

Future work will aim to separate solids from liquids prior to TF testing to avoid these above complications. Alternative methods that do not rely on sample thermal transformation are also under consideration, since we know that thermal transformation of PFAS is difficult. We are investigating particle-induced gamma-ray emission (PIGE) spectroscopy as an alternative method for TF measurement.[6] PIGE provides a non-destructive and quantitative measurement of TF on solids. Since it does not require destruction, it may more accurately measure TF in robust difficult-to-destroy samples. Additionally, using the total oxidizable precursor (TOP) analysis could provide insight into the non-targeted PFAS that were not measured in this test but do contribute to TF. PFAS defluorination and transformation mechanisms continue to be an area of investigation of other programs, but notably was not an objective of this program.

1.3.6 Metal in the effluent samples

The effluent samples contained metals at low concentrations (Table 26), all of which were below the discharge limits set by the City of San Diego (Table 27). Among the detected metals, titanium, nickel, chromium, and tungsten were present indicating some corrosion/degradation of reactor within the operation. However, these effluent concentrations are within range of what is typically seen on this SCWO reactor [7]. This is important especially for the RO reject waste because high salt concentrations can result in increased corrosion. These results show that General Atomics' SCWO reactor is a robust system for processing solid and liquid PFAS waste, even with high salt concentrations.

Sodium concentrations were at high levels across all effluent samples due to the addition of sodium hydroxide (NaOH) to the quench water, which is used to neutralize hydrofluoric acid (HF) and other acids produced during the destruction process.

Table 26 Summary of metal concentrations of all influent and effluent waste streams.

mg/L	Biosolids		AFFF wastewater		RO reject		GAC		Resin		Baseline
	Inf.	Eff.	Inf.	Eff.	Inf.	Eff.	Inf.	Eff.	Inf.	Eff.	Eff.
Barium	2.2	3.2	0.015	0.1	0.11	0.01	1.5	1.3	0.43	0.4	0.023
Cadmium	0.04	0.07	-	-	-	-	-	-	-	-	-
Chromium	-	0.4	-	0.42	0.23	0.18	0.094	0.24	0.21	0.11	0.027
Nickel	0.12	0.24	-	0.15	0.65	0.48	0.11	0.23	0.14	0.31	0.13
Lead	-	0.25	-	-	0.08	0.04	-	-	0.04	-	-
Sodium	51	1,400	23	4,000	2,000	1,700	100	840	180	470	480
Selenium	-	-	-	0.02	-	-	-	-	-	-	-
Titanium	0.17	1.10	0.014	0.16	0.02	0.02	0.02	0.093	0.09	0.29	0.11
Tungsten	-	-	-	0.012	1.0	-	-	0.032	-	-	-

Table 27. Local Limit for Industrial wastewater from City of San Diego Public Utilities Department

City of San Diego
Public Utilities Department
Environmental Monitoring & Technical Services Division
Industrial Wastewater Control Program
Current Local Limits

POLLUTANT	LOCAL LIMIT
pH	5-12.5
O&G	500 mg/L
Dissolved Sulfides	1.0 mg/L
Flash Point	> 140 ºF
Temperature	< 150 ºF
Cyanide (Total)	1.9 mg/L
Cadmium	1.0 mg/L
Chromium	5.0 mg/L
Copper	11.0 mg/L
Lead	5.0 mg/L
Nickel	13.0 mg/L
Silver	n/a
Zinc	24.0 mg/L

Local limits are re-evaluated annually and subject to change.

Note: City of San Diego local limits are applicable to Significant Industrial Users (SIUs) only.

1.3.7 COD and TOC analysis

COD measures the amount of oxygen required to chemically oxidize organic and inorganic compounds in water. It provides an estimate of the total amount of pollutants in the water, including those that are not biodegradable. COD is often used in industrial wastewater treatment because it gives a quick and comprehensive measure of the organic load.

TOC, on the other hand, measures the total amount of carbon in organic compounds present in the water. It provides a direct measure of the organic content and is often used in both municipal and industrial wastewater treatment. TOC analysis is faster and can be more sensitive than COD, making it useful for real-time monitoring and process control.

The relationship between COD and TOC is that both parameters are used to assess the organic pollution in water, but they provide different perspectives. COD gives a broader measure of the total pollutants, while TOC focuses specifically on the organic carbon content. By using both parameters together, wastewater treatment plants can get a more comprehensive understanding of the organic load and the effectiveness of their treatment processes.

COD and TOC both decreased across all effluent waste samples in this test (Table 28), indicating that SCWO is an effective wastewater treatment method to reduce COD and TOC. In this test, COD removal efficiencies ranged from 92.8% to 99.98%, showing that SCWO is very effective at removing these oxidizable pollutants in the waste. TOC reduced similarly significantly above 99% for RO reject, GAC and resin. Biosolids and AFFF showed lower TOC removal efficiency because these wastes had at much lower starting TOCs. Overall, these reductions demonstrate the system's effectiveness in breaking down and eliminating organic matter.

Table 28 Summary of COD and TOC in the influent and effluent samples.

Waste	COD (mg/L)		Removal Efficiency	TOC (mg/L)		Removal Efficiency
	Inf	Eff	%	Inf	Eff	%
Baseline	NA	NA	NA	NA	5.5	NA
Biosolids	11,060	34	99.69	36	9	73.89
AFFF wastewater	720	38	94.69	46	34	26.09
RO reject	143,100	35	99.98	46,000	8	99.98
GAC	14,980	1,077	92.81	710	4	99.46
Resin	10,290	58	99.44	290	3	99.44

1.3.8 VOC analysis

Volatile organic compounds are another class of chemicals of concern outside of PFAS. Several VOCs were detected in the influent samples up to 21 ppm (Table 10, Table 12, Table 14, Table 16, and Table 18). However, no VOCs were detected in the effluent samples of any of the processed waste streams, indicating the complete destruction of all the VOC samples by the SCWO process. These results are significant because unlike PFAS which is not actually yet regulated in wastewater, these VOCs do have regulated discharge limits.

1.3.9 Solids analysis

One value of SCWO over other traditional wastewater treatment methods demonstrates a significant decrease in solids management required after treatment. The end products of SCWO include gas, water, salts, and minerals, so the solids production is much more minimal than incineration or landfilling. Total solids in the effluent samples were measured (Table *20*) to quantify the solids fraction present. This measurement informs discharge compliance and further analytical needs. The results indicate that the total solids percentage of all effluents samples was below 1%, with solids reduction ranging from 94-98% for the high solids content waste of biosolids, GAC and resin. Observations of white, salt like particles in the AFFF and GAC samples post-drying suggest the presence of crystallized mineral precipitates. Though out of scope of this project, further analysis is required to identify these particles such as with the PIGE method mentioned above, as they may be indicative of mineralization processes. Mineral precipitation would point to the breakdown of organic compounds, including PFAS. In depth effluent solids analysis would require long run times of

more waste than was specified for this test because the output mass of solids is so low from SCWO. For future tests, enough waste must be run to at least 10 grams of solids per sample are collected for analysis.

1.4 Discussion Data Quality Objectives and Tests

Table 29, Table 30, and Table 32 were the three DQO goals and planned tests of this program In each of these tables, the various tests planned to address each DQO is outlined, along with the inputs and the outputs. Table 31 summarized relevant input and output values and ranked the relevance of all inputs with the PFAS removal efficiency for DQO 1. DQO 2 is the main focus of this project – to demonstrate the utility of the AFS system, and is therefore presented first. DQO 2 was also tested first followed by DQO 1, then DQO 3. We note that while we aimed to test single variables effect on SCWO efficacy, the reality of working with real world waste meant that isolating single variables was not possible.

1.4.1 DQO 2 Goal Discussion

DQO 2 Goal was to design and test AFS prototype equipment that can produce a feed that met the known boundary conditions for GA's SCWO process. AFS was designed and built by AxNano at our prototyping facility in NJ. The AFS was designed to grind solid waste materials and prepare 10-15% slurry to be used for the August SCWO test. Both virgin GAC and resin materials were pretested in NJ from May to July before AFS was shipped to the pilot SCWO facility at end of July. During the August onsite SCWO test at GA-EMS facility, AFS process was further optimized to grind GAC (2,000 microns) and resin (500-1,100 microns) to < 250 microns in a much shorter period. Within less than two weeks at GA-EMS, we were able to grind the particle sizes to much less than the <250 microns target and reduced the GAC grinding time from approximately 3-6 weeks without AFS to about 2 hours. For resin, we achieved 60% of particles below 250 microns, with the remaining 40% ranging between 250 and 1,110 microns, likely due to resin elasticity versus GAC rigid structure.

Table 29 Test and Data Outputs for DQO 2 Goal.

DQO 2 Goal: Design and test AFS prototype equipment that can produce a feed that meets known boundary conditions for GA's SCWO.			
Test Number	Description	Data Input	Data Outputs
5	Motor controls	• Programmable logic control	• Power on • Run/stability • Power off
6	Valve actuation	• Programmable logic control	• Power on • Power off
7	Slurry tank load cell test and calibration	• Volume	• Electronic conversion to weight
8	Flow meter test and calibration	• Volume per time	• Power on • Power off
9	Screw conveyor speed	• Volume per time	• Mass transferred
10	Grinding test - 1	• Process time	• Particle size
11	Grinding test – 2 (Optional)	• Number of pumps • Process time	• Particle size • Capital Cost • Operational Cost

Overall, the DQO 2 Goal tests for the AFS listed in Table 29 have been successfully completed. Test 5 and 6 data outputs were shown by the system turning on and operating through the use of the PLC. Test 7 was not completed because the load cells were damaged during shipping. Instead slurries were prepared by weighing wastes outside of

the AFS and manually adding to the hopper. Load cells have been replaced and are awaiting further testing. Test 8 was completed by our ability to control the flow of slurry through the circulation pump and the particle reduction mill, developing key intellectual property and know-how. Test 9 was completed by successfully transferring solids from the hopper to the mixing tank. Test 10 was completed in pre-testing in NJ as described above. Test 11 was completed at GA facility as described above without the need for a secondary grinding pump.

Our resin tests indicate that the SCWO process did not experience significantly low PFAS removal efficiency or unstable operation despite the resin particles being larger than the required size (i.e., <250 microns). All results demonstrated that our AFS provided a crucial pathway for SCWO treatment of solid waste and achieved excellent PFAS removal efficiency, sufficiently indicating that our AFS was successful. Through process optimization and hardware improvements, we have reduced the grinding time for 100 gallons to within one hour, fully meeting the requirements for the continuous operation of the SCWO system at a feed rate of 3 GPM, where waste feed makes up a portion of this total 3 GPM. As shown in Table *2*, the highest waste throughput for solids-containing wastes was 1.8 GPM..

Our next steps are to integrate the AFS into the SCWO process as part of the DIU/ESTCP program. Longer term testing for resin and additional process optimization may also needed. We plan to pursue follow-on Phase III funding to support this resin optimization work.

Table 30 Test and Data Outputs for DQO 1 Goal.

DQO 1 Goal: Identify effects of the waste feed on PFAS destruction efficiencies and reactor stability.				
Test Number	Objective Description	Wastes Comparisons	Data Inputs	Data Outputs
1	Liquid PFAS Waste vs. Solid PFAS Waste	AFFF wastewater vs. GAC	• Particle size • Solids wt% • BTU • PFAS concentration • Reactor temperature • Reactor pressure	• PFAS removal efficiency • Reactor temperature stability • Feed rate
2	Low PFAS Concentration Solids vs. High PFAS Concentration Solids	Resin vs GAC	• PFAS concentration	• PFAS removal efficiency
3	Low BTU PFAS Waste vs. High BTU PFAS Waste	Biosolids vs GAC	• BTU • Ratio of waste:diesel:water in feed	• PFAS removal efficiency • Reactor stability • Cost per volume treated
4	Low Salt PFAS vs. High Salt PFAS	AFFF wastewater vs RO reject	• Salt concentration • PFAS concentration	• PFAS removal efficiency • Metals in effluent (Table 24)

1.4.2 DQO 1 Goal Discussion

DQO 1 goal was to identify the effects of waste feed on PFAS destruction efficiencies and reactor stability of SCWO test using various real-world wastes, including liquids and solids. The tests for DQO 1 are summarized in Table 30 and the results are shown in Figure 12, Figure 13, Figure 14, Figure 15, and Figure 16.

For Test 1, we aimed to evaluate the effect of PFAS form (i.e., solid versus liquid waste) on removal efficacy. For this test we compared dissolved PFAS in AFFF wastewater versus sorbed PFAS on spent GAC. We hypothesized that PFAS sorbed onto a solid particle may be less susceptible to oxidation and destruction. However, our tests showed slightly higher removal efficiency for the solid PFAS than the liquid waste. SCWO treatment of GAC waste resulted in 99.995% removal efficiency compared to 95.91% for liquid AFFF wastewater. This could be due to two reasons. First, the starting PFAS concentrations for GAC was two orders of magnitude higher than AFFF, so there was greater opportunity for percent destruction. Second, the PFAS effluent of GAC was less than the AFFF effluent, which could have been from: (1) residual GAC or ash in the effluent sorbing PFAS in the effluent and decreasing measurable PFAS; or (2) incomplete destruction of the PFAS in the AFFF. Other variables that changed in this test include: the two wastes were processed under different conditions with AFFF wastewater fed at a lower rate (1 lpm vs. 7 lpm) and higher pressure than GAC (227.9 bars vs. 204.8 bars), respectively. The lower feed rate of the AFFF was used as a safety precaution as we anticipated a higher PFAS concentration would increase acids in the effluent. The AFFF wastewater also had much lower COD and TOC than GAC, which meant the reactor was fed higher diesel to account for the anticipated lower waste BTU. Both SCWO tests had stable temperature and pressure in a steady state and no obvious difference was observed as discussed in Section 6. Ultimately, no correlation was observed between PFAS removal efficiency and solid versus liquid (Figure 14) or particle size (Figure 15).

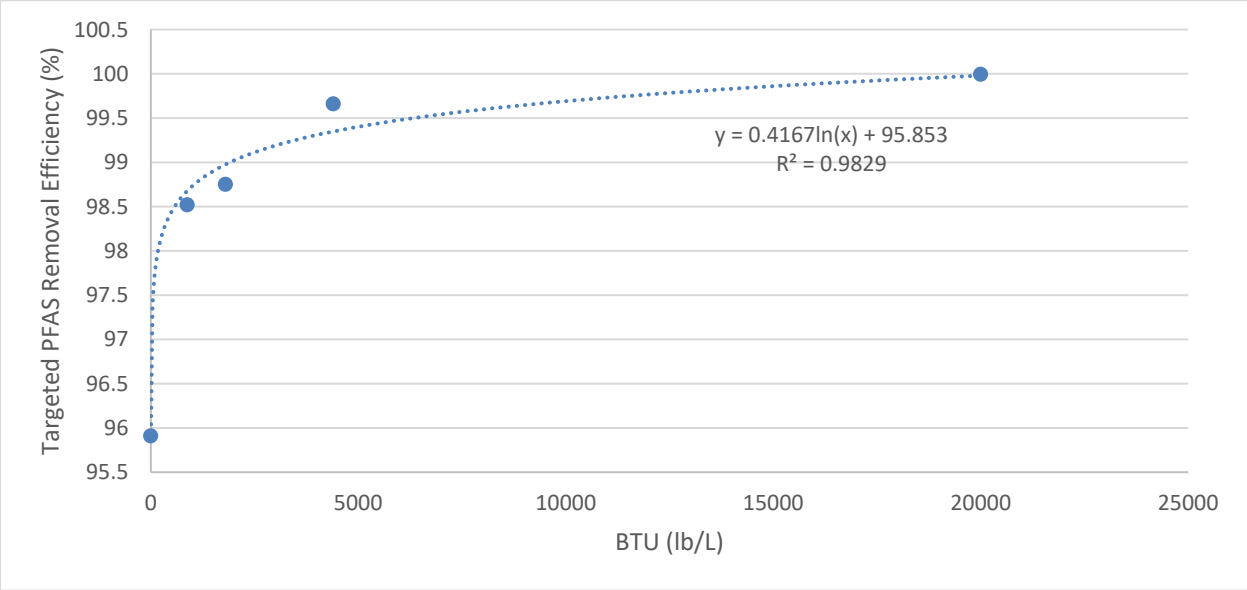

Figure 12. Targeted PFAS removal efficacy (%) compared to waste BTU (lb/L) across the various wastes tested showed a strong exponential correlation in this study with increasing BTU resulting in increased PFAS removal efficency.

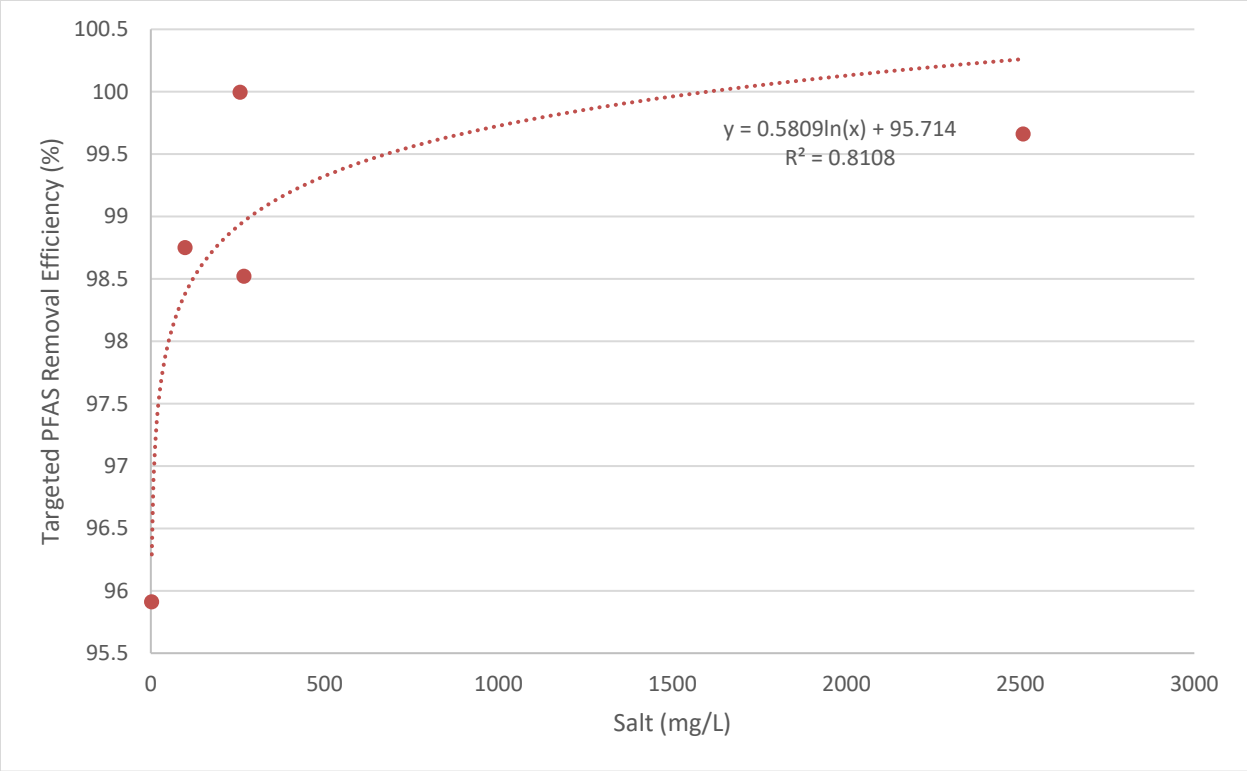

Figure 13. Targeted PFAS removal efficacy (%) compared to salt concentration (mg/L) across the various wastes tested showed a loose exponential correlation in this study with increasing salt concentration presenting increasing removal efficiency.

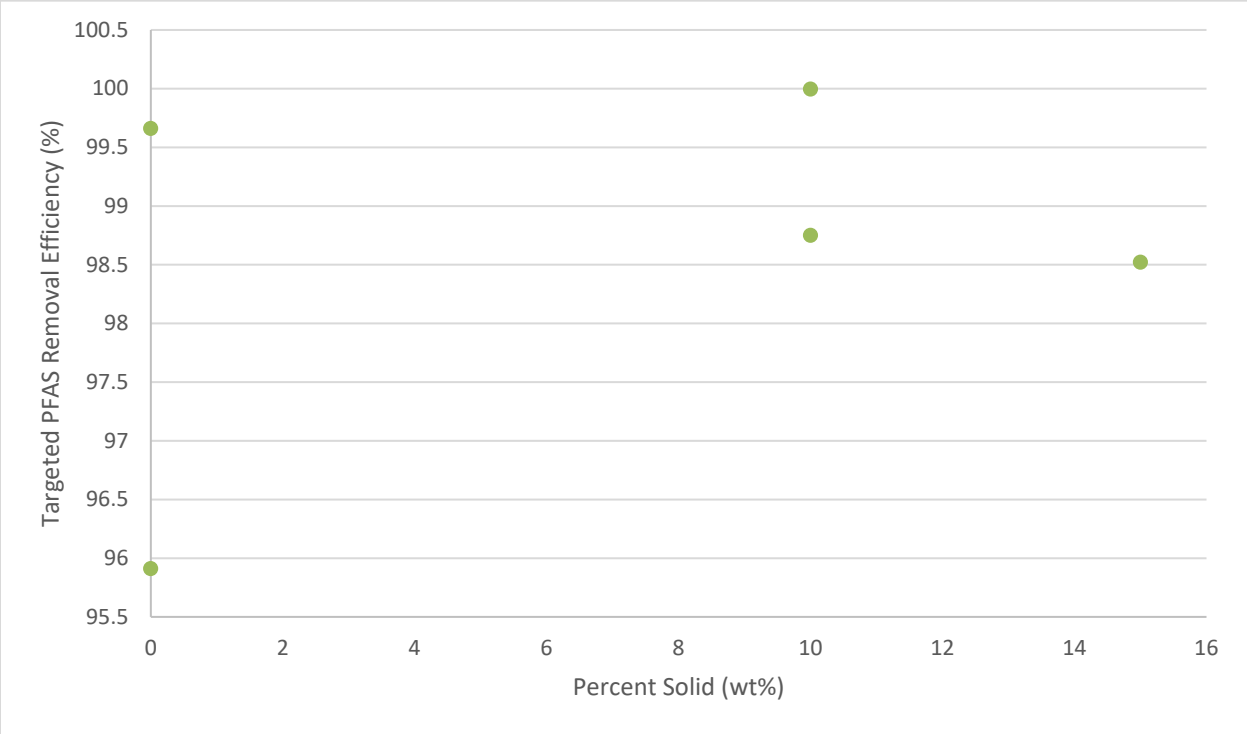

Figure 14. Targeted PFAS removal efficacy (%) compared to percent solid (wt%) across the various wastes tested showed no correlation in this study.

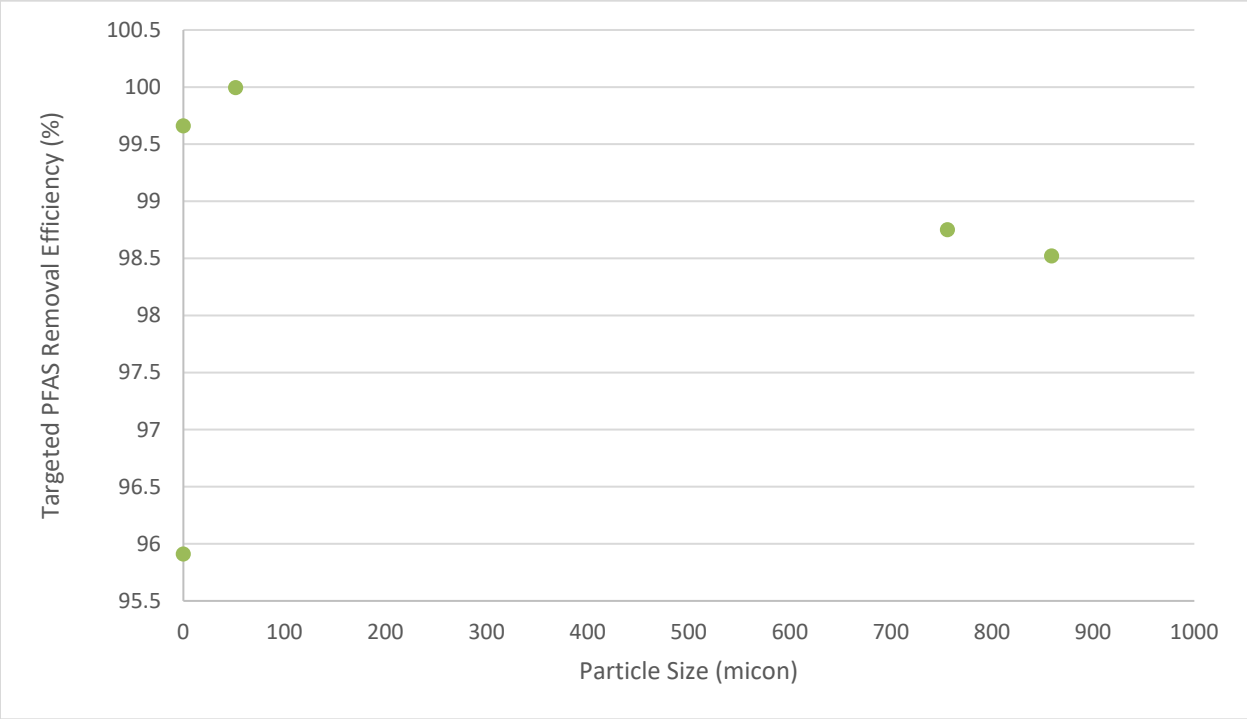

Figure 15. Targeted PFAS removal efficacy (%) compared to particle size (micron) across the various wastes tested showed no correlation in this study.

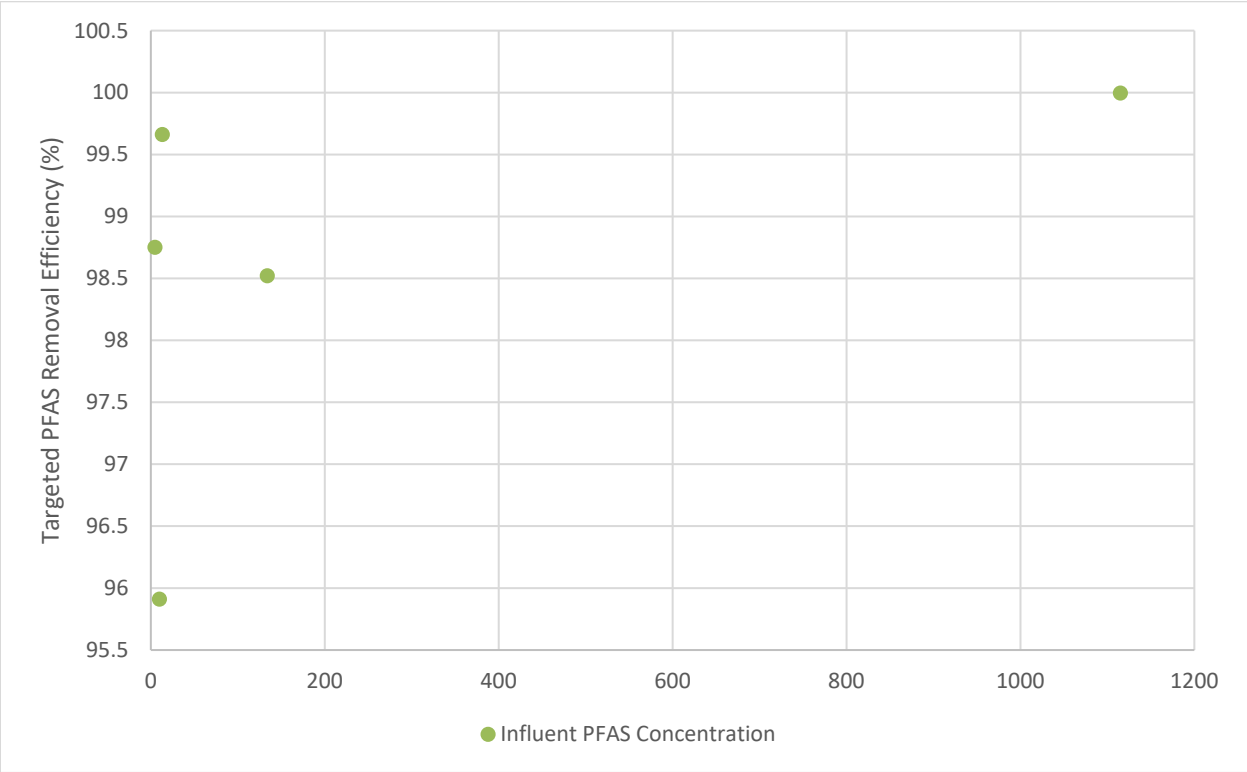

Figure 16. Targeted PFAS removal efficacy (%) compared to influent targeted PFAS concentration across the various wastes tested showed no correlation in this study.

In summary for Test 1, our results indicate that PFAS sorbed on a solid does not negatively impact removal efficiency in SCWO treatment as was originally hypothesized. The development and use of our AFS successfully prepared a stable slurry feed that allowed for the SCWO treatment of solid-phase PFAS without interrupting reactor stability. Low influent PFAS concentration of AFFF, with low flow rate and higher SCWO pressure did not result in increased removal efficiency. Therefore, compensation of lower flow rate to achieve higher PFAS removal efficiency might not be a deciding factor. Lower pressure (i.e., 204.8 bars) than 220 bars was not critical at 650 ^{o}C because we are well above the supercritical temperature requirement of 374 ^{o}C. The COD and TOC inputs for GAC were 93% and 95% higher, respectively, compared to AFFF, suggesting that higher energy content may support greater PFAS destruction.

For Test 2, the objective was to examine the effect of PFAS concentration on SCWO treatment of solids. This test compared spent GAC to spent resin because they varied in 10x order of magnitude starting PFAS concentration (though we note an uncertainly in the starting resin PFAS concentration, see discussion in Section 1.3.3). PFAS removal efficiency was higher for spent GAC (99.995%) than for resin (98.52%). The two wastes had very different chemical and physical properties, including particle size, modulus of elasticity, COD, and analytical limitations, making definitive conclusions difficult. Resin is highly effective at removing PFAS from water, resulting in strong binding that makes desorption challenging [8]. This likely caused measurement inaccuracies, leading to underreporting of starting PFAS concentration, skewing the comparison. To address the analytical limitations, future work will collaborate with contract laboratories and universities to refine our approach as described above. It was observed that resin has a higher modulus of elasticity than GAC, meaning it can deform reversibly under stress. This affected the ability of the AFS to achieve the target particle size, however, the resin was mostly destroyed in the SCWO process as indicated by the 98% solids destruction. COD was also 60% lower in the resin as was BTU. Ultimately, no correlation was found in starting PFAS concentration on eventual removal efficiency in this test (Figure 16).

For Test 3, we aimed to evaluate the impact of waste BTU on SCWO treatment efficacy (Figure 12), but it turns out that AFFF and RO were better comparisons. AFFF had a BTU of non-detect (<1 lb/L) and total PFAS of 10.1 ug/L. RO reject had 4,400 BTU and total PFAS of 13.4 ug/L. AFFF PFAS removal efficiency was 95.91%, whereas RO reject was 99.66%. AFFF wastewater had 25% lower PFAS concentration, much lower COD, lower TOC, and lower BTU than RO reject. Although AFFF wastewater was tested at lower feed rate (i.e., 1 lpm vs. 8.2 lpm) and higher pressure (215.8 bars vs. 200.1 bars), AFFF wastewater still resulted in much lower PFAS removal efficiency than RO reject (95.91% vs. 99.66%). This suggests that waste with lower BTU and PFAS concentrations tends to have lower removal efficacy, regardless of variation in flow rate and pressure. In some cases, higher BTU concentrations are associated with higher PFAS levels which may explain why BTU appears to influence removal efficiency [9]. Both SCWO tests maintained stable temperature and pressure in a steady state with no obvious difference was observed. When comparing across all wastes, a positive correlation is found where increasing BTU supports improved PFAS removal efficiency (Figure *12*). Higher BTU wastes will release energy as the oxidation occurs in the reactor zone, which could support PFAS destruction.

For Test 4, we aimed to evaluate the impact of low salt PFAS versus high salt PFAS as summarized in Table 30. Salt concentration is an important parameter for SCWO because elevated salt levels, especially chloride and carbonate, can cause salt precipitation onto the reactor lining and corrosion. Salt plugging can occur, where salt precipitates build up in a horizontal pipe reactor and restrict flow. Note that the General Atomics SCWO reactor is not a horizontal pipe reactor and is less susceptible to salt build up and corrosion because of this positioning. A loose correlation was found with increasing salt concentration in the waste corresponding to increased PFAS removal efficacy (Figure 13), however, we believe this is correlational and not causational. We will continue to monitor this parameter to see if this correlation exists in future testing. The RO reject had nearly 1,000x higher salt concentration than AFFF wastewater. The elevated salt concentration did not negatively impact PFAS removal efficacy. The high salt RO reject also did not result in increased titanium, tungsten or nickel in the effluent, indicating that the higher salt did not induce increased reactor corrosion.

Table 31 Ranking of DQO 1 inputs and outputs.

	Biosolids	AFFF wastewater	RO reject	GAC	Resin
PFAS Removal Efficiency (%)	98.75	95.91	99.66	99.995	98.52
Energy Content (BTU)	1,800	ND < 1	4,400	20,000	880
COD (mg/L)	1,1060	720	143,100	14,980	10,290
TOC (mg/L)	36	46	46,000	710	290
Feed Rate (lpm)	7.0	1.0	8.2	7.0	7.0
Reactor Pressure (bars)	215.8	227.9	200.1	204.8	191.9
Salt (mg/L)	99	2.7	2,508	257	268
Influent PFAS Concentration (µg/L)	5	10.1	13.4	1,115.2	134.1
Reactor Temperature (C°)	645±5	649±12	646±4	646±5	647±2
90% Particle size (µm)	< 756	NA	NA	< 51.8	< 859
Solids wt%	10%	NA	NA	10%	15%
Approx. Waste: diesel: water ratio	4:1:2	1:1:4	14:1:1	7:0:4	7:0:6

Table 31 summarizes the results of our DQO 1 testing by comparing various inputs (feed content and reactor conditions) with PFAS removal efficiency across all the wastes streams. The data suggests positive correlation between high BTU and PFAS removal efficiency; a loose correlation between initial PFAS concentration and removal efficiency; and a positive correlation between salt concentration and PFAS removal efficacy, though we think this is not causal. Certainly, with a higher starting PFAS concentration one can calculate a higher removal efficiency if all PFAS is removed in the effluent. This is evident in the GAC results where the very high starting concentration and low effluent concentration result in a high removal efficiency. We note that uncertainty remains regarding the accuracy of the starting resin concentration. The other four inputs did not appear to predict removal efficacy. Solid versus liquid, flow rate, and SCWO pressure did not correlate with output. Lower pressure (i.e., 204.8 bars) than 220 bars was not critical because we are operating at temperatures (at 650 ^{o}C) well above the supercritical limit (374 ^{o}C). Further experiments for validation and replication will be done in subsequent testing programs to support conclusions around PFAS removal efficacy.

Table 32 Test and Data Outputs for DQO 3 Goal.

DQO 3 Goal: Test if AxNano's process (AFS prototype + SCWO reactor) meets technical (throughput, discharge limits) and economic requirements (cost, logistics) for integration into DAF workflow.			
Test Number	Description	Data Input	Data Outputs
12	Effluent concentrations versus disposal requirements	• PFAS concentration, metals concentration	• Waste disposal categorization
13	Cost per volume treated versus current methods	• Volume waste treated per unit time • Chemical consumption • Utility usage • Equipment costs • Labor effort • Effluent chemical composition	• Capital costs • Operating costs • Adherence to current DAF disposal guidance

1.4.3 DQO 3 Goal Discussion

DQO 3 assessed the potential integration of AxNano's process into DAF hazmat treatment. The goal was to use the results of this Phase II testing to estimate if AxNano's process (AFS prototype + SCWO reactor) could meet technical

(throughput, discharge limits) and economic requirements (cost, logistics) for integration into DAF workflow at scale. Details of the DQO 3 Goal Tests are summarized in Table 32.

In Test 12, we evaluated whether the effluent from the AxNano process -consisting of water, salts, and minerals -could be discharged into the city stormwater system leading to a wastewater treatment plant (WWTP) or if it must be managed as a hazardous waste. Key data inputs included effluent analytical results for metal and PFAS concentrations. Effluent samples contained low concentrations of metals, all of which were below the discharge limits set by the City of San Diego. PFAS concentrations were measured across all effluent samples. Currently there are no discharge limits on PFAS, though WWTPs may be reluctant to accept PFAS-impacted waste. The baseline effluent sample, which was collected before introducing waste to the SCWO reactor, had low levels of PFAS compounds (1.1–6.9 ng/L), except for PFBA, which was measured at 200 ng/L. These findings suggest potential background PFAS contamination from tap water mixing with the effluent in the quench tank, making it difficult to draw a definitive conclusion on the true effluent PFAS concentrations. While the waste from this test was not directly discharged, improved PFAS analysis and removal of background PFAS would likely allow for direct discharge.

In Test 13, we assessed the operating cost per volume treated for AxNano's process (AFS + SCWO). This test was essential in addressing the overall objective: determining whether AxNano's Automated Feed System (AFS) combined with SCWO could create additional value for Air Force hazardous substance treatment beyond SCWO alone. Key data inputs included actual parameters from AFWERX-supported tests on various solid and liquid waste streams.

The cost estimation process involved measuring the quantities of chemicals used, such as diesel, pH adjusters, and additives. These costs were estimated with the waste treatment volume to determine the cost per unit of waste treated, informing the Cost of Goods Sold (COGS) and pricing structures for AxNano's process. Performance and pricing were then compared to incineration, using the EPA interim guidance document as a reference due to the lack of a current DoD comparison, and to landfilling costs, which were sourced from Shaw AFB, Biddle ANG (AFCEC, Environmental Flights, and contractors). Incineration and landfilling were assumed to be off-site practices and therefore included transportation costs, whereas SCWO cost model assumed on-site treatment with no shipping costs.

COGS were estimated for three waste treatment scenarios listed below. Data from our pilot test was used as representatives for these three scenarios where scenario 1 was a slurry of spent GAC in clean water; scenario 2 was a slurry of spent GAC in RO reject wastewater; and scenario 3 was RO reject wastewater.

- Scenario 1: A slurry waste feed where the feed is composed of 10% solid waste and 90% clean water. (e.g., clean water is used to create the slurry waste feed)
- Scenario 2: 100% wastewater (no solids)
- Scenario 3: A slurry waste feed where the feed is composed of 10% solid waste and 90% wastewater (e.g., other wastewater is used to create the slurry waste feed)

Inputs into operational costs were obtained from actual data from pilot testing as well as GA's experience in operation and maintenance of their pilot system as well as with customer support of industrials systems. Assumptions included: the system was assumed to run 24/7 for 300 days annually; flow rate inputs (gpm) of waste, diesel, and process water; power consumption; and quench water flow were set as those used in the pilot testing for Scenarios 1 and 2. Scenario 3 was estimated based on combining Scenarios 1 and 3. Tons reported are metric tonnes. Cost of electricity, diesel and caustic were based on those sourced for pilot testing in San Diego. The results of this operational cost analysis for the three different waste scenarios is shown in Table 33.

Equipment purchase costs, including capital expenditures and annual operation and maintenance are estimated based on our teams' experience with the pilot system and in GA's experience from customer-operated SCWO systems. Disposal cost inputs for landfilling ($840/ton) are those paid for landfilling disposal of spent resin from Biddle Air

National Guard (ANG) at Willow Grove. Transportation cost inputs ($706/ton) are those paid for landfilling disposal of spent resin from Biddle Air National Guard (ANG) at Willow Grove. Disposal costs for incineration were not able to be determined from Air Force bases due to the incineration moratorium, so the cost input ($2,822/ton) was obtained from 2024 INTERIM GUIDANCE ON THE DESTRUCTION AND DISPOSAL OF Perfluoroalkyl and Polyfluoroalkyl Substances and Materials Containing Perfluoroalkyl and Polyfluoroalkyl Substances— Version 2.[10] The EPA document does not include transportation, thus all transportation costs used were those cited from Biddle ANG.

The AFS adds only 3% to total CapEx, but provides up to 93% reduction in operating costs by being able to manage feeds to an optimal formulation, increasing throughput and decreasing diesel costs. Current OpEx range from $88.46-$1,346.16/ton depending on waste stream. In the case of Scenario 1, SCWO has highest OpEx due to low throughput of 172 ton per year. In scenario 1, the feed is made of a 10 wt% slurry where spent GAC is mixed with clean water. In Scenario 2, the throughput increases up to 3512 because the feed is 100% wastewater. In Scenario 3, the throughput is further increased to 3900 ton per year because the feed is a slurry of spent GAC and wastewater. As shown in Table 33, this feed management in Scenario 3 significantly decreases the per ton operating cost. Diesel consumption and labor are the highest cost drivers for OpEx, thus our ability to mix wastes and decrease diesel as well as automate the system to decrease labor provides value.

Table 33. Estimated operating costs based on parameters from this test for two actual wastes run (Scenario 1 and 2) and one theoretical mixed waste stream (Scenario 3).

Description of waste	Scenario 1 10% GAC Slurry in clean water	Scenario 2 RO Reject	Scenario 3 10% GAC Slurry in RO Reject Mix	Units
Btu Content	632	1,640	2,108	Btu/lb
Waste flow rate	1.85	2.16	2.98	gal/min
Diesel flow rate	0.00	0.16	0.02	gal/min
Water flow rate	0.98	0.16	0.00	gal/min
Power Consumption	400	400	400	KW
Basis	1	1	1	hr
Flow rate of waste	0.12	8.19	11.29	L/min
Cost of Diesel fuel	0	1.00	1.00	$/L
Cost of Electricity	3.70	0.12	0.12	$/kWhr
Diesel flow rate	0.00199	0.599544	0.0636134	L/min
Process DI water flow rate	0.00	0.60	0.00	L/min
Cost of DI water	0.00199	0.00199	0.00199	$/L
Caustic flow rate (35%)	0.05	0.05	0.05	L/min
Caustic (35%)	1.51	1.51	1.51	$/L
DI Quench Water Flow	21	20.6	21	L/min
Total waste processed in 1hr	41.968	677.48	491.6261	Liters
Cost of electricity in 1hr	48	48	400	$
Cost of fuel in 1hr	0.00	3.82	35.97	$
Cost of caustic in 1hr	4.53	4.53	1,271.97	
Cost of soft-water in 1hr	2.95	2.51	491.63	$
Total costs in 1hr	55.48	58.85	91.03	$
Cost per liter	1.32	0.09	0.19	$/L
Cost per ton feed	1,346.16	88.46	188.56	$/MT
Cost per tote (330 gal)	1,651.16	108.51	231.29	$/tote
Estimated Cost per gallon	5.00	0.33	0.70	$/gal

Table 34. Assumptions used in Table 35 cost analysis of 10-year use of SCWO on-site versus off-site incineration or landfilling.

Assumptions: (Based on High BTU)		
Waste Feed Processed by AxSure™ Per Year[a]	1,296,000	gallons
Tons Waste Treated per Year - Scenario 1 (10% solid waste and 90% clean water)	172	tons solid-waste (GAC) treated per year
Tons Waste Treated per Year - Scenario 2 (100% wastewater)	3900	tons wastewater per year
Tons Waste Treated per Year - Scenario 3 (10% solid waste and 90% wastewater)	3512	tons of waste slurry treated per year of solid-waste (GAC)
Landfill Cost Per Ton[c]	$840	
Incinerator Cost Per Ton[d]	$2,822	
Waste Transportation Costs Per Ton[e]	$706	

a 2.90 gallons per minute waste throughput based on August testing of RO reject. Year round operation with 27% down time (running 300 out of 360 day operation based on estimate from consolidated waste processing facility operator estimates and currently operating GA-made SCWO systems)
c From actual disposal cost Biddle ANG base landfill disposal of resin, not including transportation
d From 2024 INTERIM GUIDANCE ON THE DESTRUCTION AND DISPOSAL OF Perfluoroalkyl and Polyfluoroalkyl Substances and Materials Containing Perfluoroalkyl and Polyfluoroalkyl Substances— Version 2. Does not include transportation. (Appendix D page D-2)
e From actual disposal costs from Biddle ANG GAC and resin roll-offs disposal
f Calculated from actual operational parameters used in August test for GAC waste
g Based on General Atomics operational experience for simpler waste streams. Lower O&M may be possible with less corrosive wastes. This is a worst case scenario estimate.
h Calculated from estimated operational parameters based on actual parameters of August testing of GAC waste and RO reject separately
i Calculated from actual operational parameters used in August test for RO reject waste

Each of the three waste scenarios were analyzed over a 10-year cost projection using the above mentioned operating costs, capital costs, and comparison to incineration and landfilling (Table 35 with assumptions listed in Table 34). For customers generating sufficient wastewater to support continuous operation, an on-site government-owned and contractor-operated AxSure™ system can reduce costs by 77–83% over a 10-year period compared to landfilling. However, for customers producing only solid PFAS waste, landfilling remains the most economical option, though long-term liability considerations were not factored into the model. In scenario 1, the cost over a 10 year period for SCWO is $8,313,896 compared to $2,657,195 for transportation and incineration disposal or $6,063,910 for transportation and landfill disposal. The higher cost of SCWO in Scenario 1 is due to the low throughput described above. Transportation costs for off-site treatment is the greatest overall waste management cost-driver. Most of the cost savings from SCWO in Scenario 2 and 3 arise from on-site treatment eliminating disposal costs. In scenario 2, 10 years of SCWO treatment is estimated at $12,928,827 compared to $60,285,660 and $137,576,187 for transportation and incineration or landfilling, respectively. The final scenario 3 provides the most cost savings where SCWO costs $9,106,761 over 10 years, compared to $54,298,622 and $123,892,800 for incineration and landfilling, respectively. These very high numbers for off-site incineration and landfilling are a key pain-point for waste generators.

On-site treatment proves economically viable for customers generating at least 930,000 gallons of liquid waste per year, ensuring continuous operation. For Air Force restoration activities, systems could be procured or leased by AFCEC and operated by contractors, while recurring wastewater treatment systems for Air Force bases could be

Table 35. Cost of ownership analysis of SCWO for onsite disposal versus off-site disposal with incineration and landfilling over a 10-year period.

Cost of Ownership Analysis Table												
Scenario 1 (10% solid and clean water)												
		Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10	Total
Purchasing AxSure™ Equipment	Equipment Cost	$ 4,000,000										
	Non Recurring Engineering	$ 1,000,000										
	Fixed Operating Cost Per Year[f]	$ 231,390	$ 231,390	$ 231,390	$ 231,390	$ 231,390	$ 231,390	$ 231,390	$ 231,390	$ 231,390	$ 231,390	
	O & M[g]	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	
	Cost of Equipment and Setup	$ 5,331,390										
	Cost of Operating (Post Year 1)		**$ 331,390**	$ 331,390	$ 331,390	$ 331,390	$ 331,390	$ 331,390	$ 331,390	$ 331,390	$ 331,390	**$ 8,313,896**
Scenario 1 (10% solid and clean water)	Annual Landfill + Transportation Cost	$ 265,720	$ 265,720	$ 265,720	$ 265,720	$ 265,720	$ 265,720	$ 265,720	$ 265,720	$ 265,720	$ 265,720	**$ 2,657,195**
	Annual Incinerator + Transportation Cos	$ 606,391	$ 606,391	$ 606,391	$ 606,391	$ 606,391	$ 606,391	$ 606,391	$ 606,391	$ 606,391	$ 606,391	**$ 6,063,910**
Scenario 2 (100% wastewater)												
Purchasing AxSure™ Equipment	Equipment Cost	$ 4,000,000										
	Non Recurring Engineering	$ 1,000,000										
	Fixed Operating Cost Per Year[h]	$ 310,676	$ 735,350	$ 735,350	$ 735,350	$ 735,350	$ 735,350	$ 735,350	$ 735,350	$ 735,350	$ 735,350	
	O & M	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	
	Cost of Equipment and Setup	$ 5,410,676										
	Cost of Operating (Post Year 1)		**$ 835,350**	$ 835,350	$ 835,350	$ 835,350	$ 835,350	$ 835,350	$ 835,350	$ 835,350	$ 835,350	**$ 12,928,827**
Scenario 2 (100% wastewater)	Annual Landfill + Transportation Cost	$ 6,028,566	$ 6,028,566	$ 6,028,566	$ 6,028,566	$ 6,028,566	$ 6,028,566	$ 6,028,566	$ 6,028,566	$ 6,028,566	$ 6,028,566	**$ 60,285,660**
	Annual Incinerator + Transportation Cos	$ 13,757,619	$ 13,757,619	$ 13,757,619	$ 13,757,619	$ 13,757,619	$ 13,757,619	$ 13,757,619	$ 13,757,619	$ 13,757,619	$ 13,757,619	**$ 137,576,187**
Scenario 3 (10% solids and 90% wastewater)												
Purchasing AxSure™ Equipment	Equipment Cost	$ 4,000,000										
	Non Recurring Engineering	$ 1,000,000										
	Fixed Operating Cost Per Year[i]	$ 310,676	$ 310,676	$ 310,676	$ 310,676	$ 310,676	$ 310,676	$ 310,676	$ 310,676	$ 310,676	$ 310,676	
	O & M	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	$ 100,000	
	Cost of Equipment and Setup	$ 5,410,676										
	Cost of Operating (Post Year 1)		**$ 410,676**	$ 410,676	$ 410,676	$ 410,676	$ 410,676	$ 410,676	$ 410,676	$ 410,676	$ 410,676	**$ 9,106,761**
Scenario 3 (10% solid and 90% wastewater)	Annual Landfill + Transportation Cost	$ 5,428,962	$ 5,428,962	$ 5,428,962	$ 5,428,962	$ 5,428,962	$ 5,428,962	$ 5,428,962	$ 5,428,962	$ 5,428,962	$ 5,428,962	**$ 54,289,622**
	Annual Incinerator + Transportation Cos	$ 12,389,280	$ 12,389,280	$ 12,389,280	$ 12,389,280	$ 12,389,280	$ 12,389,280	$ 12,389,280	$ 12,389,280	$ 12,389,280	$ 12,389,280	**$ 123,892,800**

procured or leased by the base host Wing and managed by Civil Engineering Squadrons or contractors. Air Force customers unable to support an on-site AxSure™ system could still utilize SCWO if a system were available at a TSDF. In cases requiring off-site disposal, specifications would be outlined in a restoration Performance Work Statement (PWS) or as a contract line item (CLIN) under a DLA contract.

This work supports DoD's evaluation of emerging technologies, such as SCWO, based on applicable criteria identified in EPA's April 2024 guidance technology evaluation framework. However, for customers producing only solid PFAS waste, landfilling remains the most economical option, though long-term liability considerations were not factored into the model. AxNano's Automated Feed System is critical for AxSure™ technology adoption, as integrating solid and liquid waste streams is necessary to make SCWO cost-competitive with landfilling.

The DoD is currently reviewing the EPA's April 2024 "Interim Guidance on the Destruction and Disposal of Perfluoroalkyl and Polyfluoroalkyl Substances and Materials Containing Perfluoroalkyl and Polyfluoroalkyl Substances-Version 2". This interim guidance encouraged permitted Class 1 non-hazardous industrial or hazardous waste underground injection wells, permitted hazardous waste landfills, and permitted hazardous waste combustors that operate under certain conditions, and interim storage with controls. The EPA also provided a technology evaluation framework to help practitioners evaluate safety and effectiveness of new destruction and disposal technologies. The guidance focused on four approaches, including SCWO as well as mechanochemical degradation, electrochemical oxidation, gasification and pyrolysis. The results of this Phase II provide data inputs in the "Materials" Category of Information by expanding the inputs to:

- What PFAS-containing material has been used to test the performance of the technology of interest?
  Results from this Phase II expand testing to reverse osmosis reject, PFAS-laden granular activated carbon, and PFAS-laden ion exchange resin
- Which PFAS, and at what concentration ranges, have been used to test the performance of the technology of interest?
  Results from this Phase II expand concentration ranges to those of spent sorption media.
- Are there non-PFAS constituents that may affect the performance of the technology of interest?
  Results from this Phase II expand inputs to show impacts of solid particle size and calorific value (BTU/lb) that will impact performance. Our results also show that salt concentrations, previously believed to negatively impact SCWO, does not have a performance impact on our system.

2 Conclusions

2.0.1 DQO 1 Goal Test Conclusions

- The DQO 1 Goal tests indicated that:
  - The data suggests a positive correlation between BTU and removal efficiency, and a loose correlation between salt and initial PFAS concentration. (Salt is expected to be a coincidence and not causal).
  - High salt concentration does not negatively impact PFAS removal efficacy nor does it result in reactor corrosion (for the duration of this test)
  - The form of waste (solid versus liquid) did not negatively impact PFAS removal efficacy because our AFS system was able to successfully produce a stable waste feed regardless of waste type.
- Total PFAS removal efficiency ranged from 95.91% for AFFF wastewater to 99.995% for GAC slurry.
- The correlation of higher BTU levels in waste materials and increased PFAS removal efficiency suggests that the energy content and organic load of the waste are crucial for effective treatment.
- Conversely, neither solid weight percentage (0-15%) nor particle sizes within the SCWO-operable range (e.g., 0-500 microns) noticeably affected PFAS removal efficiency or reactor stability.

- Operating SCWO at a lower pressure (e.g., 200 bars) instead of 220 bars was not detrimental at the SCWO operating temperature of 650°C, though maintaining higher pressures could be necessary to achieve ultra-high removal efficiencies (e.g., >99.999%).
- Reducing the flow rate by eight times did not prove effective in enhancing PFAS removal efficiency, indicating that flow rate optimization may not be the key lever for improving efficiency, and attention should be given to other operational parameters, in particular maximizing waste feed BTU.

2.0.2 DQO 2 Goal Test Conclusions

- AxNano successfully designed and built the AFS, enabling consistent slurry preparation that supports stable SCWO operation.
- The AFS demonstrated the ability to process previously unmanageable solids (GAC and resin), expanding the range of materials treatable by the SCWO system.
- The AFS reduced waste processing times from months to days.
- With a slurry throughput that meets the 3 gpm continuous SCWO operation requirement, the AFS ensures continuous and reliable SCWO performance.

2.0.3 DQO 3 Goal Test Conclusions

- Effluent met all Industrial Wastewater Control Program Current Limits the City of San Diego
- Currently there are no discharge limits on PFAS and WWTPs may be reluctant to accept PFAS-impacted waste. Effluent from this test contained PFAS concentrations ranging from 43 ppt to 1,700 ppt. Baseline measurements were 219 ppt PFAS. Therefore, all effluent was disposed of as hazardous waste.
- The AFS adds only 3% to total CapEx but provides up 93% reduction in operating costs.
- Current OpEx range from $88.46-$1,346.16/ton depending on waste stream.
- Diesel consumption and labor are the highest cost drivers for OpEx.
- For customers generating sufficient wastewater to support continuous operation, a government-owned and contractor-operated AxSure™ system can reduce costs by 72–77% over a 10-year period compared to landfilling. However, for customers producing only solid PFAS waste, landfilling remains the most economical option, though long-term liability considerations were not factored into the model.
- This work supports DoD's evaluation of emerging technologies, such as SCWO, based on applicable criteria identified in EPA's April 2024 guidance technology evaluation framework.

3 Report Summary

In summary, this Phase II resulted in a successful design, build and test of our prototype AxSure™ system. This program specifically tested our proprietary automated feed management system (AFS) that integrates with a SCWO process to efficiently reduce contaminants like PFAS, which traditional wastewater treatments fail to remove. Over 1,445 gallons of five different PFAS-containing waste including solids and liquids were processed. We successfully demonstrated the value of our AFS to achieve destruction goals beyond use of SCWO alone. The development of the AFS reduced waste processing time for typical SCWO pre-treatment from 10's of gallons per day to 0.33 gallons per minute. We also achieved 99.995% PFAS reduction efficiency in a GAC waste that was previously not able to be processed by SCWO. Moreover, AFS adds only 3% to total CapEx but provides up 93% reduction in operating costs by maximizing the waste stream for SCWO operation. The successful completion of this Phase 2 program confirms that the AFS system meets necessary quality and operational standards, validating its effectiveness for long-term SCWO applications and highlighting AxNano's engineering achievement in advanced waste processing.

Our work also produced a technoeconomic analysis demonstrating that leasing the AxSure™ system is the most cost-effective solution over the 10-year period. Leasing offers a lower overall expenditure, eliminates sizeable upfront capital

costs, and reduces financial risk for organizations managing PFAS waste. Predictive leasing expenses provide financial flexibility while ensuring access to cutting-edge waste treatment technology. Additionally, leasing removes concerns related to long-term maintenance and depreciation costs, making it an attractive option for entities that require efficient PFAS destruction without significant capital investment. This report equips environmental consultants, contractors, and decision-makers with key insights to determine the most technically sound and financially viable approach according to EPA's technology assessment framework.[10]

4 References

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(9) Zhang, H.; Chen, Y.; Liu, Y.; Bowden, J. A.; Tolaymat, T. M.; Townsend, T. G.; Solo-Gabriele, H. M. Relationships between Per- and Polyfluoroalkyl Substances (PFAS) and Physical-Chemical Parameters in Aqueous Landfill Samples. *Chemosphere* 2023, *329*. https://doi.org/10.1016/j.chemosphere.2023.138541.

(10) EPA. *Interim Guidance on the Destruction and Disposal of Perfluoroalkyl and Polyfluoroalkyl Substances and Materials Containing Perfluoroalkyl and Polyfluoroalkyl Substances— Version 2 (2024)*; 2024.

5 List of Attachments

Attachment 1: Sampling Design and Rationale Plan
Attachment 2: Standard Operating Procedure for Sampling Liquid and Gas Samples Provided by Eurofins
Attachment 3: Standard Operating Procedure for Total Fluorine Method ELLE developed by Eurofins

Appendix B

Stakeholder Mapping for Integration into Department of Air Force Work Flow

Contract FA864923P1125
AxNano, LLC

1 Introduction and Scope

This work was supported by AxNano's SBIR Phase 2 contract FA864923P1125, entitled "Sustainable, Non-Incineration-based Destruction of Air Force Hazardous Waste Stockpiles." Successful commercialization of military and dual-use technologies requires a strategic approach to stakeholder engagement, ensuring alignment with defense priorities, regulatory requirements, and operational needs. This report presents the results of our stakeholder mapping activities as a critical tool for identifying key influencers, decision-makers, and end-users within the military ecosystem. By systematically analyzing stakeholders, technology developers can mitigate risk, streamline acquisition pathways, and enhance adoption within defense environments. The justification for this report lies in the need for a structured approach to navigate complex military procurement processes and optimize technology deployment for mission success. We intend for this stakeholder mapping report to serve as a resource for other technology developers in the military environmental space, helping them refine commercialization strategies, clarify procurement pathways, and understand the various stakeholders to engage in their commercialization efforts.

2 Methods

Our process was informed by guidance from the following DOD sources:

- Office of the Assistant Secretary of Defense (EI&E)[1]
  Interim Guidance on Destruction and Disposal of Material Containing (PFAS) in the United States
- Defense Logistics Agency (DLA)[2a,b]
  Disposition Services- Hazardous Waste Disposal Contract Model
- Defense Acquisition University (DAU)[3]
  Stakeholder Analysis & Engagement Decision Framework
- Stakeholder interviewing with AFCEC and Base Civil Engineering Flights
  Stakeholder Analysis & Engagement Decision Framework

3 Stakeholder Mapping

Civil and Environmental Flights for Air Base Wings

Numbered wings on bases. The host Wing's Civil Engineer Group is responsible for providing, operating, maintaining, restoring, and protecting the managed assets of the base infrastructure and facilities, including environmental stewardship, energy and utility systems, for the entire base (host Wing and tenants). Host Wing's CE Environmental Group (CEG) or Squadron (CES) will hold the RCRA permits for handling or processing hazardous materials or wastes. Their buy-in is important because they will be the "end-users" if reactors were to be integrated into base wastewater, drinking, and process treatment facilities. Environmental Flights answer to the Air Wing Commander that hosts the base. This Commander is responsible for environmental activities. Commander has ultimate authority as waste generator, but may delegate manifest signing to CEGs.

Air Force Civil Engineer Center (AFCEC)

Provides advisory and technical support to the Environmental Flights on compliance, restoration, NEPA, Legal and Reg. Remediation Project Managers (RPMs) can be tenants on a base, or located at HQ in San Antonio. AFCEC Branch manager (geographically), then reports to AFCEC/CZR (San Antonio). HQ Tyndall has AFCEC subject matter experts (e.g., fire, construction).

Their buy-in is important because they are recommenders of technology for base CEG and customers for restoration technologies. AFCEC is a direct reporting unit. Commander can sometimes delegate waste generator authority to AFCEC tenant. AFCEC is responsible for the Air Forces CERCLA response to PFAS, so their buy-in is important for PFAS-specific treatment technologies.

Contractors and Environmental Engineering Consultants

Are contracted to perform clean-up activities, transport and dispose of waste off-base. These are channel partners who will be contracted by CEGs (on-base civil work), AFCEC (restoration), or DLA (waste transport).

For restoration work, waste disposal (on-site treatment, off-site landfill or off-site incineration) is responsibility for restoration contractors as specified in the PWS. These stakeholders are important as they would lease and/or operate reactors for on-site restoration activities. In some cases, AFCEC may lease the unit and contractors would operate and maintain (O&M).

Transfer Storage and Disposal Facilities (TSDF)

Facilities that can receive hazardous and nonhazardous waste for disposal (incineration, landfilling, or other emerging methods). TSDFs are important as these are likely to procure reactors for off-site disposal activities.

For base operations/waste disposal, waste disposal is a contract line item for TSDFs through Defense Logistics Agency (DLA). CE Hazardous Material/Hazardous Waste manager will coordinate disposal with DLA according to waste contents and associated CLIN. These stakeholders are important to have a SCWO on their facility as an offering for DLA.

Air Force Materiel Command (AFMC) (MAJCOM)

Acquires technologies. AFSC is one of six centers in AFMC. This group is separate from AFCEC and Base CE, but can be an champion customer of equipment solution because AFSC/EN (environmental) is tasked with developing hazard reduction.

4 Conclusions

For customers generating enough wastes to support Scenario 2 and 3, procurement and contract operations of an AxSure™ will provide a 10 year 72-77% savings compared to landfilling or incineration. For customers unable to provide enough wastewater, then landfilling is currently the most economic option. However, we note that long-term liability is not included in these cost estimates.

AxNano's Automated Feed System is essential to AxSure™ technology adoption because mixing solid and liquid waste streams is necessary to make the SCWO process cost competitive with landfilling. On-site treatment is economical if customers are generating at least 930,000 gal liquid waste per year to support continuous operation. Systems for Air Force restoration activities could be procured or leased by AFCEC and operated by contractors. Systems for Air Force reoccurring wastewater sources could be procured or leased by the base host Wing and operated by Civil Engineering Squadrons or contractors. Air Force customers that cannot support on-site AxSure™ could use SCWO only if a system was available at a TSDF. Off-site disposal requirements would be specified in a restoration PWS (performance work statement) or according to a contract line item (CLIN) of a DLA contract

- AxNano's Automated Feed System is essential to AxSure™ technology adoption because mixing solid and liquid waste streams is necessary to make the SCWO process cost competitive with landfilling
- On-site treatment is economical if customers are generating at least 930,000 gal liquid waste per year to support continuous operation
- Systems for Air Force restoration activities could be procured or leased by AFCEC and operated by contractors

- Systems for Air Force reoccurring wastewater sources could be procured or leased by the base host Wing and operated by Civil Engineering Squadrons or contractors
- Air Force customers that can not support on-site AxSure™ could use SCWO only if a system was available at a TSDF
- Off-site disposal requirements would be specified in a restoration PWS (performance work statement) or according to a contract line item (CLIN) of a DLA contract

5 Summary

In summary, this document provides a comprehensive cost analysis of the AxSure™ system, highlighting the financial considerations of ownership versus leasing. The AxSure™ system integrates an automated feed management system (AFS) with a supercritical water oxidation (SCWO) process to efficiently eliminate contaminants like PFAS, which traditional wastewater treatments fail to remove.

The analysis demonstrates that leasing the AxSure™ system is the most cost-effective solution over the 10-year period. Leasing offers a lower overall expenditure, eliminates sizeable upfront capital costs, and reduces financial risk for organizations managing PFAS waste. Predictive leasing expenses provide financial flexibility while ensuring access to cutting-edge waste treatment technology. Additionally, leasing removes concerns related to long-term maintenance and depreciation costs, making it an attractive option for entities that require efficient PFAS destruction without significant capital investment.

This report equips environmental consultants, contractors, and decision-makers with key insights to determine the most financially viable approach. By leveraging this analysis, stakeholders can confidently choose leasing as the optimal strategy, ensuring effective PFAS destruction, regulatory compliance, and long-term cost savings.

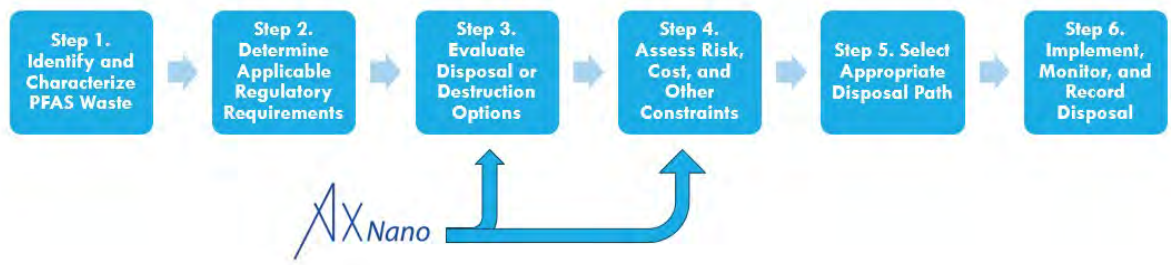

Figure 1. DAF Process for PFAS disposal. AxSure(TM) SCWO for disposal inserts into steps 3 and 5 as indicated.

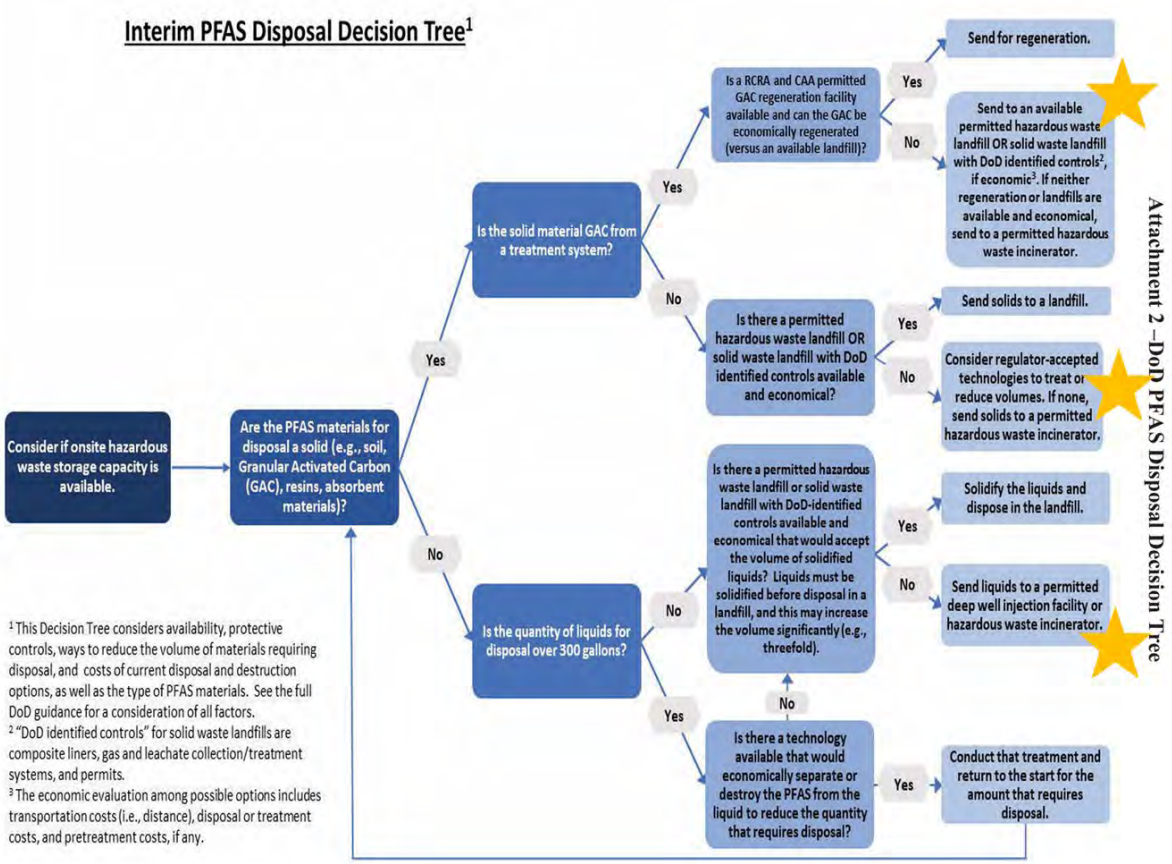

Figure 2. AxSure™ inserts into the PFAS disposal decision tree at three points, providing an alternative to incineration, landfilling and deep-well injection.

6 References

https://www.epa.gov/pfas/interim-guidance-destruction-and-disposal-pfas-and-materials-containing-pfas

Appendix C

AxSure™ Process and Equipment Information to Support Future Customer Permitting

Contract FA864923P1125
AxNano, LLC

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Acronym List

°C	Degrees Celsius
µ	Micron
µg/L	Microgram per Liter
AFFF	Aqueous Film Forming Foam
AFS	Automated Feed System
As	Arsenic
Avg	Average
Be	Beryllium

BTU/lb	British Thermal Unit per Pound
CA	California
$CaCO_3$	Calcium carbonate
Cd	Cadmium
cfm	Cubic Feet Per Meter
CFR	Code of Federal Regulations
CO	Carbon Monoxide
CO_2	Carbon dioxide
COD	Chemical Oxygen Demand
Conc	Concentration
Cr	Chromium
DRE	Destruction and Removal Efficiency
Eff	Effluent
FID	Flame Ionization Detector
Inf	Fahrenheit
IX	Ion Exchange
GAC	Granular Activated Carbon
GLS	Gas-liquid Separator
gpm	Gallons per Minute
H	Height
HC	Hydrocarbon
HCl	Hydrochloric Acid
HF	Hydrofluoric Acid
HFPO-DA	Hexafluoropropylene oxide dimer acid
HP	High pressure
HPAC	High Pressure Air Compressor
HRMS	High-Resolution Mass Spectrometry
Hz	Hertz
Inf	Influent
L	Length
lb	Pounds
LP	Low pressure
mg/DSCM	Milligrams per Dry Standard Cubic Meters
mg/L	Milligram per Liter
min	Minutes
MJ/kg	Mega Joules per kilogram
MPa	Mega Pascal
N_2	Nitrogen gas
NAAQS	National Ambient Air Quality
NaOH	Sodium hydroxide
NESHAP	National Emission Standards for Hazardous Air Pollutants
ng/DSCM	Nanograms per Dry Standard Cubic Meters
ng/L	Nanogram Per Liter
ng/m^3	Nanogram Per Cubic Meter
NO	Nitric oxide
O_2	Oxygen gas
Pb	Lead
PFAS	Per- And Polyfluoroalkyl Substances
PFBS	Perfluorobutanesulfonate
PFHxS	Perfluorohexanesulfonic Acid
PFNA	Perfluorononanoic Acid

PFOA	Perfluorooctanoic Acid
PFOS	Perfluorooctane Sulfonate
PID	Photoionization Detector
ppm	Parts per million
ppmv	Parts per million by volume
psig	Pounds per Square Inch Gauge
s	Seconds
Sb	Antimony
scfm	Standard cubic feet per meter
SCWO	Supercritical Water Oxidation
SO_2	Sulfur dioxide
SVOCs	Semi-Volatile Organic Compounds
TEQ	Toxic Equivalents
TDS	Total Dissolved Solids
TOC	Total Organic Carbon
TSS	Total Suspended Solids
USEPA	Unites States Environmental Protection Agency
V	Volts
VOCs	Volatile Organic Compounds
W	Width
wt%	Weight percentage

1 Introduction and Scope

AxNano's AxSure™ system combines an automated feed management system (AFS) with a supercritical water oxidation (SCWO) process to effectively destroy contaminants that persist through traditional wastewater treatment methods. The AFS is designed to optimize waste feed preparation and ensure a uniform and consistent input into the SCWO reactor. The SCWO reactor then uses high temperatures and pressures to completely break down organic contaminants into carbon dioxide, water, and inorganic salts, providing a clean and effective waste treatment solution as an alternative to landfilling and incineration.

This document provides detailed information on AxSure™ system components, operational requirements, and process outputs in support of site preparation and permitting efforts. While it does not outline specific permitting requirements, it offers essential technical data that aids in regulatory compliance, project planning, and system integration into existing or new waste treatment sites. This information is intended to assist environmental consultants, project managers, and general contractors in preparing for AxSure™ installation.

This work is supported by a Small Business Innovative Research contract FA864923P1125 where the objective of the program is to build, design and test our AFS prototype and demonstrate that it is a valuable component of an overall SCWO system by supporting continuous operations with a range of waste feed types. This document has been created to help support future technology commercialization and customer acquisition, which is a key mission of the Small Business Innovative Research program. Future customers will likely be environmental contractors or consultants that want to use SCWO for destruction of PFAS-containing wastes on behalf of their clients. The customers will have to apply for appropriate permits based on site location, waste type, and current regulations. Components of the permit application will be specific to our technology, and therefore those customers will need our input. This document includes representative prior data intended to provide insights into expected system performance, destruction efficiency, and operational parameters under real test conditions. These metrics support future customer permit processing, compliance documentation, and operational planning. By understanding these components and requirements detailed in this document, customers can better anticipate site logistics, address permitting challenges, and ensure successful deployment of AxSure™.

2 Technology

2.0 Technology Overview

AxSure™ technology utilizes Supercritical Water Oxidation (SCWO) to achieve complete destruction of organic contaminants by leveraging the unique properties of water in its supercritical state. When water is heated above its critical temperature (374°) and pressurized beyond its critical point (221 bar), it enters a supercritical phase where it

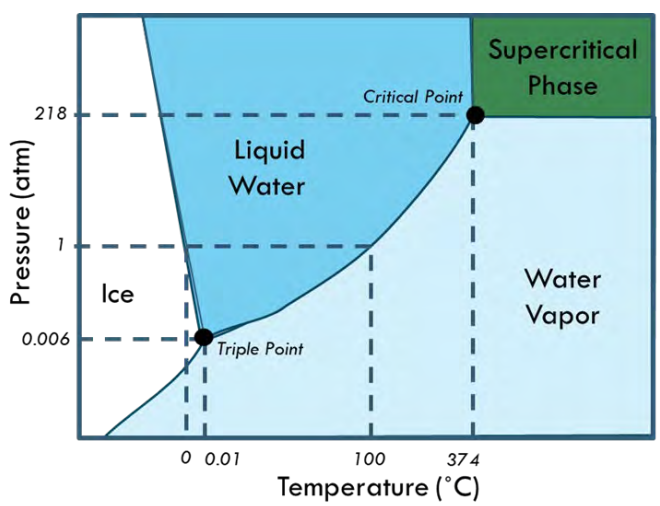

Figure 1. Water phase diagram showing the supercritical water region.

behaves neither as a liquid nor a gas but as a dense fluid with exceptional solvating and diffusive properties (Figure 1). When this state is achieved, water exhibits some unique characteristics such as reduced hydrogen bonding, high diffusivity, zero surface tension, small dielectric constant, low viscosity, controllable dissociation constant, and excellent transport properties. Under these conditions, substances that are typically insoluble in water, such as oils and other organics, mix thoroughly with supercritical water, along with the oxidant and reaction byproducts. Any salts present will become non-soluble and fall out of solution.

The unique chemistries in the supercritical phase allow SCWO to destroy organic materials with higher efficiency than typical thermal treatment methods, such as incineration. The process involves introducing organic waste, water, an oxidant, and sometimes a fuel into a reactor operating at temperatures between 650°C and 700°C and pressures around 221 bar (~3,200 psig). The oxidant is typically oxygen either in the form of oxygen gas or oxygen-enriched air. Our system uses oxygen-enriched air supply. While 374°C is the minimum temperature required for water to reach supercritical conditions, our SCWO systems typically operate at 650°C to promote higher contaminant destruction efficiency within short residence times of 10-12 seconds. (Sahle-Demessie, n.d.)

The single-phase environment of SCWO ensures rapid, homogeneous mixing, eliminating mass transfer limitations that are common in conventional processes, such as adsorption, chemical oxidation, and biological treatment. These conditions allow for the complete and efficient oxidation of any organic compound, reducing it to simple and unreactive byproducts including carbon dioxide, water, and inorganic salts. SCWO is particularly effective for disposing of hazardous substances in complex waste streams that are challenging for other technologies, making it an ideal solution for environmental remediation and industrial waste management applications. Examples of these waste streams include PFAS-impacted solids (GAC and resin), landfill leachate, reverse osmosis reject, still bottoms, and industrial waste where a mixture of contaminants may exist in high total organic carbon loading, high salt concentrations, or mixed phases of liquids and solids.

The AxSure™ unit has a total capacity of 3 gallons per minute (gpm), including water, fuel, and waste feeds combined. The feed nozzle is designed to ensure thorough mixing of the oxidant with the fuel, water, and waste feed at the reactor inlet. Throughput may be increased with modifications to the pumps and process. The system is designed for continuous 24 hours a day, 7 days a week operation, though actual runtime depends on factors such as unplanned shutdowns or changes in waste energy content.

The Automated Feed Management System (AFS) is designed to improve the operating efficiency of technologies like AxSure™ when managing solid and complex waste streams, ultimately supporting improved destruction efficiency. SCWO processes have specific boundary conditions for what the waste feed must be in terms of chemical and physical properties. The AFS focuses on manipulating waste stream properties, such as reducing particle size, controlling solid-liquid ratios, and ensuring proper mixing, to create a uniform feed stream enabling continuous flow operation and enhancing waste destruction performance. By processing multiple waste streams to meet the feed parameters required of the SCWO process, the AFS supports reliable processing of diverse waste types without energy- and time-intensive shut-downs and start-up.

2.1 System Components and Process Overview

Solid and complex waste is processed through the AFS to create a slurry suitable for destruction technologies. A process flow diagram for the AFS is shown in Figure 2. A predetermined amount of solid or granular material is placed into a hopper, which feeds the material at a controlled rate into a mixing tank. There, it is combined with a specified amount of liquid to create a slurry with a defined solid-to-liquid ratio. The slurry is processed through a particle size reduction system and returned to the mixing tank in a continuous loop, which ensures consistent particle size reduction and uniform distribution within the mixture. Adjustments of timing, pressure, and feed rates are made based on the composition and particle size of the initial material until the desired composition is achieved. Once the slurry reaches the target specifications, it is transferred to a holding tank before being introduced into the SCWO process.

Additional process equipment used for the AxSure™ process, following the AFS includes a high-pressure air compressor (HPAC), a fuel-fired preheater, a high-pressure SCWO reactor, a pressure letdown system, a low-pressure

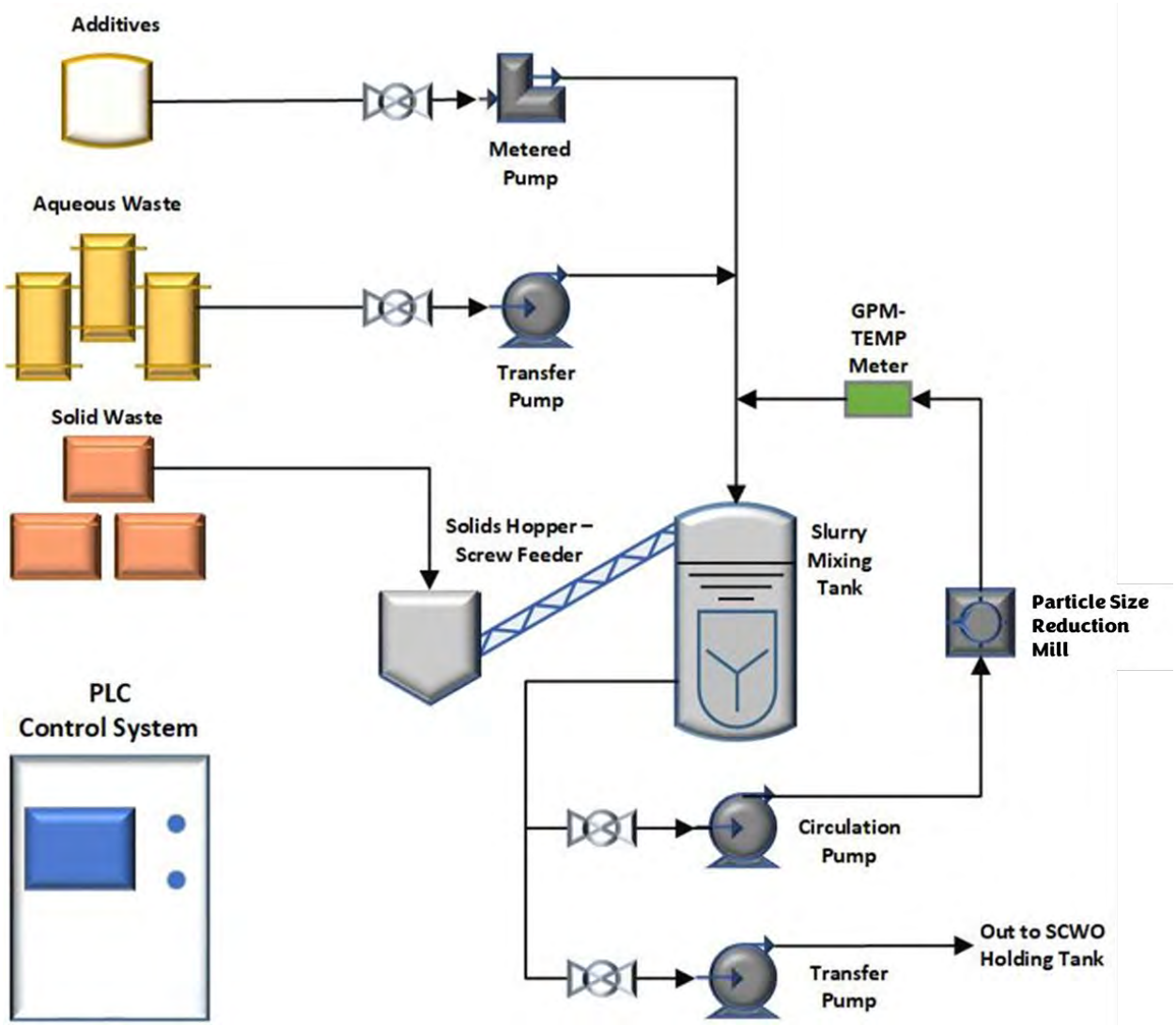

Figure 2. Automated Feed Management System (AFS) concept design for solid waste transferring, griding, and slurry preparation.

gas-liquid separator, chemical-resistant piping and connections, and holding tanks for influent waste and treated effluent. The system flow diagram is shown in Figure 3.

During normal AxSure™ operations, waste feed and water and/or fuel are injected into the high-pressure reactor through a top feed nozzle using a chemical injection pump for liquids and a slurry pump for solids (10-15%) prepared by the AFS system. The nozzle also supplies an oxidant, typically air. The air, waste, and diesel are mixed in a reaction zone maintained at 650°C, with the temperature regulated by an automated system that adjusts diesel flow as needed. Fuel such as a low-sulfur diesel is added when the waste's heating value is below ~5.1 MJ/kg (2,200 BTU/lb) to maintain sufficient reactor temperatures for high destruction efficiency. Alternatively, a second compatible waste stream can be blended to maintain the target energy content. If the energy content is higher, water will be added manually using the AFS to dilute the feed and prevent overheating of the reactor due to exothermic reactions. The reactor inlet pressure is set to 230 bar to ensure operating pressure remains above 200 bar, with variation influenced by flow rate, reaction kinetics, and temperature fluctuations.

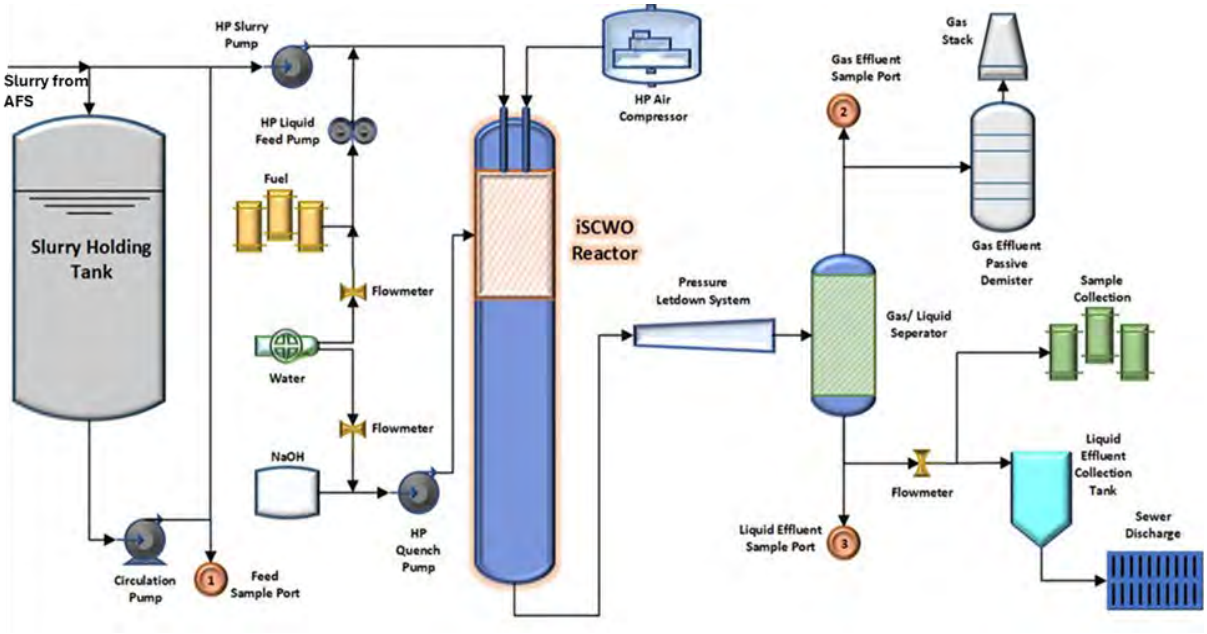

Figure 3. AxSure™ flow diagram for current SCWO process.

Quench water flowing outside of the reactor, containing a 5-30% sodium hydroxide (NaOH) solution, is mixed with the effluent near the reactor's base to cool the liquid to ~300°C before entering the pressure let-down system. The NaOH neutralizes inorganic acids such as hydrofluoric acid (HF) based on the input material analysis and real-time pH monitoring.

The quenched effluent exits through the reactor's pressure letdown system and then flows into a gas-liquid separator (GLS), where gas, mainly N_2, CO_2, O_2, and steam, is vented. The liquid is further cooled through a heat exchanger. The resulting liquid effluent, typically clean water with dissolved salts or metal oxides, is discharged from the system at around 55°C and collected in a storage tank. Permits required for corresponding air and water discharge are addressed in section 5 below.

3 Site Requirements

The customer is responsible for obtaining all required permits, preparing the site, and supplying the necessary infrastructure, utilities, and equipment to ensure the proper installation and operation of the AxSure™ system. This applies to a permanent setup; any differences for a temporary setup will be specified as needed. Below is a detailed breakdown of the requirements:

Site Preparation

- Equipment Dimensions:
  - Assembled AxSure™ system including SCWO and HPAC: 23 ft (L) × 8 ft (W) × 20 ft (H), weighing ~16,000 lb and 32,000 lbs, respectively (Note, height depends on stack requirements of site)
  - Assembled AFS system: 10 ft (L) × 6 ft (W) × 8 ft (H), weighing ~3,000 lb.
- Land and Access: Preparation of land and roads. The system should be installed on a level surface.
- Concrete Infrastructure:
  - Preparation of concrete for forming and pouring as required.
  - Approximately 6,000 sq ft of concrete:
    - Level 27 ft W x 75 ft L concrete pad for the AxSure™ system with an environmental canopy or metal shed for weather protection.
    - 30 ft W x 50 ft L concrete base for control room (small trailer).

- Roads and Traffic Control: Roads, fencing, signs, and traffic control.
- Lighting:
    - Indoor and outdoor lighting.
    - Lightning protection (grounding, lightning rods, or wire cables).
- Enclosures:
    - Metal housing or shed over the AxSure™ system and associated equipment, with at least 30 feet of vertical clearance or an easily removable roof for maintenance.

Utilities

- Electrical:
    - Supply of 480V, 3-phase, 60 Hz, 600 amps.
    - Supply of 240V and 120V for permanent power source (generators for temporary setup if needed)
    - Breakers, disconnects, conduit, and wiring.
    - Distribution system for power to the AxSure™ system and associated equipment.
    - Standby diesel generator with automatic transfer switch (as needed).
- Water:
    - Water softener (if needed)
    - Supply of 10 gallons per minute of softened or demineralized water, conditioned to contain:
        - <250 μm filtration, <500 ppm total dissolved solids, and hardness <5 ppm as CaCO3.
    - Storage tanks and piping to maintain continuous operation.
- Fuel:
    - Diesel fuel (low-sulfur #2) supply for the AxSure™ reactor and preheater.
    - Flow rate requirements:
        - AxSure™ reactor: Up to 0.4 gpm.
        - Preheater: 0.3 gpm during startup (approximately 30 minutes).
    - Fuel delivery line at 100 psig, -15 to 52°C (5 to 125°F).
    - Propane gas for the preheater's pilot flame (100-liter tank recommended).

Waste Feed and Effluent

- Feed:
    - Feed holding tank(s) (can be drums, totes, bladders, or frac tanks for liquids; drums, supersaks, or roll-offs for solids)
    - Waste feed input capable of delivering up to 3 gpm at 40±2 psig.
    - Waste must be maintained above freezing but below 52°C (125°F).
- Liquid Effluent:
    - Storage tanks with capacity to hold effluent volumes (if needed)
    - Stacks must have a discharge drain at the bottom of stack.
    - Gas-liquid separator drainage to effluent storage tanks.

Emissions and Compliance

- Stack System:
    - Two 20-ft stacks (or per local regulations), with discharge drains and demisters (if needed), extending above the edge of the canopy roof.
        - Stack diameter 12 inches
        - Air flow: dry air, 650 to 700 scfm
        - Gas velocity: 2,600 to 2,800 feet per minute
    - Scaffolding for inspection and sampling stations.
        - Air stack sampling requires two test ports at 15-20 ft at 90 degrees from each other.
        - Platform for stack inspections to take gas measurements (if needed)
    - Gas scrubbers for additional treatment (if needed)
- Gas Effluent Emissions Monitoring (as a Testo Add-on):
    - Continuous emissions monitoring systems for CO, CO_2, O_2, NO_x, SO_x, HC, HF, HCl, and VOCs.

    - Calibration gases for gas detectors such as FIDs and PIDs.

Infrastructure and Housing

- Control Room:
    - A small trailer serving as a control room, equipped with a workstation, desktop computers, and monitors supplied by the customer.
    - Positioned no more than 50 feet from the AxSure™ skid.

Process Air

- Air Supply:
    - High-pressure air compressor capable of delivering 150 psi, 20 cfm demand.
    - Air dryer with <40 µm particle size, <1 ppm lubricant content, and dew point < -30°C.

Neutralization (if needed)

- NaOH Supply:
    - Storage for up to 50% sodium hydroxide solution to neutralize acids.
    - Metering pump system for NaOH delivery.

Equipment Design, Fabrication, and Inspection Codes

- Before the system is delivered to the client, a factory acceptance test is conducted, which involves powering the system and verifying that all components function properly at a basic level. Following this, the system customer validation test is performed, including coupling to the HPAC and operating the system for several hours using water and diesel fuel to verify the system is functioning as designed.
- The AxSure™ system and its components will comply with the following codes and standards:
    - Pressurized Equipment: American Society of Mechanical Engineers (ASME) Boiler & Pressure Vessel Code (BPVC).
    - Materials: American Society for Testing and Materials (ASTM)/ASME or fabricator standards.
    - Electrical Equipment and Wiring: National Electrical Code (NEC), National Fire Protection Association (NFPA) 70.
    - Rotating Equipment Safety: Occupational Safety and Health Administration (OSHA) standards.

4 Permitting and Compliance

The customer is responsible for acquiring all necessary federal, state, and/or local permits for installation and operation of the AxSure™ system. Regulations regarding PFAS are ever changing, so customers should consult licensed and knowledgeable professionals. For example, CERCLA requires expeditious response to clean up contaminated areas. To ensure responses to identified release(s) of hazardous substances, pollutants, and contaminants occur quickly, CERCLA exempts response actions from the requirement to obtain federal, state or local permits related to any activities conducted completely on-site (USEPA, 1992). The U.S. Environmental Protection Agency (EPA) has clarified that surrogate or permit equivalency procedures should not slow the progress or increase the cost of CERCLA site remediation. However, if RCRA applies, then federal, state, and local permits would be required. Some examples of permits that may be required are listed below; however, this is not an exhaustive list, and it is the responsibility of the customer to ensure all applicable federal, state, and local permits are obtained for the system prior to installation and operation.

- Stormwater Permit
- Electrical Permit
- Grading Permit
- Heating Permit
- Building Permit
- Hazardous Waste Storage Permit
- Concrete footing/foundation/slab permit(s)
- Air Permit
- Plumbing Permit
- Fencing/Signage Permit
- AST Installation Permit
- Erosion Control Permit
- Construction General Permit
- National Pollutant Discharge Elimination System (NPDES) permit

5 Analytical Measurements and Methods

The following tables provide recommended standard measurements and analytical methods for measuring AxSure™ inputs and outputs. Table 1 details liquid parameters and corresponding analytical methods and Table 2 covers gas parameters and their respective analytical methods. Each measurement serves a specific purpose but is not limited to those listed and may vary according to federal, state, and local regulations. Specific measurements and methods depend on waste stream characteristics and compliance obligations for discharge and emissions. It is the responsibility of the customer to monitor all outputs required to comply with standards.

Table 1. Analytical methods for AxSure™ liquid inputs and/or outputs.

Parameter	Analytical Method	Purpose
Total Fluorine	Direct CIC	Efficiency, mass balance, regulation; important to know approximate F content for neutralization
Target PFAS	USEPA Method 1633	Contaminant of interest, mass balance, efficiency, regulation
Non-Target PFAS	HRMS	Contaminant of interest, mass balance, efficiency, regulation
TOP Assay	USEPA Method 1633	Contaminant of interest, mass balance, efficiency, regulation
VOCs	USEPA Method 8260C	Contaminant of interest, efficiency, regulation
SVOCs	USEPA Method 8270D	Contaminant of interest, efficiency, regulation
TOC	USEPA Method 9060A	Performance, process stability
COD	USEPA Method 410	Performance, process stability, proxy for waste energy content
Anions	USEPA Method 9056A	Efficiency, mass balance
Cations	USEPA Method 6010C	Performance, waste management
Metals	USEPA Method 6010C	Waste management, reactor corrosion
TDS	SM 2540C	Waste management
TSS	SM 2540D	Waste management
pH	USEPA 9040C	Waste management, process stability

Table 2. Analytical methods for AxSure™ gas outputs.

Parameter	Analytical Method	Purpose
Polar Semi Volatile PFAS	OTM-45 / TO-13 modified	Contaminant of interest, mass balance, efficiency, regulation
Non-Polar PFAS	OTM-50	Contaminant of interest, mass balance, efficiency, regulation
Non-Polar Semi Volatile PFAS	OMT-55	Contaminant of interest, mass balance, efficiency, regulation
VOCs	TO-15	Contaminant of interest, air regulation
Carbon dioxide	USEPA Method 3A	Compliance monitoring, carbon trading
Carbon monoxide	USEPA Method 10	Regulated by the Clean Air Act's NAAQS
Chlorine	USEPA Method 0050	NESHAP compliance monitoring
Hydrofluoric acid	USEPA Method 26	Compliance monitoring
Hydrogen sulfide	USEPA Method 11	Compliance monitoring
Lead	40 CFR Part 50	Regulated by the Clean Air Act's NAAQS
Methane	40 CFR Part 53	Compliance monitoring
Nitric oxide	USEPA Method 7E	Air regulation
Nitrogen dioxide	USEPA Method 7E	Regulated by the Clean Air Act's NAAQS
Oxygen	USEPA Method 3A	Performance, process stability
Ozone	40 CFR Part 53	Regulated by the Clean Air Act's NAAQS

Parameter	Analytical Method	Purpose
Particulate matter	USEPA Method 5	Regulated by the Clean Air Act's NAAQS
Sulfur dioxide	USEPA Method 6C	Regulated by the Clean Air Act's NAAQS

6 Representative Prior Data

The following representative data from prior pilot system operations highlights typical influent and effluent characteristics. These values reflect performance under standard operating conditions for processing waste. The data will vary depending on specific waste streams and system configurations.

Typical gaseous effluent concentration for processing wastes primarily composed of carbon, hydrogen, oxygen, and nitrogen are listed on Table 3. Waste containing dissolved metals, halogens, or sulfur can produce oxides in the gaseous effluent, which may require additional treatment or control measures. Volatile organic compounds will likely be degraded in the reactor; however, if feasibility studies indicate otherwise, then gas phase abatement accessories can be added to the stack.

Table 3. Typical gas analysis from SCWO gaseous effluent.

Type of Emissions in Effluent Gas Stream	Characteristic Performance (no gas cleanup)
Dioxins/furans, ng/DSCM (TEQ)	< 0.006
Particulate matter, mg/DSCM	< 4
Toxic metals, mg/DSCM	< 0.015 for Cd + Pb < 0.015 for Sb + As + Be + Cr
Organics DRE, %	> 99.9999
HCl, ppmv	< 0.4
NO_x, ppm	< 1
SO_x, ppm	<1
CO, ppm	< 10
Hydrocarbons, ppm	< 10

A validation test was conducted at the General Atomics facility in San Diego, CA in March 2021(McDonough et al., 2022). Two continuous-flow test runs were performed, each lasting approximately 120 minutes. A 1000-fold diluted AFFF solution was used as the test material for this validation. The test parameters and associated measurements for both runs are summarized in Table 4. Note that total system effluent volumes are higher than influent volumes (including waste and diesel) due to the quench water system used for neutralization.

Table 5 shows the analytical results for PFOS (the major component of AFFF), PFHxS (the second highest constituent), and PFOA (another widely used PFAS). Concentrations are shown for the influent, quench, and liquid and gaseous effluents of Run 1 and 2. Also shown is the average destruction and removal efficiency (DRE), which is a measure used to evaluate the effectiveness of a treatment process in breaking down and removing contaminants from a waste stream. DRE is calculated as:

$$Destruction\ and\ Removal\ Efficiency\ = \frac{W_{in} - W_{out}}{W_{in}}$$

Where W_{in} and W_{out} are mass feed rate of a particular pollutant.

Table 4. Test parameter and associated measurements for SCWO test on AFFF destruction. (McDonough et al., 2022).

7	Test Parameter	8	Run 1	9	Run 2
10	Diesel consumed (L)	11	107	12	106
13	Diesel feed rate (lpm)	14	0.89	15	0.88
16	AFFF concentrate (L)	17	1.1	18	1.0
19	Influent AFFF (L)	20	1,085	21	997
22	Influent AFFF feed rate (lpm)	23	9.1	24	8.3
25	Liquid effluent volume (L)	26	1,359	27	1,301
28	Liquid effluent flow rate (lpm)	29	11.3	30	10.9
31	Quench volume (L)	32	2,844	33	2,768
34	Quench feed rate (lpm)	35	23.7	36	23.1
37	Influent air flow rate (lpm)	38	19,255	39	19,255
40	Influent air volume (L)	41	2,310,651	42	2,310,651
43	Run time (min)	44	120	45	120
46	Residence time (s)	47	10-11	48	10-11
49	SCWO temperature (°C)	50	649.1	51	648.5
52	SCWO pressure (MPa)	53	22	54	22
55	% Excess oxygen	56	10.6%	57	12.4%
58	pH post-Quench	59	6.16	60	6.05
61	Corrected stack flow rate (dscm)	62	20,181	63	19,512
64	Metered gas volume (dscm)	65	2.2	66	2.4
67	Water collected in sample train (L)	68	1.9	69	2.3

Table 5. Analytical results for the influent, quench, and liquid and gaseous effluents.

PFAAs	Liquid Influent results (ng/L)	Liquid Effluent results (ng/L)	Quench (ng/L)	Gaseous Sum ng/train	Gaseous Conc. ng/m^3	Avg DRE % (Run1 & 2)
Run 1						
PFOA	72,210	2.2	2.1	2.6	1.21	99.99434%
PFOS	10,780,000	5.6	4.3	3.4	1.57	99.99996%
PFHxS	717,100	1.67 J	1.52 J	1.6	0.72	99.99969%
Run 2						
PFOA	83,220	1.79 J	1.48 J	4.4	1.87	
PFOS	12,590,000	3.4	3.1	3.7	1.57	
PFHxS	855,600	1.44 J	0.914 J	1.9	0.80	

J, results are less than the reporting limit but greater than or equal to the method detection limit and the concentration is an approximate value.

Another pilot test of the SCWO system was conducted in August 2024 at the General Atomics facility in San Diego, California. The demonstration utilized four different waste streams: RO reject water, AFFF wastewater, spent GAC, and IX resin. The resulting test parameters and analytical results from the test are summarized in Table 6 and 7. Note that for GAC and resin processing, no diesel fuel was used because the waste contained a high enough heating value to maintain the reactor temperature.

Table 6. Test parameters and associated measurements for various waste streams tested in a second pilot SCWO test.

Test Parameter	AFFF Wastewater	RO Reject	GAC	Resin
Diesel consumed (L)	149	144	0	0
Diesel feed rate (lpm)	1.1	0.6	--	--
Influent waste (L)	135	1,968	315	140
Influent waste feed rate (lpm)	1.0	8.2	7.0	7.0
Liquid effluent volume (L)	270	2,016	369	166
Liquid effluent flow rate (lpm)	2.0	8.4	8.2	8.3
Quench volume (L)	1,444	4,944	945	426
Quench feed rate (lpm)	10.7	20.6	21.0	21.3
Run time (min)	135	240	45	20
SCWO temperature (°C)	649.1	645.0	646.5	647.3
Reactor pressure (bar)	227.9	200.1	204.8	191.9
Process water (lpm)	4.3	0.6	3.7	5.6
pH post-Quench	8.57	7.55	7.69	6.82

Table 7. Analytical results for the various waste influent and liquid and gaseous effluents.

Test Parameter		AFFF Wastewater	RO Reject	GAC	Resin
Waste Description		Unknown diluted AFFF composition	RO reject from industrial wastewater treatment	Reactivated carbon waste from groundwater treatment	Purofine™ PFA 694E waste
Total PFAS (ng/L)		10,124	13,443	1,115,224	134,112
Normalized Effluent Total PFAS (ng/L)		414.4	45.4	50.4	1,991.3
Total Salt (mg/L)		2.7	2,508	257	268
Particle Size (μ)		Liquid	Liquid	98% <2,000	500-1,110
Total Effluent solids (wt%)		1.0%	0.5%	0.4%	0.2%
PFOA (ng/L)	Inf	120	68	69,000	7,100
	Eff	1.6	< 0.96	0.9	62
PFNA (ng/L)	Inf	< 48	< 5	1,400	230
	Eff	< 0.5	< 0.54	< 0.5	0.62
PFBS (ng/L)	Inf	170	2,200	3,900	2,600
	Eff	7.7	2.4	2.1	3.9
PFHxS (ng/L)	Inf	490	69	400,000	29,000
	Eff	6	1.9	< 0.8	8.7
PFOS (ng/L)	Inf	1,400	200	450,000	< 48
	Eff	20	4.3	5.2	< 0.61
HFPO-DA (ng/L)	Inf	< 120	< 12	< 50	< 120
	Eff	1.5	< 0.54	< 1.2	< 0.61
TOC (mg/L)	Inf	46	46	710	590
	Eff	34	8.0	3.8	3.3
Sodium (mg/L)	Inf	23	2,000	100	180
	Eff	4,000	1,700	840	470
Cadmium (mg/L)	Inf	< 0.001	0.0054	< 0.0052	< 0.0052
	Eff	< 0.001	0.003	< 0.0052	< 0.0052
Chromium (mg/L)	Inf	< 0.0031	0.23	0.094	0.21
	Eff	0.42	0.18	0.24	0.11
Nickel (mg/L)	Inf	< 0.0022	0.65	0.11	0.14
	Eff	0.15	0.48	0.23	0.31
Lead (mg/L)	Inf	< 0.0073	0.08	< 0.037	0.044
	Eff	< 0.0073	0.04	< 0.037	0.11

Test Parameter		AFFF Wastewater	RO Reject	GAC	Resin
1,4-Dioxane (µg/L)	Inf	< 3.0	6,500	< 8.5	< 8.5
	Eff	< 1.7	< 1.7	< 0.17	< 0.3
2-Butanone (µg/L)	Inf	890	21,000	< 250	< 25
	Eff	< 5.0	< 5.0	< 0.50	< 0.50
Benzene (µg/L)	Inf	<150	220	< 150	< 15
	Eff	< 3.0	< 3.0	< 3.0	< 0.30
Chloroform (µg/L)	Inf	< 3.0	< 150	170	< 15
	Eff	< 3.0	< 3.0	< 0.30	< 0.30
Trichloroethene (µg/L)	Inf	< 3.0	< 150	1300	130
	Eff	< 3.0	< 3.0	< 0.30	< 0.30
CO stack gas (mg/L)		1.17	0.50	1.14	1.67
NO stack gas (mg/L)		0.01	0.01	0.02	0.01
O_2 stack gas (mg/L)		18.06	14.6	14.24	14.11
NO_2 stack gas (mg/L)		0.01	1.10	0.04	1.47
SO_2 stack gas (mg/L)		0.02	0.01	0.02	0.02
CO_2 stack gas -Not measured					
Methane stack gas -Not measured					
Ozone stack gas -Not measured					
Particulate matter stack gas -Not measured					
Lead stack gas -Not measured					
Hydrogen sulfide stack gas -Not measured					
Chlorine stack gas -Not measured					

7 Summary

In summary, this document provides a comprehensive overview of AxSure™ system, detailing its components, operational requirements, and process outputs to support future customers efforts towards site preparation, permitting and system integrations efforts. The audience for this document includes environmental consultants or contractors that are overseeing AxSure™ installation on behalf of their clients. The AxSure™ system combines an automated feed management system (AFS) with a supercritical water oxidation (SCWO) process to effectively destroy contaminants, including PFAS, that persist through traditional wastewater treatment methods.

8 References

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